



GOOD PRACTICE GUIDE:

Approaches to Commissioning and Qualification of Pharmaceutical Water and Steam Systems

Third Edition

Streamlined Delivery Model
Quality Risk Management
C&Q Plan
Process Analytical Technology
Rapid Microbial Monitoring



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ISBN 978-1-946964-94-6

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Preface

The *ISPE Good Practice Guide: Approaches to Commissioning and Qualification of Pharmaceutical Water and Steam Systems* focuses on the engineering approaches and practices involved in providing cost effective water and steam systems in a cost and timely manner that are specified, designed, installed and operated to meet their intended purpose. Specifically, this Guide addresses the lifecycle process of designing, constructing, commissioning, and qualifying water and steam systems regulated by international healthcare authorities.

This Guide describes how a capital project delivers process, commissioning and qualification (verification) activities, and on-going monitoring work together through the validation lifecycle for GMP water and steam systems.

The main benefit of this Guide is that it describes project execution, commissioning and qualification (verification) approaches for GMP water and steam systems based on process understanding with the possible usage of Process Analytical Technology (PAT). The required quality of water or steam is achieved without unnecessary project expense due to duplication of effort and non-value added activities.

This Guide describes current, established, good practice, and references other Baseline and Good Practice guidance documents published by ISPE, where applicable. The focus is on achieving cost and time effective compliance with existing international regulatory mandates and guidances.

Acknowledgements

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Special Thanks

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The Guide Leads would like to express their grateful thanks to the many individuals and companies from around the world who reviewed and provided comments during the preparation of this Guide; although they are too numerous to list here, their input is greatly appreciated.

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Table of Contents

1	Introduction	7
1.1	Overview	7
1.2	Purpose.....	8
1.3	Objectives	9
1.4	Scope.....	10
1.5	Rationale for the Third Edition	12
1.6	Benefits	12
1.7	Guide Structure.....	12
1.8	Key Concepts.....	13
2	The Validation Lifecycle.....	17
2.1	Project Execution Planning	21
2.2	Design Phase Items.....	24
2.3	Commissioning and Qualification Plan	27
2.4	Procurement and Construction	28
2.5	Commissioning and Qualification.....	29
2.6	Traditional Approach to Qualification	43
2.7	Science and Risk-Based Approach to Qualification.....	44
2.8	Sampling	54
2.9	Performance Qualification.....	56
2.10	Project Completion and Document Delivery	58
2.11	Performance Qualification Final Reports	58
2.12	Ongoing Quality Monitoring	59
2.13	Periodic Performance Evaluation or Periodic Review (PR)	59
3	Process Analytical Technology (PAT)	61
3.1	Introduction	61
3.2	Definition of PAT.....	61
4	Rapid Microbial Monitoring	63
5	Appendix 1 – Typical WFI Generation and Distribution System Component Classification Example	67
6	Appendix 2 – Process Water, Pure Steam, and Process Steam	73
6.1	Non-Monographed/Non-Compendial Water	73
6.2	Steam.....	75
6.3	Pure Steam Quality [56].....	77
6.4	Steam Quality Testing [56]	78
6.5	Steam Dryness [56]	79
6.6	Non-Condensable Gases [56].....	80
6.7	Superheated Steam [56]	82
7	Appendix 3 – System Classification	89

8	Appendix 4 – Typical Specifications for Water and Steam Systems.....	91
8.1	Introduction	91
8.2	User Requirements	91
8.3	Detailed Design Specification/Functional Design Specification	93
9	Appendix 5 – Sampling and Testing Plan	95
10	Appendix 6 – Change Management During the Operation of Pharmaceutical Water and Steam Systems.....	97
10.1	Considerations	97
10.2	System Validation	98
11	Appendix 7 – Sample Turnover Package Requirements	101
12	Appendix 8 – References	105
13	Appendix 9 – Glossary.....	109
13.1	Acronyms and Abbreviations	109
13.2	Definitions	111

1 Introduction

1.1 Overview

Water and steam are used in cleaning operations, laboratory activities, and the manufacture of pharmaceutical and biopharmaceutical products. Water and steam may contact product or process contact surfaces or may be used in product testing.

Processes using Good Manufacturing Practices (GMP) produce drug substances or drug products. Water also may be used as a final product, e.g., as US Pharmacopoeia (USP) [1] packaged waters, European Pharmacopoeia (Ph. Eur.) [2], Japan Pharmacopoeia (JP) [3], and other pharmacopeial equivalents (e.g., World Health Organization (WHO) [4], and as a combination drug product, including:

- Sterile Water for Injection
- Sterile Water for Irrigation
- Sterile Water for Inhalation
- Sterile Purified Water
- Bacteriostatic Water for Injection
- Water for Hemodialysis

Systems that generate and distribute pharmaceutical water and steam are subject to commissioning, qualification, and/or verification to ensure that they will consistently deliver output meeting the specifications.

Commissioning and Qualification (C&Q) confirms that the system is installed and operating reliably with the necessary support systems in place, ensuring that the output will meet the specifications. This allows the system to be handed over and used to support operations.

This Guide describes the science and risk-based approach in alignment with the *ISPE Baseline® Guide: Volume 5 – Commissioning and Qualification (Second Edition)* [5]. An overview of the traditional approach is included.

Legacy systems may be qualified using a traditional approach, but it is highly recommended and preferential to qualify new systems using a science and risk-based approach.

The approach described in *ISPE Baseline Guide: C&Q (Second Edition)* [5] incorporates the foundational work required to implement such a science and risk-based model using Good Engineering Practices (GEP) to:

- Develop a concise User Requirements Specification (URS) ensuring an efficient process.
- Incorporate Quality by Design (QbD) into the design development lifecycle through design review/design qualification.
- Define an engineering quality system to allow engineering to deliver the documented evidence that a system is installed and operating to meet the specifications (installation and operational verification) using GEP.
- Conduct a risk assessment to ensure that a proposed design has an acceptable risk profile.

- Qualify the system by confirming that:
 - Design risk controls have been tested and found to be working as specified
 - Operational verification has been successfully completed to ensure operations across the entire intended range
 - Procedural risk controls are included in the relevant procedures
 - Engineering changes have been satisfactorily closed
 - Parts and spares lists are available
 - There is a list of operational and maintenance procedures
 - There is a list of personnel trained on the procedures
 - The system is incorporated in the maintenance and calibration programs
 - The commissioning summary report has been completed and signed by all responsible entities
 - Manuals, drawings, Piping and Instrumentation Diagrams (P&IDs), and any additional documentation are available
 - Conformity declarations are completed and delivered

This enables the system to move to Performance Qualification (PQ); the company will define at which PQ stage it is considered suitable for use operationally and what controls are required. For example, a system may be used in PQ stage 2 provided no product produced using the water is released prior to satisfactory completion of all Quality Control (QC) testing. An alternative is the use of Process Analytical Technology (PAT) (see Chapter 3), where water can be released as soon as all compendial mandates, URSs, and relevant Standard Operating Procedures (SOPs) are met.

1.2 Purpose

This Guide addresses pharmaceutical water and steam systems¹ C&Q by both science and risk-based and traditional approaches.

This Guide aims to assist in the integration of the capital project management process, the C&Q process, best practices for defining the PQ approach and the ongoing operation into the Quality Risk Management (QRM) validation lifecycle.

Ultimately, the goal is to increase understanding of GMP water and steam systems in the pharmaceutical and biopharmaceutical industries.

The main emphasis of this Guide is on high purity water and pure steam. Discussion and information on other pharmaceutical waters and steam are also provided. The USP [1], Ph. Eur. [2], and JP [3] define different types of compendial high purity water, and steam for bulk usage including:

- Water for Injection (WFI)
- Purified Water

¹ **Note:** Other approaches also may be used for commissioning and qualification as accepted by the appropriate regulatory authority.

- Pure Steam

The type of water and steam used for a specific pharmaceutical and biopharmaceutical process is dependent on a technical evaluation of which water is appropriate for that process. The appropriate water should align with expected classification of the water system as defined in the relevant pharmacopeia with enforcement by designated authorities such as the US Food and Drug Administration (US FDA), European Medicine Agency (EMA), China FDA (NMPA), etc.

Different compendial pharmaceutical waters have different specifications.

The requirement for a specific project must be defined in the URS – typically this is based on the most onerous requirement of the local pharmacopeia, and the pharmacopeial requirements of the markets served by the facility.

Harmonization of pharmacopeial mandates for WFI and Purified Water classifications between the USP [1], Ph. Eur. [2], Chinese Pharmacopoeia (CP) [6], Brazilian Pharmacopoeia (FB) [7], and JP [5] is an ongoing endeavor. At the time of publication of this Guide, there are still small differences and nuances between these pharmacopeias.

The intended audience for this Good Practice Guide is corporate, global, and local within an organization and in the following disciplines:

- Critical Utilities
- Facilities
- Engineering, Process and Project Management
- Business Process Owners
- Validation Team
- QC
- Quality Assurance
- Subject Matter Experts (SMEs)

1.3 Objectives

This Guide has seven principle objectives:

- Provide understanding on the importance of C&Q and how it fits into both a validation and a project lifecycle
- Explain and demonstrate an example of a science and risk-based risk management process for water and steam systems and what are typical system boundaries and critical aspects
- Discuss Critical Quality Attributes (CQAs), Critical Process Parameter (CPPs), Critical Aspects (CAs), and Critical Design Elements (CDEs)
- Discuss business essential parameters – key business considerations for design, construction, commissioning, and qualification. Some of these relate to non-product quality items like Health, Safety, Environmental (HSE) considerations, Information Technology (IT), or other “commissioning only” project scope items.
- Provide identification of and discussion on project scope activities included in the qualification of water and steam systems in the context of full project delivery

- Provide information on the URS, Functional Specification (FS) for automation, and how a science and risk-based QbD approach may be implemented.
- Ultimately, provide a robust science and risk-based method that can be applied to the qualification of GMP water and steam systems. PQ and operational management can follow a QRM strategy.

1.4 Scope

This Guide covers practices and activities associated with the C&Q or verification of pharmaceutical water and steam systems. Discussion is centered on those items that directly affect water or steam CQAs during production, storage, and distribution.

The scope includes pharmaceutical, biotechnology, medical device, and ATMP manufacturing as well as compounding pharmacies.

This Guide can be used as a standalone document as well as in conjunction with the *ISPE Baseline® Guide: Volume 4 – Water and Steam Systems (Third Edition)* [8], the *ISPE Baseline Guide: C&Q (Second Edition)* [5], and *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)* [9].

Note: The terminology used in this Guide is aligned with *ISPE Baseline Guide: C&Q (Second Edition)* [5]. However, the terminology is not critical and installation and operational qualification may be used, if aligned with company practices.

Installation Verification (IV), Operational Verification (OV) and the commissioning summary report should confirm that:

- The system commissioning was completed confirming that the installation and operation complies with the design specification.
- Engineering change management was done with quality engagement where appropriate. The installation is considered fit for the intended purpose with the installation and operation meeting the final revision of the specifications.
- The risk controls identified in the risk assessment are in place and operating as specified.
- The commissioning was completed following appropriate GEPs (see the *ISPE Good Practice Guide: Good Engineering Practice (Second Edition)* [10]), with defined deliverables.
- The scope of the engineering testing and inspection covered requirements for business, safety, and compliance with specifications (from a contractual and quality perspective).
- Typical documents delivered include:
 - IV and incorporate vendor site Factory Acceptance Testing (FAT) as appropriate
 - OV and incorporate Site Acceptance Testing (SAT) owner site as appropriate
 - Performance Verification and Qualification Summary
 - Commissioning/Verification Summary Report
 - Turnover Package

1.4.1 Regulatory Aspects

This Guide addresses compliance with international regulations and guidelines for water and steam systems, and considers:

- New systems and extensions to existing systems
- Existing validated systems

A wide range of healthcare requirements related to water and steam system compliance have been taken into account, including:

- GMPs
- ICH Q2(R2) [11], ICH Q7A [13], ICH Q8(R2) [14], ICH Q9 [15], and ICH Q10 [16]
- ASTM E2500 [70]

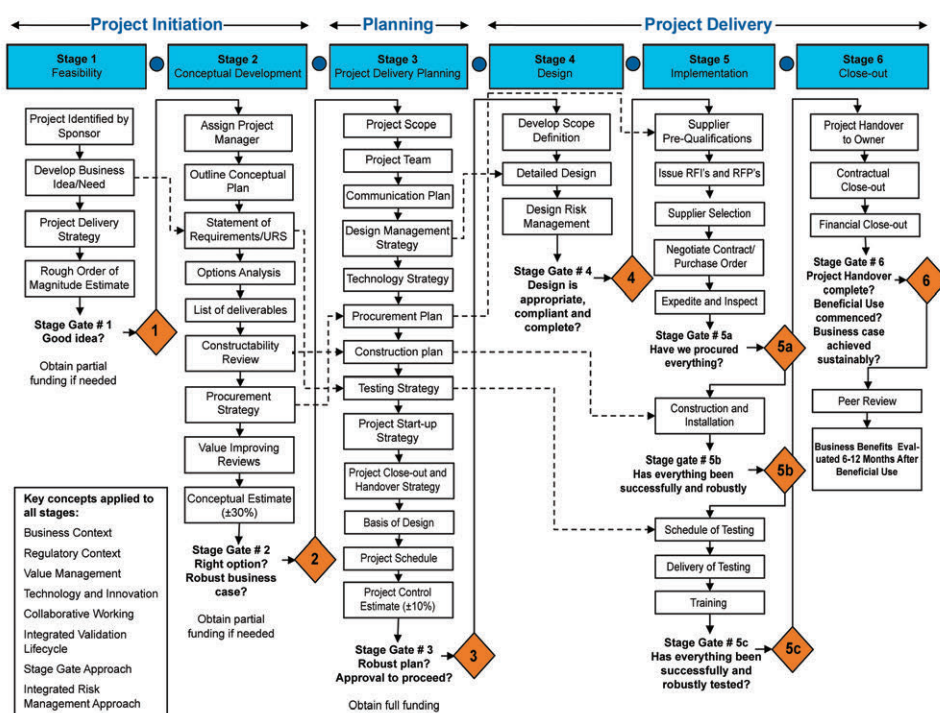
The following regulations and guidelines have been specifically considered in developing this Guide:

- FDA regulations and guidances, compliance policy guides, and 21st Century Initiatives [17]
- EudraLex Volume 4 GMP Guidelines [18], e.g., Annex 15 [19] and others
- Relevant sections of global pharmacopeias: Ph. Eur. [4], USP [1], JP [5], CP [6] FB [7]

The C&Q of such a system is typically considered an element of project execution to provide this system.

A typical project lifecycle is shown in *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry* [20]; aligned terminology is used in this Guide.

Figure 1.1: Project Management Flowchart [20]



1.5 Rationale for the Third Edition

The third edition of the ISPE Good Practice Guide: Approaches to Commissioning and Qualification for Water and Steam Systems is being published to align with the general changes in C&Q approaches developed and instituted over the past 10 years. Risk-based C&Q approaches are preferential to traditional C&Q procedures. Alignment with existing *ISPE Baseline and Good Practice Guides* [21] is imperative for cohesive procedures, documentation, reporting, and regulatory compliance.

1.6 Benefits

This Guide provides guidance on current best practices; adopting these should help to ensure efficient and compliant C&Q practices and activities associated with GMP water and steam systems. Benefits and attributes of this approach include:

- A streamlined delivery model minimizing duplicate testing, utilizing engineering documentation to provide the documented record that the installation and operation meets the specification. This approach uses SMEs in the most effective manner.
- Risk-based approach to the qualification of GMP systems is employed as compared to the traditional approach that repeats the engineering work, delivering separate but often voluminous Installation Qualification (IQ) and Operational Qualification (OQ) documents.
- The risk assessment provides a useful tool to reference when considering the potential impact of proposed changes and provides the supporting rationale for the classification of quality critical instruments.
- Integration of project management good practices

This Guide can be used by industry professionals involved in the design, construction, commissioning, and qualification of new or newly renovated water and steam systems that are regulated by the international healthcare and regulatory authorities.

Other benefits of a QRM-based qualification approach include:

- Efficient and cost-effective use of resources and minimum overlap in either commissioning and traditional validation or the use of PAT in lieu of traditional validation and process qualification
- Effective focus on objectives and deliverables through the entire lifecycle
- Minimal duplication and overlap in documentation
- Efficient handling of audits, inspections, and assessments

1.7 Guide Structure

Chapter 2: The Validation Lifecycle

Traditionally, the qualification strategy for water and/or steam systems has solely relied on retrospective execution of IQ and OQ documentation to verify that all system specifications are met. Commissioning documentation traditionally has not been or only moderately been leveraged to cover these qualification activities.

The traditional approach, although is still considered acceptable if adjusted to incorporate a risk-based strategy with change control. Change control is defined in the *ISPE Baseline Guide: C&Q (Second Edition)* [5]. Depending on the severity of the change, additional testing may be needed to comply with risk-based guidance.

In contrast, there are multiple benefits of transitioning from the traditional qualification approach to a more streamlined process such as the one proposed in this Guide and in the *ISPE Baseline Guide: C&Q (Second Edition)* [5], including:

- Avoiding extensive generic testing and unnecessary testing repetition between commissioning activities and IQ/OQ protocols; instead, the qualification strategy is tailored to the individual system needs
- Better flexibility and adaptability; this approach focuses on topics where the risk is higher. With better flexibility and adaptability, the cost/time is significantly reduced compared to the traditional approach.
- Better alignment with current regulatory expectations, promoting a QbD strategy
- Product quality as a central point of the strategy, instead of exclusively focusing on generic requirements

Chapter 3: Process Analytical Technology (PAT)

Chapter 4: Rapid Microbial Monitoring using Online Water Bioburden Analyzers

Chapters 5 to 7: Appendices

The appendices are intended to provide examples of documents such as component classifications, FDSs, URS, etc. The sample documents show the type of information typically included and tasks to be conducted. These examples are not intended to be absolute guidelines that must be explicitly followed. Equipment, processes, temperatures, flows, pressures, and other parameters are examples of current methods; however, other equally effective alternate methods and processes are also in operation.

1.8 Key Concepts

Key concepts and principles of a global implementation as used within this Guide include:

Commissioning

Commissioning is a well-planned, documented, and managed engineering approach to the start-up and turnover of facilities, systems, and equipment to the owner/user that results in a safe and functional environment that meets established design requirements and stakeholder expectations.

In this Guide, the term commissioning describes the GEP activity used to provide documented evidence that a system is installed and operates to meet the specifications.

Critical Aspects (CA) [5]

Are typical functions, features, abilities, and performance or characteristics necessary for the manufacturing process and systems to ensure that critical process parameters are consistently controlled, and other requirements are met (e.g., regulatory requirements). They should be identified and documented based on scientific product and process understanding.

Critical Design Element (CDE) [5]

Components, instruments, materials of construction and features of an engineered system that are necessary to consistently achieve required process controls enabling manufacture of products meeting Critical Quality Attributes (CQAs). Examples of automation features include ability to alarm and to manage data. CDEs are identified and documented based on technical understanding of the product CQAs, process CPPs, and equipment design/automation. CDEs are verified through Commissioning and Qualification (C&Q). CDEs are the outputs of the design development lifecycle.

Critical Process Parameter (CPP) (ICH Q8(R2) [14])

A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces desired quality.

Critical Quality Attribute (CQA)

A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality [14].

Attributes of the water or steam product that usually relates to the identity, purity, or quality of the water or steam product. CQAs for water and steam (condensate) include Total Organic Carbon (TOC), conductivity, microbial presence, endotoxin, etc.

A CQA is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. CQAs are generally associated with the drug substance, excipients, intermediates (in-process materials) and drug product.

Quality (ICH Q6A [12])

The suitability of either a drug substance or drug product for its intended use. This term includes such attributes as the identity, strength, and purity.

Quality by Design (QbD) (ICH Q8(R2) [14])

A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.

Quality Risk Management (QRM) (ICH Q9 [15])

A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle.

Qualification

The process to demonstrate the ability to fulfill specified requirements.

System

An organization of engineering components that have a defined operational function, e.g., piping, instrumentation, equipment, facilities, computer hardware, computer software.

A **System Process Parameter (SPP)** is a term used for process variables that have an impact on a system Quality Attribute (SQA) and, therefore, should be monitored or controlled to ensure the system product obtains the desired quality.

This is also known as an Equipment Process Parameter.

A **System Quality Attribute (SQA)** is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired system product quality.

A **System Quality Attribute (SQA)** and **System Process Parameter** are terms used in the *ISPE Baseline Guide: C&Q (Second Edition)* [5] are CQA and CPP, as defined in ICH Q8(R2) [14]. Some companies prefer to use these terms for product but not for qualification.

Where equipment is used for a number of products, or, as in this case where the system is a utility system, the CQAs/ CPPs may not be the same for all products (as defined in the regulatory filings). In this case, a platform approach to the system risk assessment may be used that address all the relevant CQAs and CPPs and the ICH definition refers to product.

For this reason, during C&Q, a company may choose to use different terminology, modifying the definitions from ICH Q8(R2) to encompass process and utility systems; typical terms seen include:

This is also known as an Equipment Quality Attribute.

User Requirements Specification (URS)

A description of the requirements of the facility in terms of product to be manufactured, required throughput, and conditions in which the product should be made.

Validation Lifecycle

Listing of a series of events within the life of a direct impact manufacturing system that are identified as important to create or maintain product quality over the life of the direct impact system when using traditional C&Q methodology.

Verification (ISO [22]) (1) (ASTM [23]) (2)

(1) Confirmation, through the provision of objective evidence that specified requirements have been fulfilled. (2) A systematic approach to verify that manufacturing systems, acting singly or in combination, are fit for intended use, have been properly installed, and are operating correctly. This is an umbrella term that encompasses all types of approaches to assuring systems are fit for use such as qualification, commissioning and qualification, verification, system validation, or other.

2 The Validation Lifecycle

Water and steam systems usually start off as capital construction projects.

Many tasks in the validation lifecycle and the project lifecycle are aligned in timing and scope. The phases within a project lifecycle and the interdependencies and interrelationships between lifecycle elements need to be understood in order to grasp the concept of a validation lifecycle.

Both lifecycles are optimized when activities are appropriately coordinated in an approved manner to eliminate unnecessary duplication of testing efforts and reporting. This is the basis for the application of a science and risk-based C&Q process where GEP based testing and related documentation are used to support GMP requirements for qualification.

Safety is critical for all water and steam systems. This is addressed through GEP and specific details are outside the scope of this Guide.

One of the key elements of science and risk-based management is the determination of CQAs and the associated CPPs. Understanding the CQAs and the CPPs provides the basis for the application of QRM.

Control of the CPPs ensures that the system output meets the CQAs. This approach is described as the system control strategy.

This chapter addresses C&Q activities specific to water systems and steam systems. These considerations are intended to address the lifecycle execution process and site-specific commissioning, qualification, and PQ processes. The information in this chapter is aligned with the approach described in the *ISPE Baseline Guide: C&Q (Second Edition)* [5].

A science and risk-based approach to validation starts with an understanding of the water or steam generation process, and the definition of what constitutes quality: the CQAs.

In addition, application of a science and risk-based approach requires an understanding of the process design in order to define the CQA and the associated CPPs for each stage of the process. Examples of typical CQA and CPPs associated with WFI systems and pure steam systems are shown in Table 2.1.

Table 2.1: Typical CQAs and CPPs Associated with Water for Injection and Pure Steam Systems

System Name	Typical CQAs	Typical CPPs
Water for Injection (WFI) System	- Conductivity at temperature⁽¹⁾ - TOC - Microbials - Endotoxin	- System water level⁽²⁾ System operating: - Pressure - Temperature - Flow rates
Pure Steam System	- Steam condensate meets conductivity at temperature¹ - TOC - Endotoxin	- Feed water temperature - System water level⁽²⁾ System operating: - Pressure - Flow rates
Notes: - Pharmaceutical specifications for acceptable conductivity limits are defined based on the temperature - Water level is the level in the steam generator; a high water level may cause increased carryover of water droplets, depending on the system design.		

Where pure steam is used for steam sterilization there are quality requirements at the point of use per EN 285 [24]:

- Superheat
- Dryness
- Non condensable gases

These are impacted by the distribution system design, operation, and maintenance, except for Non-Condensable Gases (NCGs) which are a quality of the steam generated.

Where production equipment is used for several products, the CQA and CPPs may not be the same for all products (as defined in the regulatory filings). In this case, a platform approach to the system risk assessment may be used that addresses all the relevant CQA and CPPs. Similarly, for utility systems some companies prefer to adopt different terminology.

Notes:

The terms CQA and CPP used in the ISPE Baseline Guide: C&Q (Second Edition) [5] are defined in ICH Q8(R2) [14] and pertain to product manufacture.

Where the system being considered is a utility system, a company may choose different terminology for C&Q, modifying the definitions from ICH Q8 R2 [14] to encompass process and utility systems. Typical terms used include:

A System Quality Attribute (SQA) or Equipment Quality Attribute is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired system product quality.

A System Process Parameter (SPP) or Equipment Process Parameter is used for process variables that impact an SQA and should be monitored or controlled to ensure the desired system product quality.

This Guide uses CQA and CPP.

A science and risk-based approach to validation applies the principles of QRM as described in ICH Q10 [16] by identifying, assessing and evaluating process risks and developing an appropriate risk control strategy, which mitigates risk to an acceptable level.

The control strategy is based on the CPPs and CAs, which are the inputs to the design and operation of the system to achieve the desired level of quality.

The most frequently applied controls used to manage risk are:

- Appropriate specifications for the equipment addressing system product-contact materials and construction standards
- Robust automation (with a suitable sequence of operation, alarms, and interlocks)
- Suitable instrumentation to monitor CPPs (range accuracy, alert and action alarms)
- Appropriate procedural controls (i.e., operation, maintenance, calibration, and cleaning)

Operational change control processes are used to ensure the system components remain in compliance with the specifications and are installed appropriately. Management of the data to record the system performance trends and results is also critical.

QbD, applied during design development, will identify the CDEs.

CDEs are “*design functions or features of an engineered system that are necessary to consistently manufacture products with the desired quality attributes. Examples of automation design functions include alarms and data management. Examples of engineering design features include components instruments, and materials of construction. CDEs are identified and documented based on technical understanding of the product CQAs, process CPPs, and equipment design/automation. CDEs are verified through C&Q.*” [5]

A system risk assessment may be used to help determine CDEs (see Section 2.7.10).

CDEs are identified and documented based on a technical understanding of the system or product CQA, associated CPPs and equipment design/automation. For example, the selection of a Reverse Osmosis (RO) membrane to suit a particular feedwater where the design specifies the membrane and operating parameters.

CDEs are verified through C&Q.

The primary focus of this Guide is on QRM-based C&Q and the associated verification (testing and documentation) with discussion of the major phases, including:

- Project initiation (feasibility and conceptual development)
- URSS
- Project execution planning
- System risk assessment
- Specification and design including functional requirement specification for automated systems
- Procurement and construction
- Commissioning (IV and OV)
- Performance verification
- Ongoing operations
- Maintaining the qualified state over the system lifecycle

Further details of these phases are given in Table 2.2. In general, the following sections are arranged in order of the validation lifecycle elements for direct impact water and steam systems.

Notes:

This Guide uses terminology aligned with the *ISPE Baseline Guide: C&Q (Second Edition)* [5]. To differentiate the approach used to develop the documented evidence that the system is installed and operates according to the specification (IQ and OQ), installation verification and OV are used.

Engineering is responsible for verifying and documenting that the installation and operation meets the specifications.

Table 2.2: Typical Validation Lifecycle for Water or Steam Systems

Engineering and Validation Project Requirements Defined Tasks	Project Initiation (Feasibility and Conceptual Development)	Design (Basis of Design and Detailed Design)	Procurement and Construction	Commissioning, Qualification and Project Closeout	Ongoing Operations/ Validation Maintenance
Project Execution Planning - Project Team Organization - Responsibilities - Cost Controls and Schedule - Design Management, Procurement, Construction, and Testing Plans - Project Validation Plan	X Project scope definition and preliminary planning	X Full project delivery plan by end of basis of design	X Procurement and construction plans tracked and managed	X C&Q execution and closeout plans tracked and managed	
URSS - Initial CQAs and - CPPs	X System requirements	X CQAs, CPPs, and regulatory requirements			
System Risk Assessment Critical Aspects (CAs) identified		X QRM application to confirm acceptable risk profile and identify risk controls; URS updated to include these		X CDEs/CAs Verification	
Design Phase Items - System/Equipment Boundary definition and classification - System Risk Assessment - Supplier Qualification Safety Review - Design (conceptual/Basis of Design/detailed) - Design Review/Design Qualification - Issued for Construction Drawings and Specifications - List of CDEs/CAs and Acceptance Criteria	X Feasibility and conceptual design	X Basis of design and detailed design	X Early procurement activities linked to detailed design completion		
Commissioning and Qualification Plan(s) - Project Master Plan - Construction Quality Plan - C&Q Plan(s) Project		X			
Procurement and Construction - System Fabrication and Construction - Supplier Documentation/Generation of Turnover Packages (TOPs) - Supplier Documentation and Material Certificate Review			X Procurement and construction plans delivered		

Table 2.2: Typical Validation Lifecycle for Water or Steam Systems (continued)

Engineering and Validation Project Requirements Defined Tasks	Project Initiation (Feasibility and Conceptual Development)	Design (Basis of Design and Detailed Design)	Procurement and Construction	Commissioning, Qualification and Project Closeout	Ongoing Operations/ Validation Maintenance
Commissioning & Qualification • Factory Acceptance Test (FAT/IV/IQ) • Start-Up Activities • Site Acceptance Test/OV/OQ • Standard Operating Procedure (SOP) and Maintenance Development • IOQ Traceability Matrix (Fit for Purpose) • Sampling • Training			X GEP-based lifecycle testing and documentation	X C&Q plan delivered (verification) Acceptance and Release/ Qualified	
Performance Qualification (PQ) • PQ Execution (may be negated by PAT) • Acceptance and Release via Qualification Summary Report • Verification Testing and Performance Testing					X Extended monitoring and testing
Project Completion and Document Delivery • Project Handover and Closeout • Project Lessons Learned				X Project closeout plan delivered	
Ongoing Quality Monitoring • Water Chemistry • Microbiological • Endotoxin • Facility-Specific Quality and Business Parameters • Continual Improvement • Periodic Performance Review Reports					X
Maintaining the Qualified State over the System Lifecycle					X

2.1 Project Execution Planning

Project execution planning activities incorporate initial key tasks that can affect the success of a project to install a compendial water or steam system. These key tasks include:

- Project team definition
- Responsibilities of project team members, including engineering, safety, health and environmental, and the Quality Unit (QU)
- Change management (engineering change management and quality change control)

- Planning each stage of the project lifecycle
- Schedule development and project cost controls
- Documentation requirements
- Approvals both within and external to the project team
- Project risk assessment and management

For compendial water and steam systems, project teams may include:

- In-house engineers, SMEs and system operators
- Contracted Architect and Engineer (A&E – the design authority)
- Equipment suppliers
- Third-party inspection personnel
- Quality personnel
- Construction contractors
- C&Q consultants
- Maintenance and technical operations personnel

The project delivery strategy should define the need for active participation by project team members throughout the project lifecycle, typically through the use of Responsible, Accountable, Consulted, Informed (RACI) charts or a similar responsibility matrix. The roles, responsibilities and means of effective communication between project team members should be understood. The C&Q plan/project validation plan defines the project scope, which could include Phase 1 of PQ or all phases.

Change management programs should be established at the start of a project to manage engineering changes and project changes.

Elements of conceptual design and the basis of design (also called front end design) may be managed under project change management where necessary. Detailed design engineering activities are typically under a more robust form of Engineering Change Management (ECM). Design Review (DR) can occur throughout the design process to demonstrate compliance to the risk profile established through the system risk assessment. Changes encountered after design completion and acceptance through the completion of design qualification and system risk assessment must be managed by a change management system to ensure proposed changes are adequately assessed for the impact of implementation; and if they are implemented, all relevant documentation is assessed and updated to reflect the final installation.

The *ISPE Good Practice Guide: Project Management* [20] describes the typical activities during project delivery planning and tasks in each stage of the project lifecycle.

The *ISPE Baseline Guide: C&Q (Second Edition)* [5] and *ISPE Good Practice Guide: Good Engineering Practice (Second Edition)* [10] describe typical C&Q documents and roles and responsibilities that are appropriate for many systems. The information provided in this Guide is based on these ISPE Guides but adds information specifically related to water and steam systems.

The role of Quality Assurance (QA) is to ensure that the organization's engineering quality program is established and appropriately utilized to manage the application of GEP activities to satisfy GMP requirements. This is referred to in the *ISPE Baseline Guide: C&Q (Second Edition)* [5] and the *ISPE Good Practice Guide: Good Engineering Practice (Second Edition)* [10] as the Engineering Quality Process (EQP). The role of the QU includes ensuring that staff are appropriately qualified for their roles, that the supporting site Quality Systems (QS) are in place, and that the appropriate reviews and approvals are identified and completed. Minimally, Quality should review and approve:

- Requirements that impact product quality and/or the assurance of patient safety, and their acceptance criteria
- Deliverable documents in support of the science and risk-based C&Q process: system classification, direct impact system URS(s), system risk assessment, and Design Qualification (DQ) (CDE/CA listing and acceptance criteria)
- Plans (usually the C&Q plan) describing how the project will verify requirements and associated design specifications have been met
- Acceptance and release: results that demonstrate the above requirements have been achieved and the conclusion that systems are fit for intended purpose and ready for next steps (e.g., process validation)

Documentation

This Guide provides guidance on the types of documents usually created during the implementation of a water or steam project.

Approval by the QU versus the Project Team

Because the content of the C&Q plan, which incorporates all the previously approved quality impacting deliverable documents (URS, SC, System Risk Assessment (SRA), DQ), is prepared by an SME, and a suitable engineering quality program is in place, it is not necessary for the QU to approve the lifecycle GEP testing/technical operations documents (e.g., FAT, SAT, IV, OV).

There should be QU involvement whenever a QU-approved document is changed. For all other changes, the project team should have an established project document change management procedure. For science and risk-based C&Q, the associated deliverables are the quality impacting user requirements (i.e., URS for direct impact systems), system classification, SRA, and DQ.

Turnover Packages (TOPs)

A vendor audit should be completed early in the project lifecycle to determine what vendor-supplied documentation will be received and what the testing requirements for equipment, components, or systems are. This is considered best practice as it ensures that only competent suppliers are asked to bid to supply the project.

Based on the audit results, the project team determines the requirements for verification activities in support of qualification. The requirements for system turnover documentation should be defined together with the personnel responsible for delivering them.

Each system may have unique requirements for TOP documentation.

A sample turnover package is in Appendix 7.

TOP requirements should be included in the equipment or system procurement documentation (i.e., as contractual requirements in the purchase order).

Draft TOPs are typically created in the latter stages of fabrication or construction; these documents contain information generated during project execution that will be used to commission, qualify, validate, and maintain the facility. Final TOPs are approved by the system owner and appropriate SME(s). Projects may consist of multiple systems, which could be commissioned and qualified at separate times.

2.2 Design Phase Items

The design phase of either a water or steam project should follow a QbD development process and be appropriately documented. These documents may include the URS, the Basis of Design (BOD), preliminary engineering design, detailed design, and the issued for construction drawings and specifications. Depending on the proposed project execution strategy and size of the project, design stage lifecycle documents may be combined.

Technical or operational requirements identified by users but are not related to water/steam quality should be included in the URS. Health, safety, and environmental considerations and other non-product quality related items should be documented to meet local requirements. Verification of these requirements may occur as part of the turnover package, commissioning test plans, or reports.

2.2.1 URS

The URS is a GEP-based document that describes the requirements of the compendial water or steam system in terms of product to be manufactured, stored, and distributed. The URS should describe the source of the feed water, the output quality (CQAs) and quantities required from each Point of Use (POU), when the output is required, and the total volume needed in a 24 h period. This allows the designer to specify the system and distribution pump sizing after factoring in future changes in the requirements calculation error as a safety factor, as well as any specific operational or maintenance requirements.

The URS should specify what is required of the compendial water or steam system but not how it is intended to operate. Quality impacting requirements recorded in the URS are the basis for C&Q verification activities. For science and risk-based C&Q, the quality impacting requirements are usually categorized as:

- CQA (product)
- CPP (process)
- Regulatory (e.g., water specifications, data integrity, etc.)
- Company quality system approved

URSs should be brief documents; revisions to the URSs should be made through change management to help to ensure that the project team is aware of any changes. Post application of QRM, focused on system CPP control failure, additional controls/CAs may need to be added to the system to mitigate unacceptable risk. These identified CAs are added to the URS. QbD is utilized to identify the CDEs to be incorporated in the system design for fabrication to enable the controls/CAs.

The URS should provide a description of the system including operational and control aspects. They are maintained as lifecycle documents and should be approved by appropriate parties to ensure agreement on stakeholder and project needs. The URS is normally developed prior to the start of the design development lifecycle. Revisions to the URS may be necessary after completion of QRM so that controls added to mitigate the risk of CPP control failure are incorporated into the design (QbD).

The URS is used as a reference during the design review/DQ process to ensure that the design satisfies user expectations/design intent and is fit for intended purpose. The results of the design review/DQ process should be summarized in a DR summary. A DQ is generated from the completion of design DR summary. The DQ includes a list of the CDEs/CAs and their acceptance criteria and is approved by the QU.

2.2.2 Basis of Design (BOD)

The BOD should specify how the compendial water or steam system will operate and be designed. A DR verifies traceability of the chosen system design to the URS. For automated systems, a user requirement/function specification is prepared and approved as a deliverable associated with the GAMP software development process. The approved BOD defines the scope of work for design development leading to fabrication/construction.

The BOD may include elements such as:

- Specific capacity requirements and flow rates for the water or steam system
- Water generation system feed water quality
- Alarms and messages
- POU: water flow, temperature, and pressure
- Sanitization techniques to be used for the generation system
- Sanitization techniques to be used for the storage and distribution system and POU
- General Human-Machine Interface (HMI) layout
- Process control system strategy including input/output and interlock configurations
- Electronic data storage and system security

Notes:

The content varies depending on company practices. Hence this is not a prescriptive or an exhaustive list but is included to portray what is typically represented in a BOD.

The design authority is responsible for identifying and complying with local/national design and safety codes. Any company-specific requirements that are particularly onerous may be included here.

(For further information see *ISPE GAMP 5 (Second Edition)* Appendix D2 [9]).

2.2.3 Issued for Construction (IFC) Drawings and Specifications

The IFC drawings and specifications are a set of documents that specify a system for fabrication. They include the component and material specifications and associated drawings for the compendial water or steam system.

If the system is a vendor standard system, there will be a Functional Design Specification (FDS).

If the system is a custom system, a GAMP Detailed Design Specification (DDS) will be generated and approved and become the basis for software configuration.

It is considered good practice to define which drawing will be used as a compliance reference document. It is usually a Process Flow Diagram (PFD) or P&ID; other drawings may then be categorized as engineering reference drawings. An understanding of the CPPs helps determine which drawings are critical to quality.

The isometric drawing showing system slope for a compendial water system was considered a compliance reference document. This was because historically the sloping of pipework for compendial water systems was considered critical as systems were steam sanitized and the slope was needed to ensure complete drainage. It is currently unusual to steam sanitize a system, but slope is still considered a GEP to allow draining of a line in the event of modifications. There are other ways to drain and dry a line, so it is not considered critical. The isometric drawing may still be considered a compliance reference drawing as it shows valve and sample point locations while the P&ID/PFD drawing does not.

Note: Confirmation of slope is required for pure steam systems to ensure removal of condensate.

- A mechanical floor layout drawing showing system sample points may be considered a compliance reference drawing as it is used to support qualification. The drawing should include the following requirements (see Section 2.5.2.1):
- Title block including drawing number and revision
- Legend/symbols clearly defined
- Major equipment labeled with location shown including identification number
- All points of use identified and labeled with location (room number)
- All sample points identified and labeled with location (room number)
- All instruments identified and labeled
- Utility flow directions should be shown
- Should define the system boundary

The IFC drawings and specifications should reference and align with the FDS so that the system follows the design control logic. This is to ensure that the design can produce a system that will provide a reliable delivery of water or steam to end users. The IFC drawings and specifications may include:

- Materials of construction used to build the water or steam system that will ensure quality water or steam²
- Pump, heat exchanger, storage vessel, valve, and other field component/device specifications, including critical instruments
- The definition of the installation requirements for equipment³
- Documentation requirements for the system:
 - Materials/construction specifications
 - Construction inspection requirements
- Storage vessel vent filter operation (e.g., electrically heat traced versus steam heat traced)
- Treatment system description (i.e., process flow sheet, etc.) in relation to source water quality and seasonal variations

² Were alternative materials to be used, problems with contamination, corrosion, or leachables could occur.

³ Incorrect installation of equipment, such as reverse osmosis units or other deionization equipment, could lead to damage or problems with equipment performance.

- Electrical drawings: these drawings should allow both a construction check and fault diagnosis throughout the system
- Hardware specification: architecture and hardware of the control system (see *ISPE GAMP 5 (Second Edition)* [9])
- P&IDs

Depending on the quality of the system feed water, water generation systems usually require unique items to reliably meet the URS.

2.2.4 QbD PAT Considerations

The FDA has encouraged QbD based on using PAT as the means to control the production of a product. TOC and conductivity measurements can be used for online monitoring, but are not typically used on control loops. Online microbial detection instruments are available. These Online Water Burden Analyzer (OWBA) instruments can count the microbial content but cannot capture and save the information on microbial Colony Forming Units (CFUs) or identify the type of contaminant at the time of publication.

USP standards for pharmaceutical grades of water in USP <1231> [1] denote the CFUs found in the water sampled are incubated and identified. However, USP <1223> and Ph. Eur. 5.1.6 [2] state alternative methods for counting and identifying microbial content and should be referenced when using alternatives to traditional incubation and identification. Pharmaceutical water systems can constitute a fully PAT system with traditional incubation, inoculation, and online testing. See Chapter 3.

Water systems should be designed by knowledgeable designers/SMEs, as the quality of a water system design is critical. The designs should include, as appropriate, items that are known and used by system users, such as types of instruments, methods of microbial control, or methods of ozone removal and detection.

Through DQ, the QU agrees that the proposed design is fit for its intended use. The QU approves the C&Q plan, confirming agreement with the process, personnel, and documentation standards, expected results, and acceptance criteria. The work is conducted following GEP, associated documentation standards, and change management processes.

At the completion of OV, a brief commissioning summary report should be produced by an independent engineer (i.e., not the engineer responsible for the commissioning) to confirm that the documentation is acceptable and demonstrates that the system is installed and operates to meet the specifications. Changes to the agreed design should be managed through the change management system. Approval of this document releases the system to have the qualification summary report developed. Chapter 3 contains information on PAT.

2.3 Commissioning and Qualification Plan

The C&Q strategy is developed concurrently with the creation of the project execution plan. The C&Q plan(s) may be generated concurrently with the project design phase. The plan(s) should be completed at the end of the design phase. This allows GEP-based testing documents (e.g., FAT, SAT, IOV) to be generated and approved during fabrication.

The approach used may influence the content and structure of the TOP documents, for example, component specifications, as-built drawings, recommended maintenance and cleaning procedures, O&M manual, as well as the C&Q requirements for the system.

The C&Q plan should define the activities and responsibilities that make up the commissioning and qualification (verification) of a system within its operating environment. Typical content of a C&Q plan includes:

- Scope of the plan (including system identification)
- Description of the C&Q execution strategy
- C&Q related documentation requirements for each identified system (URS, FAT, SAT, IOV, PQ, SOPs, etc.)
- Roles and responsibilities
- Reference necessary C&Q document templates (e.g., test instructions, test scripts, reports, discrepancy form)
- Planned inspections and tests

Any requirements to witness tests:

- Documentation requirements for the inspection and test results
- Reference procedures for inspections and tests to be performed and identification of SMEs for review of the inspection and test results
- For tests to be used to support qualification, include the acceptance criteria to be met

The C&Q plan should define the transition from commissioning and qualification to PQ to maintain the system in a satisfactory state of qualification, e.g., maintenance, calibration and sampling.

Often an overall validation master plan describes the validation strategy for a complete process facility. Similarly, an overall C&Q plan may be generated and used to describe commissioning and qualification for a specific system (e.g., WFI system, Purified Water (PW) system, pure steam system).

C&Q plans should incorporate “right first time” lifecycle testing. The development of C&Q plans for compendial water or steam systems should be considered to ensure that responsibilities are clearly defined and activities that impact final water or steam quality are included in GEP commissioning activities and are not unnecessarily repeated for qualification.

The SME and owner should be involved with the development of the C&Q plan which is approved by the QU.

The design of compendial water systems is usually based on the quality of the feedwater and the average and instantaneous quantity and quality (or different qualities) of water required by an end user. Sampling (whether online using PAT or off-line) of the generation and distribution system should be dictated by the design and address any risks identified through applied risk management for C&Q and also address any regulatory expectations. It is considered good practice to sample the system as soon as it is made operational in order to obtain an understanding of its operating characteristics.

2.4 Procurement and Construction

Procurement and construction activities should ensure that purchased equipment complies with the specifications created during the project design phase with any variations from the specifications reviewed and accepted or rejected by the design authority⁴ and organization prior to installation.

⁴ For example, the system designer's representative.

A user/system designer that has a lot of experience with local installations can add a lot of value during the procurement and construction stage, suggesting changes/improvements based on that detailed experience.

The most efficient process does not repeat tests unless there is a logical reason to do so, e.g., reconfirming control wiring is correct after it has been disconnected for shipping purposes, then reconnected.

In addition to the traditional approach of purchasing, installing and operating a water and steam system, organizations also can contract for the supply of water from a third party that installs, operates and maintains the system, generally charging based on the utility consumption. The utility consumption should be discussed prior to the project delivery strategy phase, which is the first stage in procurement planning. This is a risk-based decision function. Refer to the *ISPE Good Practice Guide: Project Management* [20] for general procurement principles.

A generation system can be purchased from a specialist supplier and the procurement, installation and QC of the storage and distribution system project managed within an organization. Alternatively, the complete installation of the storage and distribution system can be purchased from a specialist contractor.

An assessment of a supplier Quality Management System (QMS) allows organizations to use appropriate supplier data and testing (such as FATs/SATs to support C&Q).

The commissioning strategy should consider the results of any supplier assessments when developing the system receipt and installation verification approaches, as well as the calibration strategy to be used within a project. The defined approach is documented in the C&Q plan.

Procurement and construction specifications should define:

- Responsibility for verification that the equipment received/installed is verified to meet the specifications
- Roles and responsibilities for construction
- Related QC activities during construction
- Performance and/or timeline-based payment schedule

Installed systems should have documentation that is acceptable to the owner or the owner's representative.

Installation verification should provide documented evidence that installed equipment complies with design specifications and incorporates agreed engineering changes.

The completion of procurement and construction is usually identified through the review and acceptance of construction TOPs, verification of as-built record drawings, walk down, and punch list of the installed system.

2.5 Commissioning and Qualification

This section is intended to indicate project activities or items typically included in the commissioning of compendial water and steam systems.

The process of commissioning should provide documented evidence ensuring that the system owners or systems users receive a system that can produce or deliver a supply of water or steam in a manner consistent with the system owner's requirements.

During the commissioning process, the ECM system should be followed in order to control, capture and document changes to compendial water and steam systems.

2.5.1 *Start-Up Activity*

A system release memo should be prepared and submitted before beginning start-up, commissioning or PQ. The memo should include the pre-start-up safety report.

Before starting the main activities for commissioning, a walk down inspection (possibly via a check list) of a system should be performed. Responsible personnel should be notified of any problems observed in:

- General cleanliness
- Missing components/instruments
- Visibly sagging or unsecured piping/tubing especially in indirect impact system utilities
- Disconnected piping/tubing
- Improper air breaks to floor drains
- Disconnected wiring/pneumatic lines
- Disconnected/broken instruments
- Disconnected/broken valves
- Disconnected/broken pumps
- Leaks
- Threaded fittings in hygienic piping/tubing systems
- Damaged or poor quality welds
- Missing tagging/identification labels
- Lack of identification for process piping or system components
- Confirmation that all sample valves are marked and safely accessible
- Dead legs
- Certification/correct installation of safety features such as pressure relief valves

2.5.1.1 *Safety Reviews and Checks*

Prior to system start-up, several safety reviews and checks should be performed on the compendial water or steam system, including:

- A safety review of ports to be sampled to ensure that sampling technicians have the appropriate Personal Protective Equipment (PPE) needed to safely take samples.
- A safety review of the equipment to ensure that the mechanical associates have the appropriate PPE needed for the safe operation of the equipment. The owner and/or owner's representative should provide authorization before turning on the utility supplies to the system (e.g., electricity, water, compressed air).

- A documented check that the system is in a safe condition; the procedures and scope of this check usually depends on the project, but typically includes a safety focused document check, e.g.,:
 - Pipework hydraulic pressure testing completed satisfactorily
 - Compressed air pipework pressure testing completed satisfactorily
 - System valves in the specified condition, i.e., open or closed
 - Warning labels and signage are in place
 - E-stops accessible
 - Insulation review
 - Relief devices check

A walk down of the electrical connections should be performed by an SME or authorized person to ensure they are acceptable to the owner or the owner's representative and that fuses/overload ratings are as specified.

2.5.2 Commissioning Activities

The on-site commissioning process usually begins when construction of a water or steam system nears completion. System elements are put into operation through a series of engineering activities. This work is usually incorporated in the C&Q plan. Major elements of the system, fabricated at a vendor site, are verified as part of the FAT scope.

Throughout the commissioning process, the principles of GEP (see the *ISPE Good Practice Guide: Good Engineering Practice (Second Edition)* [10]) should be applied when putting system components into operation. GEP principles include:

- Compliance with pressure vessel codes
- Electrical codes
- Plumbing codes
- Other local building codes and practices

Compendial water and steam systems require a sequential commissioning process to verify conformance to specifications and design intent prior to operation.

The following is a representative list of typical commissioning activities that bring a compendial high purity water system into operation:

- Walking down and checking that P&IDs, general arrangement, and isometric drawings are representative of the installed system.
- Checking system for maintainability, e.g.,:
 - Verifying those components (such as pumps, vent filters, gasket replacement, gauges, and sensors) that will require maintenance and/or calibration have sufficient room for mechanics to safely service components.
 - Checking that all the parts to be replaced during preventive maintenance are clearly identified.

- Checking that installation records (construction turnover packages, etc.) for the system equipment and components have been created. In addition, check vessel internally for items such as filters, spray balls and vortex breakers.
- Checking that the water system has been installed in accordance with the design and installation drawings, and that planned backflow prevention (including air breaks for drains) equipment has been correctly installed.
- Checking that the pipe flushing with soft or deionized water to remove construction debris, or use of mechanical cleaning methods, as necessary, has been verified as complete and documented properly.

Note: In-line filters and/or strainers that had been installed during the flushing phase, but are not part of the permanent installation, should have been removed after the flushing phase, and this should also have been documented.

- Checking that the pipe system pressure tests have been verified as complete and appropriately documented.
- Checking that the cleaning and passivation of stainless steel equipment and piping systems has been verified as complete and appropriately documented.
- Checking that a documented and approved material joining program (e.g., welder's certificates, procedures, isometric drawings, and non-destructive tests including joining of metals used on product/process contact surfaces) has been used during construction of the system and that appropriate joints have been examined/inspected as required by the specifications with supporting documentation.
- Checking that specified piping/tubing material was received and installed, as specified.
- Construction materials in contact with media must be documented:
 - With certificates that contain measured material composition (for example, corresponding to EN 10204 type 3.1 [27])
 or alternatively,
 - With test reports (for example corresponding to EN 10204 type 2.2 [27])
 - With supplier declarations (for example corresponding to EN 10204 type 2.1 [27]) that are supplemented with measurements of material composition identifying the construction material

Polymers can be documented by supplier declarations alone (for example corresponding to EN 10204 type 2.1 declarations [27]). The declarations must describe the construction material in question and also refer to which standard it meets (for example, reference to 21 CFR Part 177 [28] or similar). Where certification is not available, the approach used to confirm appropriate materials must be agreed to by the QU.

- Checking heat numbers of materials (if applicable).
- Polymer materials, such as Polyvinylidene Fluoride (PVDF) and Polypropylene (PP), should be documented by conformity declarations.
- Checking received materials of construction versus materials specifications for components (e.g., instruments) that will be in direct contact with the compendial water or steam.

- Checking that the piping/tubing system specified to be sloped and drained was constructed to achieve drainage to low points. A compendial water system should be sloped to aid the drainage of the system. It may not be possible to slope some sections of piping/tubing. These sections should be identified on the drawings. Alternative methods (i.e., blowing with air or removable spool pieces) may be used in these areas of piping to achieve drainage when required.
- Checking that steam traps and condense systems are installed per specification.
- Checking that instrumentation loop checks have been performed.
- Checking that instrument inspection loop tuning and instrument calibration was performed per the specification and according to the manufacturer's recommendations.
- Testing "usability" of each sample point, by use of correct sample bottles (size and volume).
- Confirming that area ergonomic and environmental conditions (e.g., acceptable ventilation, sample point access) are suitable to conduct QC sampling. Inadequate area ventilation can lead to water sample contamination. QC samples should be taken in the same way that the POU will be used.
- Checking that a master list was created between non-critical and critical instrument components. See Section 2.7.10 or the *ISPE GAMP® Good Practice Guide: A Risk-Based Approach to Calibration Management (Second Edition)* [29].
- Performing start-up and checks of support utilities, electric power, pump motor run, and pump hot alignment and lubrication checks.
- Checking for adequate pressure and fluid flow per design within the system, including spray ball operation.
- Checking the compendial water generation system CPPs such as feed water flow, quality, pressure, and temperature.
- Performing engineering runs and tests to confirm specified operation of the unit operations in pretreatment and final treatment, including sanitizing, shutdown, start-up and alarming. Included in these tests, are specific automation system tests, e.g., valve sequencing, interlocks, loop tuning, and connections between water systems and process systems (e.g., connection to glass washer or other processing pieces of equipment). This data can be compared to the FAT to verify system performance.
- Performing engineering runs and tests to confirm the operation and capacity of heating and/or cooling exchangers that may exist within the water or steam system (e.g., run hot water sanitization cycle and cool down for operation after sanitization).
- Performing system safety tests (relief, emergency shutdown, continuity, etc.).
- Checking that applicable construction codes (pressurized equipment, electrical, mechanical codes, etc.) and building permit conditions have been followed (including seismic and electrical requirements).

A TOP should be referenced when developing the initial SOPs/work orders for system operation and maintenance.

TOPs may be created on a "by system" basis to allow systems to be commissioned in an organized fashion within a larger facility construction project.

Note: In the EU, a technical dossier describing the system/equipment is required to cover the same items recommended in this chapter. If the technical dossier is not present, the system is of "less" use.

Typical items that contribute to product quality are confirmed during commissioning to support qualification are discussed in Sections 2.5.2.1 – 2.5.2.19.

2.5.2.1 *As-Built Drawings*

Drawings can depict the flow of materials, critical features or sampling locations. Drawings IFC may include submittal drawings (for approval), P&IDs, equipment drawings, and PFDs.

Approved as-built drawings of the generation and the storage and distribution systems should verify features that may be relevant to product quality or maintenance. Components shown on the drawings should be physically inspected (field verified) to confirm that the drawings accurately reflect features such as:

- Appropriate installation
- Major equipment location and identification (tagging)
- Location
- Direction of flow
- Sample points and POU location/identification
- Online quality monitoring devices or other instrumentation in the process flow
- Slopes
- Diameter
- Total loop length

Typically, an organized set of as-built drawings that includes isometric drawings is created for compendial water or steam systems. Where modifications are planned for a system, as-built isometric drawings can be used to determine the correct technique to drain the system. As-built designated drawings should be signed off by the owner or owner's representative and copies made available to the owner or owner's representative.

Example 1: Sample POU Drawing

A drawing should all POUs in a system as well as all sampling points and their designation/identification for the purpose of commissioning the system, and so that the chemists or microbiologists know where to collect samples.

Example 2: P&ID Drawing

A P&ID drawing should show the sequence of the equipment, which can directly impact product (water) quality and key piping and instrument characteristics, such as:

- Materials of construction
- Line sizes
- Valve types
- Nozzle orientations
- Pipework labeling showing contents of pipework

- Insulation
- Flow direction

2.5.2.2 Equipment or Piping Component Installation

Water and steam generation and distribution equipment should be installed so that the equipment can operate per design. Checks should be performed to confirm whether equipment (e.g., RO units, stills, pure steam generators, and continuous deionizers) is installed per design. The installation should be compared to construction drawings and specifications. Any discrepancies between the design and the installation of water and steam generation and distribution equipment should have a documented assessment to determine the impact on the water or steam quality and/or the need for correction.

For equipment and piping components, as-found conditions should comply with installation drawings and specifications.

2.5.2.3 Welding and other Material Joining Procedures and Documentation

Documentation on welding or other material joining techniques, such as ASME 31.3 [30] is required. The weld procedure should be approved by the system owner or the owner's representative (e.g., the third-party inspector) before execution of the weld procedure.

Appropriate welding or other material joining methods and procedures and documentation practices should be established and used for all applicable piping/tubing components.

Many compendial water and pure steam systems are largely constructed of 316L stainless steel (DIN 1.4435 [31] and/or DIN 1.4404 [32]). Where appropriate, plastics and other metals may be used for these systems.

An appropriate welding or material joining program should be defined prior to construction activities and include at a minimum:

- Routine inspections by qualified or certified SMEs throughout installation to verify that requirements are being met
- A defined weld or material joint inspection program. This should be performed for all applicable piping and tubing in the compendial water or pure steam system.

For example, for compendial water distribution systems constructed of 316L SS, weld joint quality is typically determined based on the internal and external features of completed welds. See Section MJ of the ASME 31.3 [30] or EN ISO 3834 [33] to determine documentation requirements. See BS EN ISO 9712 [34] to determine third-party requirements.

Examination of the external surface of all welds (both manual and automatic machine) is normally performed by the installer of the piping/tubing system. Examination reports should be completed by the installer on weld log sheets. Internal welds may be inspected by indirect visual methods (e.g., borescope or using mirrors) by the system owner or the owner's representative at a typical minimum of 20% of total daily completed welds for automatic machine welds and 100% of daily completed manual welds. For welds produced by an automatic machine, the number (or percentage) of daily internal welds inspected may be increased based on the number of rejected welds. See the European Pressure and Machinery Guidelines (EU PG) [35] for pressure and x-raying of welds for pure steam lines that have special safety requirements.

- Labeling welds or other connections and appropriate inspections during construction. Inspection reports should be included with supplier documentation.

- A weld map drawing showing weld locations. The heat numbers of metals joined on either side of a weld should be included in weld logs. The heat numbers should be traceable to material certificates of analysis. (The heat number corresponds to the chemical composition of the material (e.g., 316L SS or similar grade such as DIN 1.4435 [31] or DIN 1.4404 [32]) used to construct the system. The heat number map should include product/process contact surfaces of tubing, valves, fittings, etc.)
- Review of the qualification records of welders. The qualification records of a welder should be assessed to ensure that they are current and adequate prior to the welder performing any work. Copies of the qualification records should be included in TOPs. ASME BPE [36], ASME B31.3 [30], EN 287-1 [37] or EN 1418 [38] can be used as a guide to determine welder certification requirements.
- Confirm that the field samples taken to represent the total good samples and bad samples of incremental pieces of pipes are in alignment with the claims attested in the inspection reports.
- Verification should be obtained that approval signatures (including installation contractors, owner or owner's representative) have been provided on weld documentation, including the weld maps.

2.5.2.4 Materials Certification

Materials used to construct compendial water systems should preserve the quality of the water or steam product. Materials should be used that are not reactive, additive or adsorptive to the water or steam product. Engineers that design systems may specify materials, such as, metals, plastics or elastomers, which should ensure the quality of water or steam products.

316L stainless steels and non-leaching plastics/elastomers are typically specified for compendial water and pure steam systems. Other materials may be used. Although the surface of these materials is usually included in a piping component material specification, material surface finish is not, and is typically critical to water quality maintenance.

2.5.2.5 Line Slope

Line slope in water and steam systems is considered a GEP activity. (See Section SD of the ASME BPE [36]).

The usefulness of line slope can vary depending on the type of system, e.g.:

- In pure steam distribution systems, sloping of distribution lines is used to ensure condensate in the system flows to drain traps, and to prevent cold spots (where condensate may collect) or preserve the physical properties of pure steam delivered to a POU.
- In one-way compendial water distribution systems or for pipework downstream of a POU valve where pure steam is used for sanitizations, sloping is intended to achieve adequate system sanitizations.
- In recirculating compendial water distribution systems, slope verification documentation may be non-value added because the system is continuously recirculated.

Line sloping is usually intended to:

- Aid in system drainage to low points for future system shutdowns
- Assist in the removal of passivation chemicals

Recirculating compendial water distribution systems are not usually drained, but if they are, compressed air can be used to supplement the system drainage procedure where it is of appropriate quality. Where recirculating compendial water distribution systems are shut down and drained, they are usually flushed and sanitized before being placed back into service.

For the BOD development, engineers should make a determination whether line slope verification should be included in the project, based upon requirements and risk assessment. The line slope verification can be verified using a calibrated level device and documented on the isometric drawings. Slope verifications should be performed prior to the addition of insulation.

For systems where commissioning has been deemed necessary, the acceptance criteria are as-built isometric drawings indicating the slope is available and that an approved slope measurement procedure has been used. Suppliers and engineering units should produce slope information.

If line sloping is specified as a GEP, commissioning should be performed to verify the extent of line sloping included in a project. Organizations may choose to install piping/tubing level and plumb, e.g., where line sloping has been determined to be non-value added in relation to water quality.

2.5.2.6 Hydrostatic Pressure Test

Water or steam systems should be free of leaks, which can lead to contamination of the water or steam. From a system operating consideration (it may also be a local code requirement), hydrostatic pressure tests should be performed and documented to ensure this. Water (e.g., deionized water) used for hydrostatic testing of compendial water systems should be of appropriate quality to avoid the introduction of contamination, which could cause problems during construction or start-up.

Hydrostatic pressure tests may be conducted in selected sections of the system, such as sections of piping that can be isolated and pressurized at varying times during construction. Tests can be performed by making reference to a particular section of the system P&ID. For further information on the requirements for hydrostatic pressure tests; see ASME B31.3 [30].

2.5.2.7 Cleaning and Passivation

Stainless steel product-contact surfaces should be maintained in a passive condition (by passivation) in compendial water systems to help to maintain the quality of water.

Passivation usually involves introducing an acid solution into the system to remove elemental iron and to maintain or enhance the chromium to iron ratio of a weld and produce a passive layer on the surface of the metal. Completed welds should undergo passivation. The first step is usually a chemical cleaning process to prepare the metal of the completed weld for passivation. Specified cleaning and passivation procedures should be followed and documented. Documentation usually includes:

- Cleaning and passivation procedures/process sequence
- Cleaning agents used
- Cleaning temperature and times
- Passivating chemical solutions used
- Temperature and contact times
- Testing of treated surfaces and associated acceptance criteria
- Identifying line sections
- Components passivated
- Approval signatures

Cleaning and passivation activities may be delayed to:

- Prevent the need to repeat cleaning and passivation steps if piping or tubing modifications are needed during start-up and commissioning
- Provide a chemical sanitization just prior to the start of water quality testing

The project team should plan when to perform the system cleaning and passivation, which may be performed sequentially on specific segments of equipment or piping/tubing sections.

Cleaning and passivation of a system should be performed only on an as needed basis.

For further information on passivation see ASME BPE [36], ASTM A380 [39] and ASTM A967 [40].

Note: Passivation is not usually required for pure steam systems.

2.5.2.8 Vent Filters

Vent filters are used on compendial water storage vessels as a particulate and microbial barrier between the surrounding environment and the water. Pressure safety devices (e.g., rupture disks) should be used to protect against vessel overpressurization or vacuum.

A method should be identified to replace vent filter cartridges. The method should minimize the risk of contaminating the system (e.g., Steam-In-Place (SIP) with pure steam or autoclaving the filter in its housing prior to installation. The process/procedure should be documented.

Hydrophobic vent filters (e.g., 0.2 µm) are suitable for microbial retentive gas filtration. Areas of concern for gas filtration are blockage of the filter caused by condensed water vapors or leakage caused by improper filter installation or defective filters.

Control measures to avoid accumulation of condensate in the vent filter housing include:

- Electrical or steam heat tracing
- A self-draining housing orientation
- Placement of the housing at a distance far enough from the tank such that vapors do not accumulate

Where heat tracing⁵ (such as electric jacket or steam) is used, the temperature should be set above the normal operating temperature of the water in the storage vessel so that the temperature is above the dew point temperature and below the manufacturer's maximum temperature rating for the vent filters.

Vent filters should be constructed of specified and verified materials. Filters should be integrity tested prior to initial use and after use. The chosen integrity method and filter change procedure must not be a source of contamination.

Post integrity testing can be used to verify that the filter is not plugged or has not been leaking at the time the filter is removed from service. Post integrity testing can provide assurance that preventive maintenance schedules and procedures are appropriate. Filter integrity testing methods include:

- Bubble point
- Diffusion

⁵ Heating by flexible electric heating elements.

- Water intrusion

Filter integrity testing methods can vary, depending on the filter manufacturer, application and type of filter.

Vent filter integrity testing equipment may not be available at some non-sterile product manufacturing sites. Acceptance of a supplier's certification of vent filter integrity may be considered sufficient based on an audit of the supplier's manufacturing site and QMS.

Commissioning documentation should contain information that confirms that the method selected for avoiding condensation is appropriate, that the filter orientation allows for drainage, and that the filter integrity testing has been performed and successful results obtained.

2.5.2.9 Support Utilities

It should be determined whether support utilities are direct or not direct impact systems. Not direct impact utilities (e.g., instrument air, drains, power, cooling water, plant heating steam) should be commissioned. These systems should be connected to the water or steam system in accordance with the specifications.

Other system utilities (e.g., compressed air blow, N2 blanket) may be required for use in the compendial water or pure steam system.

2.5.2.10 Loss of Utility

Loss of utility testing can be specifically planned for systems based on the likelihood of occurrence and potential water or steam quality risk. A risk assessment on the loss of utilities should be performed at the beginning of the project and documented. This should include the proposed testing to be performed or why testing does not need to be performed. The risk assessment should be approved by the system owner. The C&Q plan should include a strategy on how the system will be restored to service after loss of utilities, such as whether the system will need to be drained, flushed, sanitized, steam sanitized or passivated, how it will be released for use, and who will release it for use.

The system's response to loss of support utilities to the generation equipment and distribution system should be investigated. Response to the loss of electrical power should be considered. This includes retention of critical data, as well as safe equipment and how the system will respond.

Equipment and system responses to the loss of other utilities (e.g., compressed air, hot/cold process water, glycol, and N2) should be documented ensuring that removal of a utility does not create a safety hazard and that adequate protection is provided.

For short downtimes (e.g., due to loss and resumption of a given utility) equipment and system behavior should be adequate to prevent contamination of the system. Except for system-specific extended downtime periods, a procedure for loss of electrical power should be developed, including the steps to perform in the event of a power loss. Upon power restoration, the system should automatically return to the service mode/cycle.

A procedure should be developed based on the loss of a utility. The procedure should identify the time period for which a system may go above or below CPPs without having an impact on product CQAs. For example, a compendial water system where temperature (which was deemed to be a CPP) decreases for a defined short time (minutes or few hours), is tested to confirm that the water microbial quality is not affected.

Loss of utility tests, such as electrical power or instrument air, may take into consideration its impact on water quality and define the actions to be taken after a specified time. For example, it is acceptable to restart and continue to use the system after a shutdown of 30 minutes. Microbial samples should be taken from the loop return and generation system output to confirm the quality of the water in the system. System shutdown exceeding, for example, 30 minutes requires a system sanitization.

Sampling is a way to determine the effects on the system if support utilities are lost for any period of time. Water quality risk should be considered when determining sampling requirements to put the system back into operation.

Loss of utility tests may be included in commissioning testing activities in an effort to finalize system SOPs.

2.5.2.11 Automated Control System

Each engineering design element/CDE required for the control system should be installed and commissioned according to specifications. Verification testing for the control systems for computerized or Programmable Logic Controller (PLC) automated equipment/system is governed by guidelines, which are specific to the type and manufacturer of the controlling system. For further information see the *ISPE GAMP® Good Practice Guide: A Risk-Based Approach to GxP Process Control Systems (Second Edition)* [41].

2.5.2.12 Automated Control System Operation

Performance testing of the automated control system is an important part of the commissioning of the controls. In most cases, the testing is specific to the type of automated control system (i.e., PLC based or computer based). Typical tests include:

- Alarm is generated for a defined alarm condition situation and message is provided.
 - Hard-wired alarms and switches are functional and appropriately connected.
- Note:** life safety systems are tested during commissioning and considered outside the scope of qualification.
- Automated control system and set points are adequately guarded by password access or procedural method.
 - Automation system has a procedure for generation of critical data and storage, as well as recovery of data.

Set points and operating ranges should be developed for control set points. Critical alarms and interlocks of the control system are functional and appropriate password security level or procedure is present.

For further information on the OV of the automated control system, and data storage and management refer to the relevant *ISPE GAMP Good Practice Guide: A Risk-Based Approach to GxP Process Control Systems (Second Edition)* [41].

2.5.2.13 Instrumentation Calibration

All instrumentation should be assigned to an instrument category, which is determined using one of the four criticalities defined in this section. These definitions should be clearly understood by the representatives before any assessments are made.

Note: In this case, product is the system output.

Product Critical Instrument

- A product critical instrument is an instrument whose accuracy/failure may have a direct effect on a product quality attribute (or CQA). Product critical instruments are usually associated with a CPP. An example is the distribution loop return conductivity/temperature monitor.

Process/System Critical Instrument

- A process/system critical instrument is an instrument whose accuracy/failure may have a direct effect on process or system performance (not CPP), without affecting final product quality or safety. An example of this is the placement of a hardness monitor after use of a softener.

Safety/Environmental Critical Instrument

- A safety/environmental critical instrument is an instrument whose accuracy/failure may have a direct effect on safety/environment. An example of this is the temperature monitor used to control cooling water flow to a steam condenser line.

Non-Critical Instrument

- An instrument whose accuracy/failure is deemed to have no effect on a CQA, process/system performance (not CPP), safety, or the environment. An example of this is the chilled water supply temperature to a system heat exchanger.

It should be verified that all product critical instruments have calibration certificates traceable to national standards (e.g., National Institute of Standards and Technology (NIST), National Accreditation of Measurement and Sampling (NAMAS)). Traceable calibration records must be on file, confirming that the instruments are calibrated within the appropriate ranges (design, allowable operating, and normal operating) in which they are to be operated.

Calibration should be documented as part of the qualification program. Calibration documentation should include verification that calibration has been completed. Instruments are usually tagged with a calibration date and calibration due date. Product critical instruments should be calibrated at the factory and calibrated on site.

Non-critical instruments may not be subject to the same calibration frequency and reference standards as critical instruments. However, non-critical instruments should have at least one verified calibration at the beginning of their use, as reference of their operation.

For additional information see the ISPE GAMP Good Practice Guide: A Risk-Based Approach to Calibration Management (Second Edition) [29].

2.5.2.14 Sequence of Operation

Sequence of operation testing should verify that the automated control system, software programming, and equipment and instrumentation operate according to the intended design sequence. The information from this testing is helpful to use when developing the system SOPs.

Normal functional and cycle testing is intended to confirm and document the ability of the components to function together as a system in a successful manner which simulates normal operation.

The system functionality and sequencing of various modes of operation is dependent on the design of a specific installation. The FDS or the supplier design specification should be referenced to determine the functionality and sequencing of the system.

There may be several different automation systems containing sequences of operations that need to be tested (e.g., generation system PLC, storage and distribution control system).

The concept of when these systems will be tested, and the documentation needed, should be outlined in the C&Q plan. For example, during commissioning the sequence of operations may be completely tested. If the sequence of operations was tested successfully during SAT or FAT and the test was documented, verification that the test was completed may be all that is needed according to an approved project C&Q plan.

2.5.2.15 Water Generation

Testing should be performed to ensure the ability of a water generation system to:

- Provide compendial water meeting specifications under peak demand conditions
- Continuously generate acceptable quality product water in a flow range defined in the system URS

The engineering runs and testing should be performed prior to PQ to ensure that the water generation equipment is capable of producing water to meet specification. Testing can provide early assurance that extended operation at minimum and maximum demand conditions will not have an adverse effect upon the quality of the product water. Extended water quality testing should be performed during PQ. The successful completion of engineering runs and tests can be used to document the system and be approved by the QU as the operational configuration used to conduct PQ.

2.5.2.16 Water Distribution

The ability of a water distribution system to provide quantities of product water (of a quality specified in the URS), under the variety of demand conditions to the individual POUs while maintaining specified pressures in the system, should be tested to ensure proper turbulent flow (where applicable) is maintained in the system. System design calculations should be checked during the engineering DR (or DQ). The system should be tested during OV test and adjustment to verify the calculations.

During these tests, the goal is to gain assurance that the defined worst case usage scenario (maximum demand) will not result in contamination of the distribution system due to conditions such as:

- Backflow
- Unacceptable pressure differentials across system components (e.g., heat exchangers)
- Distribution water flow that falls below minimum acceptable rates
- Pressures that fall below minimum values

The water distribution system should be capable of maintaining a minimum pressure at all points within the distribution system piping during the peak water demand for the system. Confirmation of turbulent flow in the recirculating loop may be verified during this testing and should include documented verification that the flow rate in the system does not drop below the minimum flow rate required for fully turbulent flow, especially if this strategy is being used to prevent biofilm growth.

Where temperature is used as the biofilm control parameter, verification that the loop return temperature is maintained should be obtained.

Note: Short term deviations from loop return flow rate/temperature are usually considered acceptable; the limits are defined by the user and verified during PQ.

The time periods for heat up, maintenance, and cool down (the time the system is unavailable for use) should be agreed and documented in the URS specification.

During commissioning, tests that define how long periods of inactivity or delays (planned or unplanned) in the different sub-systems are included to monitor the system water quality to ensure that the system performance is not compromised during these periods.

2.5.2.17 Temperature Maintenance

Compendial water distribution systems may be designed to remain hot to maintain continuous sanitization of the piping and storage vessel networks. Other systems may be heated at a validated frequency to sanitize the system, allowing the water system to operate at a lower temperature. This lower temperature may be either controlled or uncontrolled; temperature control is preferred unless the normal operating temperature is consistent to provide repeatable system performance.

A system should be able to maintain desired temperatures (hot/cold) at defined sensing points on the distribution system.

Temperature sensors should be appropriately located and used to prove that when heat sanitization is required, the water system network can be brought up to the sanitization temperature for the required interval of time. Usually, the coldest part of the system is the distribution loop return just before the heating heat exchanger.

Where pure steam is used for sanitization, a system temperature mapping test may need to be developed.

Verification that the temperature, frequency, and duration of sanitization are adequate should be confirmed through water quality testing during PQ.

2.5.2.18 Ozone Level Generation and Maintenance

Distribution systems may use ozone (either continuous or intermittent) as a method to maintain specified microbial levels below appropriate alert levels within the distribution system. For additional information, refer to the *ISPE Good Practice Guide: Ozone Sanitization of Pharmaceutical Water Storage and Distribution Systems (Second Edition)* [26].

The ozone generation, distribution, and destruction system should maintain required ozone levels, as required for distribution system normal operation and sanitization.

Verification that the ozone levels, frequency, and duration of sanitization are adequate for microbial growth control should be confirmed through the distribution system operating range and confirmed through water quality testing during PQ.

2.5.2.19 SOP Development

Approved SOPs for the functioning (including use, maintenance, calibration, sanitization, etc.) of the water system are required. These documents should be developed (in draft form at a minimum) before the commissioning stage so that they can be executed, verified, and updated during the commissioning phase. Approved SOPs are usually required for PQ.

2.6 Traditional Approach to Qualification

Traditionally, the qualification strategy for compendial water and steam systems has solely relied on the use of IQ and OQ documentation (often executed retrospectively) to verify that all system specifications are met. (Appendix 1 provides a typical WFI generation and distribution system component classification example.) Commissioning documentation has not traditionally been, or has moderately been, leveraged to cover these qualification activities. Additionally, the traditional approach considered the different steps and links between design and testing, but did not incorporate a science and risk-based approach based on product quality to define the qualification strategy with a focus on system process CPPs control.

The traditional approach is still considered acceptable if it is the basis of a site quality system for C&Q and is currently accepted by the relevant regulatory authority. However, the traditional approach presents less flexibility and adaptability to new knowledge or technical changes. It can also involve repeated testing, since all aspects of the system are tested regardless of their impact on system product quality, as well as the retesting of successful activities performed during the commissioning of the system for qualification.

In contrast, there are multiple benefits of transitioning from the traditional qualification approach to a more streamlined science and risk-based approach such as the one proposed in this Guide, and in *ISPE Baseline Guide: C&Q (Second Edition)* [5]. Such benefits include:

- Avoiding extensive generic testing and unnecessary test repetition between commissioning activities and qualification, instead tailoring the qualification strategy to the individual system needs.
- Better flexibility and adaptability. This approach focuses on topics with higher risk. With better flexibility and adaptability, the cost/time is significantly reduced compared to the traditional approach.
- Following a lifecycle-based verification approach allows for efficient resolution of nonconformance when acceptance criteria are not met.
- Better alignment with current regulatory expectations, promoting a QbD strategy
- Product quality as a central point of the strategy, instead of exclusively focusing on generic requirements
- Identification of the system control strategy (through definition of the CAs and a focus for qualification on the CDEs)

2.7 Science and Risk-Based Approach to Qualification

2.7.1 Good Engineering Practice

An essential element of the science and risk-based approach is the use of GEP to provide documented evidence that the system is installed and operates according to the specification. There are many ways of implementing a science and risk-based approach; this Guide presents one example.

Quality oversight of the process is a requirement. An approach to this is the establishment of an EQP as described in *ISPE Baseline Guide: C&Q (Second Edition)* [5] and the *ISPE Good Practice Guide: Good Engineering Practice (Second Edition)* [10]. In the following example, it has been achieved through the definition of GEP in a series of procedures:

- Good documentation practice
- DR and qualification
- Commissioning planning and execution
- ECM
- Automation system lifecycle management

2.7.2 Defining the Commissioning Deliverables

The commissioning deliverables should be clearly defined in the C&Q Plan with the intent of the deliverables and the associated roles and responsibilities. C&Q plans also should define the type of test, acceptance criteria, and timing for verification. These documents may be created by an organization SME or by a specialist supplier. The C&Q Plan should be approved by the QU.

Typical commissioning deliverables associated with on-site construction may include:

- Receipt verification
- Installation Verification (IV) (including FAT as applicable for major components)
- Safety inspection/walk down
- OV (including SAT as applicable for major components)
- Transfer of custody
- Stage summary
- Development and use of the SRA
- Defining the risk assessment process
- CQAs
- CPPs
- CAs
- CDEs

2.7.3 Receipt Verification

This can be a simple process of visually inspecting the equipment on arrival at the site for damage before accepting delivery or completing a detailed check of components and functional testing/calibration of instruments.

This inspection also can be used to document instruments into a calibration database and components into the maintenance database.

2.7.4 Installation Verification (IV)

Usually, the first field effort is the Installation Verification (IV). This document provides evidence that the installation complies with approved design specifications, and that instruments are calibrated in accordance with those specifications.

It should be noted that instruments may be categorized in order to define specific calibration requirements:

- Some instruments may be acceptable with a manufacturer's certificate of conformity
- Some instruments, such as variable area flow meters, do not require routine calibration, just an inspection for wear
- Instruments critical to product quality (system output is considered the product in this instance) are usually calibrated on site to ensure that they were not damaged during shipping.

Other items generally covered in IV are the engineering quality checks carried out during construction to confirm that:

- The as-built drawing (or red line drawing if the as-built drawing was not available) accurately reflects the installation

- The correct data defining spare parts is available to put into the maintenance management system from which a bill of materials, including component documentation, is produced.
- Instrumentation checkout had been completed (point to point wiring checks and functional loop checks)
- The system has been cleaned and passivated as per specification (this may be in IV or OV).
- Fuse ratings and overload settings are as specified.
- Tags and labels are installed according to a components list.
- Pipework contents and flow direction labels have been added as specified.

2.7.5 Safety Inspection/Walk Down

Before a new or modified system is energized, and power made available, it is considered good practice to perform a safety walk down; the scope of this is generally defined in the organization's procedures and can include electrical safety, valve orientation, safety warnings, etc.

2.7.6 Operational Verification (OV)

OV provides documented evidence that a system operation meets the URS. The final stages of OV are executed after the completion of system adjustment and engineering runs are completed to establish the fully integrated and operational configuration meeting the URS. The approved operating configuration is used for PQ. For a water system, this is PQ Stage 1. Refer to PQ Section 2.9.

The scope of work may include:

- Sequence of operation testing
- Determination of adjustable set points, e.g., level settings
- Tuning of control loops and verification of controller action
- Determination and verification of control module parameters and constants
- Testing of interlocks including indication and response actions
- Verification of alarm functions
- Power loss and recovery testing
- Emergency shutdown verification (confirmation that there is no stored energy in the system)
- Testing at the specification limits, e.g., system design peak usage

C&Q lifecycle verification can be documented in separate IV and OV documents or in a combined single IOV document. FAT and SAT test documents may be used to provide data to support IV and OV. Upon completion of IV/OV, the verification testing is reviewed by an independent SME and a commissioning summary report is issued stating the system is installed and operates to meet the URS. The report should include a summary of any changes made and issues experienced during commissioning, so that the history is available to the system owner. The summary report is approved by the appropriate SME(s) and the system owner.

The commissioning summary report is referenced as part of the acceptance and release that is approved by the QU.

A Requirements Traceability Matrix (RTM) may be used by engineering to confirm specified testing has been completed.

2.7.7 Transfer of Custody

Transfer of custody to the system owner occurs after the QU has approved system acceptance and release. This may be through approval of the qualification summary report or an acceptance and release report.

2.7.8 Acceptance and Release

The qualification summary report/acceptance and release report will confirm that:

- The quality requirements in the URS have been met
- The design risk controls in the SRA have been tested and shown to operate in a manner that meets the specifications with the appropriate set points; thus, the system risk profile is confirmed as accurate.

Content may vary depending on the system but typically the qualification summary report includes:

- The approved commissioning summary report confirming that:
 - Installation/operation of the system meets specifications
 - ECM summary
- Traceability matrix showing quality requirements from the URS and where the test document can be found (these include the design controls from the SRA).
- Traceability matrix showing procedural requirements from the SRA and where they are included
- Confirmation of a parts and spares list
- List of operational and maintenance procedures
- List of personnel trained on the procedures
- Confirmation that the system is incorporated in the maintenance and calibration programs
- Acceptance and release

Approval of the qualification summary report transfers responsibility for operating and maintaining the system from the project team to the system owner.

An acceptance and release or qualification summary report is used to accept a system (rather than the traditional IQ, OQ, and PQ protocols) and release it for the relevant PQ phase as described in the C&Q plan/project validation plan.

Note: It may be considered appropriate to use the system for commercial manufacturing/product process validation on a risk basis, where no material is released until the relevant sampling testing has been completed and confirmed as acceptable. The strategy to be used should be described in the project validation plan.

2.7.9 Science and Risk-based C&Q Summary

Through the science and risk-based C&Q process Quality agrees that, from a quality perspective, the design is fit for purpose.

Quality should have approved the:

- URS
- System classification
- SRA
- DQ
- C&Q plan
- Acceptance and release
- Project validation plan

IV and OV have been completed providing documented evidence that the system is installed and operates in a manner that meets specifications.

At the completion of the C&Q, a commissioning summary report is produced. This should be approved by an independent engineer/SME (i.e., not the engineer responsible for the commissioning) to confirm that the system is fit for the intended purpose. The report should verify that there is adequate documentation confirming that the system is installed and operates in a manner that meets specifications, and that changes from the agreed design have been managed through the ECM system.

The work is conducted following GEP and the associated documentation standards and change management processes. Approval of the commissioning summary report by the SME and system owner releases the system to proceed to qualification.

2.7.10 QRM and the Risk Assessment Process

For detailed background information of QRM and risk assessment refer to ICH Q9(R2) [15] and the *ISPE Baseline® Guide: Volume 7 – Risk-Based Manufacture of Pharmaceutical Products (Second Edition)* [25].

The SRA is developed based on the proposed system design. It considers the risks to the system output based on the proposed design and anticipated controls (design and procedural controls). Assessing the risk and the controls, SMEs determine the risk level for each step of system operation. Upon completion of the SRA, the assessing team determines if the proposed design and controls are adequate to provide an acceptable risk profile for that system design.

Note: The risk being assessed is the impact to the CQAs of the system output (product). The CQAs have a direct correlation to patient safety.

If the initial assessment shows unacceptable risks, then controls are added. Upon completing the SRA, it must be mapped back to the URS to ensure all the design controls identified in the SRA are incorporated into the user requirement specifications. If the SRA process reveals additional controls not covered in the URS, the URS is updated to include these new requirements.

Note: The risk assessment may be conducted at any time during design development and is used as a tool to facilitate development of a low risk design. It becomes a quality record when it is confirmed to be representative of the final (issue for construction) design. The SRA becomes part of the validated documentation system with the supporting summary report.

The SRA is a system lifecycle document, that is, it is kept up-to-date for the life of the system.

2.7.11 Critical Quality Attributes

The CQAs described in this section are typical examples for a compendial water system.

A CQA is “A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality.” (ICH Q8(R2) [14])

CQAs should be measured with appropriate accuracy; instrumentation accuracy requirements are usually described within the relevant pharmacopeia.

For compendial water systems, CQAs are defined by the specifications in the pharmacopeia that applies to the market being supplied. Where several markets are supplied, the most demanding requirements of the relevant pharmacopeias should be met.

For compendial water systems, alert and action levels should be set for CQAs. Limits for the alert and action levels used should take account of the accuracy of the instrumentation used, as necessary.

System performance data should be trended as part of PQ; analysis of the data and review of performance trends help to provide evidence that the system is operating in a state of control, i.e., the water from the system reliably meets specifications.

Unusual or undesirable trends should be investigated and a corrective action initiated, e.g., upward trends in the system microbial levels may indicate that more frequent system sanitization is required.

Example CQAs include:

- TOC
- Conductivity (at temperature)
- Microbial levels
- Endotoxin levels

TOC

Where a TOC measurement is used for compliance purposes (e.g., release of water), the TOC measurement may be taken online or using a sample analyzed in a laboratory.

Where an online TOC measurement is for in-process control, upset indication, or information, rather than as an indication of compliance to a CQA, the online TOC instrument is not considered critical.

Conductivity

Where a conductivity measurement is used for compliance purposes (e.g., release of water), a non-temperature compensated conductivity measurement (online or laboratory value) can be used to record the CQA. Where temperature compensation is used, it should comply with the relevant pharmacopeial requirements.

Where the conductivity measurement is for in-process control or information and not an indication of compliance to a water specification attribute, an online conductivity measurement may not be considered a critical component.

Samples may be taken to measure compliance with water specifications and analyzed in a laboratory.

Temperature

Temperature is usually considered a CPPs; however, when temperature is used or reported with conductivity for water release, it may be considered a CQA.

Microbial

Where microbial tests are required, water is usually sampled and analyzed off-line in a laboratory. Microbial quality is a pharmacopeial requirement and, therefore, a CQA. Test results may be used to demonstrate conformance to the requirement.

Endotoxin

A Bacterial Endotoxin Test (BET) is normally required for WFI water distribution systems as defined by local pharmacopeia at points of use.

Examples of CQAs for a compendial steam system depend on the application. For example, there may be point of use specifications for a pure steam system per EN 285 [24] for:

- Superheat
- Dryness
- Non-condensable gas

For further information on CQAs, see the *ISPE Baseline Guide: Water and Steam Systems (Third Edition)* [8].

2.7.12 Critical Process Parameters

A CPP is defined as “A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces the desired quality.” (ICH Q8(R2) [14])

Typically, CPPs are used to create and maintain water quality. They are characterized at minimum and maximum values, which represent the qualified allowable operating range of values for the particular parameter. If a minimum or maximum value is violated, an investigation may be warranted. For example, where a WFI distribution loop has been qualified to consistently produce “in specification” WFI when operated above 70°C (158°F) and an alarm is triggered if the system temperature drops below 70°C (158°F), there is no need to trend the operating temperature of the system on an ongoing basis.

CPPs, including minimum and maximum operating ranges, should be qualified. The ability of a system to meet these parameters is usually tested during commissioning. The effects on water quality should be demonstrated as part of PQ. This includes, for example, the operating temperature range of the pretreatment and generation system for PW.

As part of gathering information to improve process understanding, CPPs for a compendial water or pure (clean) steam system should be identified and managed. For example, CPPs for a compendial water system may include:

- Water temperature
- Ozone level
- Flow rate
- Water pressure
- Water level

Temperature

Temperature levels (measured and/or controlled) may be considered:

- Critical (if directly related to a CQA (e.g., online conductivity measurement))
- Of process importance to a specific qualified system

The temperature of water (maintained or uncontrolled) may not be considered a CPPs in some sections of a compendial water system. Examples of where the temperature of water should be considered as a CPP include:

- Online conductivity measurements
- Continuous sanitization applications where a system is maintained at a specific temperature (e.g., hot or cold) to inhibit microbiological growth
- Sanitization cycle control where the proven sanitization process for a system is dependent upon the system operating at or above a specified temperature during the sanitization cycle
- Other process applications where temperature levels and, therefore, associated control, is deemed critical to a specific process

Typically, microbial contamination does not proliferate in high purity water systems above specific temperatures. Decreasing operating temperatures below ambient temperatures in cold high purity water storage and distribution systems can retard microbial growth.

Example:

For many years, a company operated a qualified PW distribution loop (storage vessel, pumps, heater, points of use, water return to vessel equipment) at uncontrolled ambient temperatures (15°C to 30°C/59°F to 86°F). Sanitization of the system included a weekly increase in loop temperature to 80°C (176°F) for two hours.

Based on process understanding gained from this existing distribution loop, a new loop was planned. The only change in the new loop was the addition of a cooler in the design. The cooler was used to maintain a normal distribution loop operating temperature of less than 15°C (59°F). Water temperature was acceptable to production operations. Based on control of the temperature with subsequent system qualification, the company was able to extend the time between sanitization from one week to two weeks, which was highly desirable from a production operation perspective. Water flow rates in both systems were very similar.

For the older system, there was uncontrolled water temperature in the system, which was acceptable to production operations. The time at temperature for sanitization was considered to be the CPP needed to keep the system in microbial control. In the newer system, however, based on qualification, both the time at temperature for sanitization and the control of distribution loop temperature to less than 15°C (59°F) were found to be CPPs needed to keep system in microbial control.

Ozone Level

The ozone level in water has a similar effect on microbial control as temperature, on both a continuous or intermittent sanitization basis. In a continuous ozonation system, there is a continuous sanitizing ozone concentration in the water storage vessel. The ozone is destroyed by UV light before the first POU in the distribution system. The presence of ozone in water is a barrier to microbial growth; however, ozone is a hazardous material. Ozone may be considered a CPP if used to control microbial levels (i.e., a CQA). For further information on ozone in water systems, see the *ISPE Good Practice Guide: Ozone Sanitization of Pharmaceutical Water Storage and Distribution Systems (Second Edition)* [26].

Flow

Water flow rate or velocity⁶ may be used to manage biofilm formation in a cold compendial water system.

In recirculating portions of a distribution system, fully turbulent flow is usually considered a minimum design requirement. Flow rate is usually verified upon system start-up and may be monitored during normal system operation. For example, flow rate may be monitored to determine the amount of water supplied to different users on the water distribution system and to confirm that turbulent flow is maintained.

In addition, the system flow rate (including the amount of water stored in a storage vessel) may be monitored and controlled as part of an overall microbial control strategy. For low water usage scenarios and where system flow is used as part of an overall microbial control strategy, water in a system may be replaced with fresh water on a routine basis.

The design flow rate and actual system usage should be demonstrated as providing water meeting the specification as part of Process Qualification. Flow rate is not usually considered a CPP as there is no proven relationship between the flow rate and biofilm formation.

Note: Where the loop return is a fixed orifice, pressure may be used as the parameter monitored to confirm flow rate.

Feedwater Contaminant Concentration

Changes in system feedwater contaminant concentrations can be either expected or unexpected.

Where the ability of pretreatment equipment to provide a quality product is affected by contaminant concentrations above or below the capacity of the pretreatment equipment to produce treatment system feedwater of a suitable quality, both minimum and maximum contaminant concentrations may be critical.

Example:

A vapor compression still is a log reduction (fractional removal) device. If there is too much microbial endotoxin in the feedwater sent to a vapor compression still, it may impact the quality of WFI produced by the still.

In this example, there is no online indication of microbial endotoxin compliance for the final pharmaceutical water produced. The concentration of microbial endotoxin in the water sent to the vapor compression still can be considered a CPP. An upper limit to the allowable endotoxin concentration in the vapor compression still feed should be considered. Routine off-line measurement of this parameter may be required.

Consideration should be given to water treatment capabilities where unexpected events (such as unusual rain levels in an area where the feedwater is sourced from surface water) may impact the ability of a pretreatment system to function adequately.

Example:

Where well water is used to source potable water for a high purity water treatment train, and wells are located primarily in agricultural areas (i.e., where fertilizers and pesticides are used), unusually high rainfall may provide an unexpected challenge to the compendial water generation equipment train.⁷

For further information on treatment system capability, see the *ISPE Baseline Guide: Water and Steam Systems (Third Edition)* [8].

⁶ The Reynolds number relates back to a specified flow rate that will result in turbulent flow, the desired operational state.

⁷ Local regulations must be complied with for the licensing and use of well water.

Water Pressure

Where a compendial water system should deliver water to a particular user at a pressure critical to downstream manufacturing operations, pressure may be identified as a CPP.

A compendial water distribution system should maintain positive water pressure relative to the surrounding outside environment at all times of operation, except for the stored water in a vessel.

Where water pressure is critical, the practice typically adopted follows the practice used for a compressed air system; a low-supply pressure alarm should be incorporated into the controls of the system using the water that has a critical supply pressure.

Where a compendial water system is susceptible to contamination because of backflow caused by inadequate pressure differentials between the system and the equipment being supplied, pressure may be considered a CPP.

Pressures in distribution systems may be needed to prevent leaks into the compendial water from the utilities and the surrounding atmosphere and are therefore considered critical. Additionally, adequate distribution loop return pressures may be needed to develop vessel internal spray ball patterns.

Alternative controls include:

- Routine pressure testing of heat exchangers, typically from the process side
- The use of sensitive system instrumentation, such as the in-line TOC measurement
- Use of block and bleed valve arrangements

These controls should confirm that there is no leakage into the compendial water from utility systems.

Water Level

Water levels in storage vessels may be monitored to control the supply of water into the storage vessels and as a control for mechanical protection and cavitation protection of downstream pumps.

The water level in storage vessels may be a determining factor in the operation of the purification train for demand of make-up and process water but water level is not normally considered to be critical to water quality.

2.7.13 Activities and Considerations for the Measurement of CQAs and CPPs

Typical activities and considerations concerning the measurement of CQAs or CPPs associated with instrumentation include:

- Verification of instrument materials of construction because of the criticality of contact with the system fluid
- Engineering and quality review of selection, configuration, and maintenance/calibration requirements
- Generation of system critical operating data – a subset of process data (typically measured values of CQAs or CPPs) that exists (measured or system generated) in electronic measurement and control systems. This data may depict the identity, purity, or quality of the water product. This data can be used during process validation.
- Strictly controlled system for tracking and replacement of the instrument
- NIST (US) or other national traceable calibration standards

- QU review of critical data
- System for calibration and maintenance of the instrument
- Measurement uncertainty analyses provide information to establish the extent of confidence limits bounding a measurable quantity

For information on data management refer to 21CFR Part 11 [42], EU Eudralex Part 4 Annex 11 [43], *ISPE GAMP® RDI Good Practice Guide: Data Integrity – Manufacturing Records* [44], *ISPE GAMP® Guide: Records and Data Integrity* [45], and *ISPE GAMP® RDI Good Practice Guide: Data Integrity by Design* [46].

2.8 Sampling

USP General Information Chapter <1231> [1] states:

“Water should be monitored at a frequency that is sufficient to ensure that the water system is in control and continues to produce water of acceptable quality ... Established sampling frequencies should be based on system validation data and should cover critical areas, including unit operation sites.”

Sampling and initial testing can commence after water flow has been established through a generation or distribution system. Typically, sampling and monitoring used to build process understanding begins during the early stages of start-up for a generation or distribution system.

Also refer to the *ISPE Good Practice Guide: Sampling for Pharmaceutical Water, Steam, and Process Gases* [47].

The sampling plan should identify which samples are taken to gain an understanding of system operation, and which are taken for compliance reasons.

Monitoring during commissioning is a GEP activity that can help to:

- Build process understanding of the system and each process step
- Define operational and maintenance practices and procedures that may be challenged as part of commissioning or PQ

The results of sampling during commissioning can help to develop plans for sampling during PQ and ongoing monitoring. Microbial testing (if required, including coliform and endotoxin) of the incoming potable water through to the end of the treatment chain ideally should begin as soon as the system is operational.

Water samples can be taken after each unit operation used to change the characteristic of the water during the pretreatment and final treatment steps of the generation system.

Typically, sampling of a storage or distribution system is performed:

- At the storage tank and at each POU servicing manufacturing, cleaning activities or a laboratory facility
- To demonstrate that the quality of water is maintained and meets the requirements of the processes being served

Sampling during the commissioning phase may be used to determine whether a system is functioning as specified. Typically, start-up samples and online chemical tests are obtained and tested to compare differences during commissioning. Microbial tests may be performed to provide start-up information.

For stainless steel systems where cleaning and passivation operations are performed to enhance corrosion control, care should be taken to begin sampling **after** these operations have been completed and the system has been rinsed and declared free of acids, bases, and chemicals.

For process engineering projects, the type of analyses performed within pretreatment steps, final steps and storage and distribution should be based on the **unit operation** (and regulatory expectations) for the desired product water.

For pretreatment steps, the typical regulatory requirement is to begin with water that meets the US Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWR) or equivalent regional (e.g., European [2] or Japanese [3]) or World Health Organization (WHO) [4] requirements for drinking water quality. Each unit operation within the pretreatment boundary (e.g., filtration, softening and dechlorination) should be sampled and analyzed for the process being performed. Sampling should occur at the inlet and outlet of each unit operation to ensure that the anticipated change takes place. Low flow rate and high flow rate design conditions may also be tested.

Table 2.3 provides examples of typical monitoring for unit operations.

Additional testing may be performed at the outlet of a unit operation to ensure that other attributes do not have an impact on the operation of downstream equipment (e.g., total heterotrophic plate count at the outlet of a carbon filter).

Additional tests also can be applied to final treatment steps, storage, and distribution. Sampling and analyses within the final treatment boundary should be selected in the same manner as performed for pretreatment, for example, sampling for total heterotrophic plate count before and after a RO step, as RO acts as a membrane barrier to microorganisms.

Further sampling and analyses may be performed to ensure continued business productivity or quality review for the next process step, which is designed for a specific feed water quality (e.g., conductivity of water supplied to an EDI unit). Subtle differences in source water quality and the local environment can later manifest as operational and/or maintenance issues.

Table 2.3: Typical Sampling Locations in High Purity Water Generation Systems

Generation System Method	Process Step to be Sampled	Monitoring
Distillation (Vapor Compression Still) Multi-Effect requires additional RO Pretreatment	Multimedia Filter	Pressure Drop, Particulate
	Softening	Hardness
	Carbon Bed	Oxidation Reduction Potential (ORP) or Free and Total Chlorine, Chloramines, Ammonia, Microbial
	Vapor Compression or Multi Effect Still	Conductivity at temperature, TOC, Microbial, Endotoxin
RO/EDI	Filter or Multimedia Filter	Pressure Drop, Particulate
	Softening	Hardness
	pH Adjustment	pH
	Dechlorination, Ultraviolet (UV), Carbon, Bisulfite	ORP or Free and Total Chlorine, Chloramines and Microbial, ORP or Free Chlorine
	Reverse Osmosis	Conductivity at temperature of Inlet, Outlet and Flow Rate (to provide % Rejection and Recovery) and Microbial
	Electrodeionization and Final filtration or Ultrafilter	Conductivity at temperature, TOC, Microbial

2.9 Performance Qualification

2.9.1 Execution of Performance Qualification

The purpose of PQ is to:

- Demonstrate that procedures, personnel, systems, and materials needed for the operation of a water system are integrated.
- Provide evidence that the system can consistently meet the water quality requirements specified in the associated URS.

Science and risk-based assessments can be performed in accordance to *ISPE Baseline Guide: C&Q (Second Edition)* [5].

During PQ, online instrumentation and PAT may be employed instead of traditional laboratory water assays to demonstrate compliance to the water specification requirements. The trend is to use online instrumentation where possible.

The use of online instrumentation with delineation for process control limits, definition of upper and lower action and specification limits, with high frequency measurement can be implemented. Integration of statistical process control and/or exponentially weighted process statistical methods can enhance the documentation and data for CPPs and CQAs and other process parameter measurements. For further information on automation, instrumentation, statistics, and process control, see *ISPE Baseline Guide: Water and Steam Systems (Third Edition)* [8], *ISPE GAMP Good Practice Guide: A Risk-Based Approach to GxP Process Control Systems (Second Edition)* [41], and *ISPE PQLI® Guide: Part 4 – Process Performance and Product Quality Monitoring System (PP&PQMS)* [48].

For water systems, PQ should demonstrate that system performance is consistent over an extended period of time. This extended period of time allows for:

- Variations in water use rate
- Initial feed water quality
- Seasonal variations, including differences in surrounding temperatures and variations in feed water quality

Dependent on the components installed in the purification system of the water system the impact of these variations should not affect the quality of the system output. Process understanding is needed in order to establish a rationale for the sampling strategy to be used during PQ.

Sampling during PQ may be either online or manual (for off-line laboratory analysis). The location and frequency of sampling is dependent upon the system design and testing phase. The chemical data collected during PQ may be obtained from online instrumentation or manually through off-line laboratory analysis. Microbial and endotoxin testing (if required) should be conducted through laboratory analysis. Sampling locations should be defined in the system design phase. The frequency of the data collection is dependent on the phase of PQ. The attributes measured are dependent upon the applicable pharmacopoeia (e.g., Ph. Eur. [2], JP [3], and USP [1]), which is usually determined by the destination of the drug product.

The sampling strategy for a distribution system should imitate the way in which the system is used during operation. For online instrumentation, the measurements should be indicative and comparable to the expected value at the POU.

An example of traditional manual sampling: if high purity water is used from a hose normally connected to the water system, water quality should be sampled from the connected hose. The strategy should have a rationale that dictates why sampling at a device (such as connected hose) or POU is warranted. The rationale for selecting the points to be sampled and frequency should be justified.

Typically, all water POU's would be sampled for microbial characteristics 'at a specified frequency or on a rotating basis. Online end of loop instrument measurements for compendial water attributes, such as TOC and conductivity, can provide accurate depictions of water quality at POU's in a distribution system. This is because physicochemical parameters are equally distributed throughout the system whereas microbial levels are not. For further information, see USP <1231> [1].

A traditional PQ sampling program is developed in three phases:

- Phase 1 is to ensure that the system is under control before entering the PQ testing phase and allowing use for production.
- Phase 2 is the first phase of PQ testing.
- Phase 3 is the remaining PQ testing period necessary to have one complete year of testing data (see Table 2.4).

Phase 1 and Phase 2 testing are typically of shorter duration with higher testing frequency.

The implementation of PAT, risk-based management, and the use of online instrumentation should measure the parameters with high frequency, eliminating the need for Phase 1, 2, and 3 regimens. Parameter data should be measured continuously, be accessible at all times, and provide complete knowledge of the process limits and values. The system can be monitored for any process anomaly or excursion, in real-time, without undue long-term data collection to verify that the process is in a state of control. For additional information see Chapter 3 in this document.

The risk-based approach to PQ may impact the sampling strategy.

A CQA for water at a specific point may be defined as needing to meet a compendial requirement. For example, the point at which conductivity may become a critical measurement would be at the critical component at the end of a PW generation system (e.g., outlet of a DI mixed bed or outlet of a Continuous Electrodeionization (CEDI) or for the storage and distribution system on the loop return to the storage tank). Sampling the conductivity at these points should occur during the C&Q process.

In the case of online instrumentation, measurements should be monitored continuously from start-up. Any deviations relating to this water CQA should be recorded according to the appropriate deviation procedure.

Non-critical attributes of water quality should be tested during commissioning, but not necessarily during PQ. Non-critical attributes of water quality may be tested to meet engineering operational needs, and to confirm adequate system process control, but do not need to be tested for quality purposes. Changes to the system or the operating procedures should be documented and managed through GMP change management. If they are included in PQ they are typically categorized for information only.

Frequency of sampling can be affected by:

- Criticality of POU
- Compendial mandates
- Adequate samples used to define expected performance levels for establishing alarm alert and action limits
- Water quality designation of a system. The robustness of the system design can counteract the need for a higher frequency of manual sampling.
- Availability of online water quality measurement data. The frequency of data recording may be set by the user, or it may be dependent on the cycle time of the instrument. Online instrumentation should use either real-time data or as near to real-time as possible.

- Stability of the sampling results from POU's. For example, during PQ, specific POU's may require extensive sampling. After demonstrating repeated results within the required specification and assessing the risk associated with not having extensive sampling results available for review, it may be justified to reduce the sampling frequency.

Table 2.4: Typical Phases of PQ for High Purity Water Distribution Systems When PAT is Not Used

Phase	Primary Objectives	Typical Duration
1	Finalize appropriate operating ranges on critical process operating parameters. Preliminary operating, cleaning and maintenance procedures (SOPs) may be further developed, but by the end of Phase 1 they must be finalized and approved. The sanitization strategy for microbial control is tested. Demonstrate production and delivery of water of the required quality. Note: Sampling must continue between phases.	Two to four weeks – continuous from start-up initiation
2	Continually review and trend all available data. Demonstrate consistent operation within established ranges when using approved SOPs. Demonstrate consistent production and delivery of water of the required quality.	Two to four weeks
3	Demonstrate extended performance. Ensure that potential seasonal variations are evaluated and treated. Evaluate effectiveness of the sanitization strategy and make adjustments if necessary.	One year
Note: When using scientific and risk-based management with PAT or advanced monitoring schemes, Phases 1, 2, and 3 are eliminated.		

2.10 Project Completion and Document Delivery

Near the end of a project, suppliers, engineering consultants, third-party inspectors, and other personnel should prepare to vacate the facility. Typically, the facility is staged for commissioning, qualification (acceptance and release), and PQ.

A project TOP summary should be provided to the facility owner summarizing what has been turned over. After project completion, there should be a project learning review with the project team including key suppliers and end users.

2.11 Performance Qualification Final Reports

Interim PQ reports may be used to provide evidence that a stage has been successfully completed or release a system to the next stage. PQ final reports should be issued to provide proof and approval of the system validation. Information typically presented in PQ final reports includes:

- All activities identified in the qualification matrix have been completed and verified. In addition, results meet acceptance criteria.
- Good documentation practices have been used.
- All test results met acceptance criteria (e.g., tolerance and range).
- All required SOPs, preventive maintenance procedures, training materials, and spare parts requirements have been developed and approved for the components and systems being qualified, or an acceptable action plan has been included to address these documentation requirements.
- Acceptance and release report(s) has been completed, including documentation of any issues identified during the execution of the qualification with a corresponding resolution.

2.12 Ongoing Quality Monitoring

During PQ, extensive data is collected to prove that the system can consistently meet specification requirements. The ongoing sampling plan may be at a lower frequency than that established during PQ, as there is a low risk of a sample failure.

For WFI systems, there is a higher risk associated with a sample failure so the ongoing sampling frequency may be very similar to that completed during PQ. All POU's should be sampled for biological contamination (microbial and endotoxin, as applicable) at a frequency determined based on the potential risk to the product. Daily sampling may be required.

For other water qualities, the frequency should be risk-based or as established by process understanding and the organization's policies and procedures. It is recommended that samples are taken at a minimum from two different POU's for each CQA so that there is additional data available in the event of a discrepancy in an assay result.

A science and risk-based approach should be taken when developing the ongoing water quality monitoring plan, continuing the logic used during PQ. There is a higher risk of microbial contamination in the water system than there is of chemical contamination. Thus, there should be a rigorous microbial and endotoxin (if applicable) testing plan to ensure the system remains in a state of control. There is less potential for chemical contamination of a water system so the monitoring plan may be less rigorous. The use of online monitoring for chemical CQAs (e.g., TOC and conductivity) is common practice with verification of the quality from the POU as determined by risk.

An ongoing quality monitoring plan should be documented in an approved SOP.

Sample points and online monitoring instrumentation should have unique identifiers that are traceable between the field valves or instruments, the ongoing sampling plan, automation system (for instrumentation), and drawings.

The water quality data should be reviewed by the QU and other key operational and technical personnel to detect adverse trends. This review should include both sample results and online water quality monitoring data and should be documented. The review can also verify that the sanitization strategy is adequate to reduce the potential for biofilm formation and ensure that the system remains in a state of control. The more frequent the review, the sooner action can be taken when an adverse trend is detected that could affect water quality.

The data collected can be used for periodic performance evaluations. The outcome of the periodic performance evaluation can justify changes to microbial alert and action levels.

For further information on process validation, see the FDA Guidance for Industry: Process Validation – General Principles and Practices [49] and ISPE PQLI Process Validation papers [50].

2.13 Periodic Performance Evaluation or Periodic Review (PR)

Pharmaceutical water and steam systems should be evaluated on a routine basis to ensure the system remains in a qualified and validated state. The frequency of evaluation should be determined based on the risk to the final drug product and is usually annually for critical systems.

The periodic performance evaluation should provide an analysis, summary, and consideration of the combined and cumulative effects of items being evaluated.

The periodic performance evaluation (also referred to as PR), may be used in conjunction with Preventive Maintenance programs. Periodic performance evaluations can include:

- An introductory section that provides information on the scope, system boundaries and the time period.

- Description of the sampling plan, including sample point identification, sample frequency, alert and action levels, and other data needed for the evaluation.
- Summary and analysis of the testing data to determine whether there are trends that justify modification to the existing sampling program or to the maintenance and operation of the system. Systems should be evaluated as a unit and with respect to control. Individual sample points also should be evaluated to detect trends.
- Identification of microbial organisms. The data obtained should be reviewed and compared to historical data to determine whether the microbial identification data results provide any useful information about the system, the environment or the sampling technique. This information can be used to determine whether objectionable organisms are present and to detect shifts in the normal flora of the system.
- Review of microbial data to determine whether microbial alert and action limits are sufficient to ensure an adequate response to system abnormalities.
- Review of parametric data (TOC and conductivity at temperature) to confirm that alert and action limits are still appropriate and that there are no inappropriate system performance trends.
- Listing of all changes performed on the system during the performance evaluation period and a holistic review of the changes to determine if an individual change or the accumulation of several changes has impacted the system's validated status.
- Listing of all deviations for the evaluation period. Review the deviations to determine whether there are any adverse trends not detected by an individual deviation investigation or if there are overlooked systemic issues that may impact the validated state of the system.
- Review of corrective maintenance performed on the system to determine whether:
 - The current maintenance program is appropriate
 - There are systemic issues that need to be addressed
- Performance of an alarm review to determine if any alarm indicates a system or equipment issues, or items that could indicate a potential water quality concern.
- System walk down to ensure the installation reflects the as-built record drawings, the tagging is complete and correct, and the system installation meets the specifications.
- Creation of attachments or an appendix containing data used to complete the periodic performance evaluation.
- A summary of recommendations and action plans, including a review of the action plans from prior periodic performance evaluations to verify that all items were completed.
- Performance of a qualification and validation assessment. Based on results from the data assessment, a conclusion should be made regarding the state of qualification or validation and whether the system is fit for its intended and continued use.
- Appropriate technical and quality reviews and validations. Approval by the QU is typically required.

The periodic performance evaluation also may incorporate the requirements and results of the associated control software periodic evaluations to alleviate the requirement for a separate document.

3 Process Analytical Technology (PAT)

3.1 Introduction

PAT is a system used to ensure process parameters denoted as CPPs and Quality Attributes, either Critical (CQAs) or non-critical, are identified in the process. These parameters show the process is under control by meeting mandated compendial parameters and values (issued by the relevant pharmacopeia or regulatory agency), applicable SOPs, and the URS.

See the FDA Guidance for Industry: PAT — A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance [51] for more information.

3.2 Definition of PAT

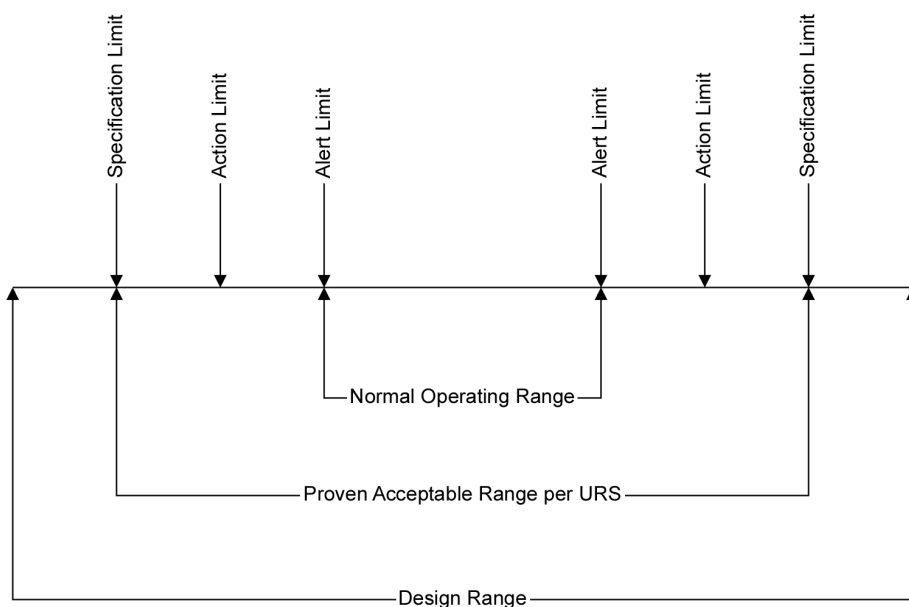
“Systems for continuous analysis and control of manufacturing processes based on real-time measurements, or rapid measurements during processing, of quality and performance attributes of raw and in-process materials and processes to assure acceptable end-product quality at the completion of the process.” [71]

Typical CPPs assigned for water consist of temperature, pressure, flow, TOC, conductivity, microbials, particulates, and entrained gases. CQAs are determined by the system owner or user and SMEs based on the quality or the required range at the POU of the water or steam system. Mandated compendial values and testing are always regarded as CQAs.

- In USP Purified Water (PW) or compendial classified Purified Water [1], TOC, conductivity, and microbials are mandated with expected limits for each parameter.
- In Water for Injection (WFI) [1], TOC, conductivity, bacterial endotoxin, and microbial limits are mandated with the limits for each parameter. Other pharmacopeia may require testing for additional parameters.
- In Pure Steam generation systems, the Pure Steam condensate is measured for all WFI mandated tests except microbials.
- PAT provides scientific principles and tools supporting innovation based on CPPs and CQAs to foster process understanding using risk-based management approaches.
- PAT sequencing includes a short duration of OQ, PQ, and validation, as product release is determined when all of the above operations are in compliance with the mandated values for each parameter.
 - If all the parameters meet the URS, compendial monographs, company specifications, and SOPs, then the in-process material can be released to either the next process step or for end-product manufacture.
 - The process itself is the validation mechanism. Understanding the process, feedback mechanisms, process knowledge, and adhering to the URS affords immediate real-time product release.
 - The use of sensors and instrumentation gives an immediate indication of the health of the system. Continuous trending and review of the data provides an ongoing health check of the system.
 - Near real-time instrumentation for endotoxin and microbial burden assessment is available in industry and can be used as an operational indicator of the health of the water system in conjunction with traditional laboratory methods.

- The use of online sensors and instrumentation, e.g., conductivity and TOC on the return loop, can eliminate manual sampling at POU, if correlation between online sensors and POU values has been verified during the C&Q process.
 - To show correlation, it is recommended to use online portable sensors/analyzers, remove sampling, and amend QC methods. Portable sensors/analyzers should, ideally, be the same type as those used in the return loop to ensure optimal correlation. Portable sensors/analyzers should follow the same calibration requirements and specifications used for sensors/analyzers on return loop. When correlations between sensors and instrumentation locations have been verified to be representative for the POU, results from sensors and instrumentation can be used for product release.
- QbD determines the design meets the design criteria specified in the URS.
 - IQ shows that the system and process are installed per design. Installation Verification (IV) verifies the installation is approved for the next phased function.
 - OQ determines that the design meets the functional requirements across the entire design range. OV verifies the operational status and is approved for the next phased function.
 - PQ shows that the process performs within the normal operational range of the process. Figure 3.1 depicts the design range and performance limits of each delineated section.

Figure 3.1 Values of Critical Parameters for Product Water [8]



4 Rapid Microbial Monitoring

Rapid Microbial Monitoring (RMM) is a broad label for many different microbial presence, quantification, and enumeration methods. RMM may be based on the real-time assessment of microbial presence, quantification, and/or enumeration.

A subset of RMM applied to pharmaceutical water systems is the Online Water Burden Analyzer (OWBA). These analyzers perform both microbial presence and quantification. OWBA units are currently used for operational data only. These units have not been approved by the US FDA for release of product, as of this publication date. Traditional sampling, inoculation on R2A media, with incubation of at least 72 hours at 30°C-35°C is the accepted methodology of the US FDA (based on USP <1116> [1]).

The USP [1] and the Ph. Eur. [2] have guidance on RMM: USP <1223> Validation of Alternative Microbial Methods and Ph. Eur. 5.1.6 Alternative Methods for Control of Microbiological Quality.

USP <1223> [1] states:

“Studies on the recovery of microorganisms from potable and environmental waters have demonstrated that traditional plate count methods reporting cell count estimates as colony forming units (CFUs) may recover 0.1%-1% of the actual microbial cells present in a sample.”

“Understanding the strengths and weaknesses of the CFU as a signal is vital in the validation of an alternative method that uses an alternative signal. The CFU cannot be considered the only unit of microbiological enumeration, because it is only an estimate of cells present rather than an absolute measure.”

“The enumerative values, given as CFU results in association with reference methods, typically cannot be used as acceptance criteria for the assessment of articles via candidate alternative methods. Instead, it is the users’ responsibility to propose values that they consider acceptable and unacceptable for the method that they have chosen; this will be done independently of existing standards expressed in terms of CFU.”

Different approaches are possible to demonstrate equivalence per USP <1223> [1]. The chapter has an interpretation concerning the demonstration of equivalence of alternative methods and traditional ones. (See Table 4.1.) Assuming that the ultimate purpose of a microbiological test is to assess the microbiological quality of products, raw materials, components, or process steps and make a decision thereon, the technologies used for the purpose may differ from those of the reference methods.

Table 4.1: Equivalence Option Matrix (USP <1223> Table 2) [1]

Option	Demonstration	Comparison to the Official Compendial Method	Based on Numerical Results or Conclusions	Number of Characteristics
1. Acceptable procedures	Acceptable	No	Results	Multiple
2. Performance equivalence	Equivalent	Yes	Results	Multiple
3. Results equivalence	Equivalent	Yes	Results	Single
4. Decision equivalence	Equivalent	Yes	Conclusions	Single
Option 1: Acceptable Procedure				
This is not strictly an equivalence option that requires direct comparison between the candidate alternative method and an official compendial method. With this option, a reference material with known properties may be used, such				

Table 4.1: Equivalence Option Matrix (USP <1223> Table 2) [1] (continued)

as a standard inoculum of a specific microorganism, a quantity of highly purified bacterial genome, an ATP level or another appropriate signal. In some cases, it could be required that the alternative method measure the signal in the presence of the test sample, with validation criteria that are consistent with the capability of the technology, as described in the scientific literature.
Option 2: Performance Equivalence Performance equivalence requires the demonstration of equivalent or better results with respect to validation criteria—such as accuracy, precision, specificity, limit of detection, limit of qualification, robustness, and ruggedness—that may be appropriate for the intended use of the alternative qualitative or quantitative method. It is possible that the alternative method may not conform to some of the validation parameters listed compared with the official method and still be acceptable because of the advantages of the alternative method. This maybe the case if the alternative method has any of the advantages stated in the General Notices, 6.30 Alternative and Harmonized Methods and Procedures. Other special circumstances would include improvements in time to obtain a result or the cost of running the test. If a candidate alternative method is suitable for assessing the quality of the material tested, it may be still acceptable, even if it differs from the official method in one or more validation parameters. The final analytical qualification criteria should reflect only the criteria that the microbiologist deems necessary to achieve performance equivalence.
Option 3: Results Equivalence When results equivalence is required, the hypothesis to be tested is that the alternative and compendial test methods give equivalent numerical results. This contrasts with the evaluation of the validation parameters, as is done in performance equivalence. Because the same sample cannot be tested in microbiology, typically a tolerance interval is established when comparing the two methods, with the alternative method determined to be numerically superior or non-inferior. Reports on the use of alternative non-growth-based methods have shown that they may produce significantly higher cell count estimates than a growth method that reports outcomes in cfu. In this case, the analyst could use a calibration curve showing a correlation between the two methods in the product specification range.
Option 4: Decision Equivalence A decision equivalence is similar to results equivalence but differs in that a numerical result is not generated; instead a pass/fail result is obtained. With this approach, the frequency of positive and negative results generated should be no worse than with the compendial method. This non-inferiority requirement is based on the long history of product quality tested by and released with the referee compendial test. For the purposes of qualification, laboratory studies involving spiking low levels of microorganisms may be considered. The following sections provide suggested approaches for demonstrating that the alternative procedure is equivalent to or better than the compendial procedure. Users may use other valid methods for demonstration of equivalence with supporting scientific justification.

Ph. Eur. General Notices [2] states:

“The tests and assays described are the official methods upon which the standards of the Pharmacopoeia are based. With the agreement of the competent authority, alternative methods of analysis may be used for control purposes, provided that the methods used enable an unequivocal decision to be made as to whether compliance with the standards of the monographs would be achieved if the official methods were used. In the event of doubt or dispute, the methods of analysis of the Pharmacopoeia are alone authoritative.”

As shown in the USP <1223> Table 2 (reproduced as Table 4.1) Equivalence Option Matrix [1], Item 2. Performance Equivalence, similar assessments need to be performed in Ph. Eur. 5.1.6 [2]: Accuracy, Precision, Specificity, Limit of Quantification, Linearity, Range, and Robustness.

Ph. Eur. 5.1.6 [2] discusses the use of alternate methods in section 4-1-2 Validation of alternative microbiological method, subsections 4-1-2-1 (Risk-benefit analysis) and 4-1-2-2 (Validation for the actual intended use).

C&Q of OWBA units should follow the USP <1223> [1] and Ph. Eur. 5.1.6 [2] guidelines, which can replace manual sampling on the distribution return loop for Process Control (PC) sampling. When using the unit for QC sampling needs, OWBA can establish QC release of the water system, but not for QC sampling at the POU for compendial mandates.

At the time of the publication, the US FDA does not accept the OWBA results for microbials for compendial release of water, as only CFU/ml values are accepted.

In instances where contamination threats are eliminated, e.g., using pure steam for SIP of fixed process equipment, OWBA may be an approved method for QC release at a POU application.

5 Appendix 1 – Typical WFI Generation and Distribution System Component Classification Example

The design of pharmacopeial water systems is based on process steps or unit operations performed by a piece of process equipment; the system is made up of a series of these steps. If the company is following the “traditional” approach (see Section 2.6), then this appendix applies.

There may be limited value in conducting a full system and component impact assessment. There is usually value in defining system functions and categorizing them in order to define the key process steps that generate the “system product CQAs” – and hence require qualification, and those that ensure the system feed to those key steps meets relevant requirements to ensure continued reliable operation of the system.

The component impact assessment can provide a useful tool to define system critical instruments, which require calibration, and those used for engineering information, which can be used for reference and/or calibrated on a less frequent basis.

An example of this approach is provided below based on the system schematic.

GEP should be used to produce documentation confirming that the system installation is completed in accordance with the specifications (installation verification), and operates in accordance with the specifications (OV).

Initial Assessment

The challenges used to determine which systems were direct impact (see *ISPE Baseline Guide: C&Q (Second Edition)* [5]) are shown in Table 5.1.

Table 5.1: Direct Impact Challenges [5]

Does the System contain CAs/CDEs or perform functions that serve to meet one or more process requirements (CQAs) including CPPs?	Yes – selection of the correct still and the associated operational settings are considered CPPs.
Does the system have direct contact with the product or process stream and does such contact have the potential to impact the final product quality or pose a risk to the patient?	Yes – system product is used to wash and provide final rinse of manufacturing equipment and primary containers.
Does the system provide an excipient or produce an ingredient or solvent (e.g., WFI) and could the quality (and compliance with the required specification thereof) of the substance impact the final product quality or pose a risk to the patient?	Yes, WFI
Is the system used in cleaning, sanitizing, or sterilizing, and could malfunction of the system result in a failure to adequately clean, sanitize, or sterilize such that a risk to the patient would result?	Yes – cleaning and feed to a pure steam generator
Does the system establish a proper environment (e.g., nitrogen blanket, closed process, exposed filling zone air quality, maintenance of temperature, humidity when such parameters are part of the product CPPs) for the process and could failure of the system to function properly pose risk to the patient?	No
Does the system use, produce, process, or store data used to accept/reject product CPPs or electronic records subject to 21 CFR part 11 [42] or EU GMP Volume 4 Annex 11 [43] or the local equivalent?	No, but system instruments are used to confirm the output meets specifications – these are categorized as critical instruments and data managed appropriately.
Does the system provide container closure or product protection, the failure of which would pose a risk to the patient or degradation of product quality?	No
Does the system provide product identification information (e.g., lot number, expiration date counterfeit protection features) without independent verification or is the system used to verify this information?	No

Use of these challenges would make all of the system process steps direct impact.

Adopting a science- and risk-based approach that utilizes GEP documents to confirm materials of construction allows a modified assessment process that can provide the degree of definition required to define which sub systems should be supported by GEP documentation, and which require additional verification (qualification).

The system shown is comprised of the process steps shown in Table 5.2:

These criteria should be used to inform a judgment based on a comprehensive understanding of the nature of the system, and not used to replace the exercise of informed judgment by appropriately qualified personnel.

Table 5.2: Process Steps

Challenge ----- Process Step	Function	The system generates or maintains a system product CQA	The system produces data that is used to accept/reject the system product	The system preserves system product quality (as defined by the CQAs)
Multimedia Filter	Remove particulate from the feed water	No	No	No
Softener	Remove calcium and magnesium salts that would coat the heat exchanger and still surfaces, reducing efficiency	No	No	No
Break Tank	Intermediate storage	No	No	No
Carbon Filter	Remove chlorine, taste, and odor	No	No	No
5 micron Filter	Remove any particle shedding from the carbon filter	No	No	No
UV Lamp	Provide microbial control of the feed water – managing microbial growth resulting from the flow through the carbon filter	No	No	No
Heating Heat Exchanger	Provides the capability to heat sanitize the pretreatment system – for biofilm control	No	No	No
Cooling Heat Exchanger	Provides an increased cooling rate after heat sanitizing, to prevent microbial growth, and maintains a consistent normal operating temperature – to assist with consistent microbial performance	No	No	No
1 micron Filter	Prevents any particulate from reaching the RO unit	No	No	No
Trailing Softener	Remove calcium and magnesium salts that would coat the heat exchanger and still surfaces, reducing efficiency	No	No	No
Pre-Heater	Improves the system efficiency by recovering heat that would otherwise be lost	No	No	No
Vapor Compression Still	Provides the chemical and microbiological control	Yes	Yes	No

Thus, the vapor compression still is considered the item requiring verification for the treatment system.

GEP documentation (installation and operational commissioning records) is considered to provide the documented record that the installation and operation complies with the specifications.

Based on the first question, the storage and distribution system (maintaining a system product CQA) also require verification.

The instrumentation on the system can be assessed using a similar approach. See Table 5.3. Note that if the science and risk based approach is used the critical instrument list is provided through completion of the SRA.

Table 5.3: Instrumentation Assessment

Process: Generation Equipment Component Classification P&ID: Owner:				Component monitors or controls critical process operating parameter.	Failure or alarm of the component will have a direct effect on quality of the product and the failure or alarm is not detected by another component downstream in the system.	Component is used for the adjustment or calibration of critical components, such as secondary/working test standard.	Information from this component is recorded as part of the product batch record or lot release data.	Critical (C) or Non-Critical (N) Instrument?
Component Item Title	Component Tag Number	Component Description	Classification Rationale					
Feed Flow Control	FE-1	Drinking Water Flowmeter	Flowmeter confirms system flow, backwash and regeneration rates	No	No	No	No	N
Pressure Indicator	P1-1, 1A	PIs for MM	Indicators are for M-M filter operation	No	No	No	No	N
Temperature Indicator	TI-1	Temperature Measurement	Gives water temperature at point in system	No	No	No	No	N
Temperature Indicator	TI-2, 3	Temperature Measurement	Temperature indicator for softener	No	No	No	No	N
Flow Element	FE-3	Water Flow Measurement	Measures water flow to break tank	No	No	No	No	N
Temperature Control Element	TE-4	Sub-loop Temperature Control Comp	Part of control system to maintain temperature during equipment sanitization	Yes	No	No	No	C
Pressure Indicator	PI-5A, 5B, 5C	Pressure Indicator on Carbon Bed	Pressure indicators for carbon bed	No	No	No	No	N
Temperature Indicator	T1-5A, 5B	Temperature Measurement	Carbon bed temperature indicator	No	No	No	No	N

Table 5.3: Instrumentation Assessment (continued)

Process: Generation Equipment Component Classification P&ID: Owner:				Component monitors or controls critical process operating parameter.	Failure or alarm of the component will have a direct effect on quality of the product and the failure or alarm is not detected by another component downstream in the system.	Component is used for the adjustment or calibration of critical components, such as secondary/working test standard.	Information from this component is recorded as part of the product batch record or lot release data.	Critical (C) or Non-Critical (N) Instrument?
Component Item Title	Component Tag Number	Component Description	Classification Rationale					
Ultraviolet Light and UV monitor	UV-1	UV Light for Microbial	Provides microbial control for carbon bed	No	No	No	No	N
Flow Element	FV-8	Feed Control to vapor compressor (VC)	Measures feed to VC – provides system capacity limit	No	Yes	No	No	C
Pressure Indicator	PT-8A	Control of the VC Operation	Indirect Impact System local plant steam pressure indicator	No	No	No	No	N
Pressure Transmitter	VC-8	Actual Equipment	Measures pressure and provides control of VC	No	Yes	No	No	C
Temperature Transmitter	FV-8A	Steam Control Valve	Measures temperature and provides control of VC	No	Yes	No	No	C
Analyzer Element	AE-8B	TOC monitor	TOC measurement alarm	Yes	Yes	No	Yes	C
Analyzer Element	AE-8B	Conductivity monitor	Conductivity measurement alarm	Yes	Yes	No	Yes	C

6 Appendix 2 – Process Water, Pure Steam, and Process Steam

The main text of this Guide concentrates on high purity water. The purpose of this appendix is to give relevant information regarding qualification items to be included in the validation lifecycle for process water, process steam, and pure steam direct impact systems.

6.1 Non-Monographed/Non-Compendial Water

Water may be used for specific early steps in the manufacturing of intermediates (e.g., antibiotic fermentation, equipment rinsing) of APIs. URS should be developed in order to understand qualification requirements for a direct impact process water system. Typically, the URS for non-monographed/non compendial water specifies that process water meets or exceeds the characteristics of drinking water.

EU guidance on water qualities is found in Guideline on the quality of water for pharmaceutical use [53].

Water meeting US EPA NPDWR regulations [54], or a regional equivalent, may meet the process needs after performing on-site processing operations to soften it and/or filtering to ensure removal of any potential particulate contamination from the storage and distribution system; softening and/or filtration may be a requirement to meet the URS. Generally, the following items are included in process water generation or distribution projects to maintain water quality.

6.1.1 *Typical Design Considerations and Verifications Performed During Commissioning/IV/OV for a Process Water System*

- The system should be designed so that it does not promote microbial growth, corrosion, or other forms of contamination. This requirement is the basis for selecting contact materials, components, and system configuration.
- The system should be capable of controlling microbial proliferation in process water generation or distribution steps. Where a sanitization method (e.g., chlorination, ozonation, flushing, or heating) is used, the relevant CPP (residual chlorine (Cl)), residual ozone (ORP or O3 monitor), flushing efficacy, or system temperature increase) should be tested/verified during commissioning.
- The system should provide backflow prevention, as needed. Backflow protection is usually required, dependent upon local codes; however, sources of potential process water contamination should be assessed as part of the project. The potential for process water contamination may determine the requirements for qualifying backflow protection methods or devices.
- Feed water quality supplied to the process water generation train should meet local potable water requirements.
- Where applicable, the system design should incorporate the capability to sample between unit operations. The system should provide documented evidence that the installation meets the specifications (IV) and the operation meets the specifications (OV).

As a minimum, documentation should include:

- As-built P&IDs
- Confirmation that the contact materials meet the specifications
- System control description and confirmation that operation meets the specifications

- Operating and maintenance procedures (including system calibration requirements)
- Third-party procedures, if outsourced

6.1.2 *Typical PQ for a Process Water System*

CQAs for sampling:

- Drinking water attributes
- Total microbial count
- Absence of coliform
- Trace elements, if applicable
- Hardness, if applicable
- Conductivity, if applicable
- TOC, if applicable
- Other attributes as defined in the URS

For a distribution system, a risk-based sampling plan to test the appropriate CQAs at POU should be established. Typically, there are three phases of PQ, and a science and risk-based sampling plan should be developed based on the properties of the water, the distribution system design, environment, considering the usage pattern, and the worst case location, which should be defined.

For example, for a system that uses chlorine as a microbial control, samples could be taken to measure the microbial quality of the water and to measure free chlorine levels to confirm that the microbial control is still effective in the worst-case conditions, for example:

- Test at the warmest temperatures
- Test the POU with the longest pipe run
- Test the POU with the longest time with the water static

A typical sampling plan could include:

- PQ Phase 1: one week – sample every POU at least once.
- PQ Phase 2: three weeks – sample beginning and end POU at least once per week.
- PQ Phase 3: sample each POU monthly.

Notes:

Where there are multiple POU on a system, and only some are for production use, the POU not used for production could be excluded from the sampling plan if the operation of them does not present a risk of contaminating the system.

Where the use of water outside specification may be a significant issue, samples could be taken more frequently. Where microbiological quality is not a process issue, the sampling plan may be adjusted accordingly, although it is recommended that a sample be taken every six months from every POU to demonstrate the system remains in a state of control.

6.2 Steam

The steps and guidelines for C&Q of direct impact steam systems should be consistent with direct impact water systems due to the close relationship between these systems. However, there are differences in the design, operation, and maintenance of a steam system that impact the C&Q efforts; for example, microbial growth is not a problem in a steam system.

6.2.1 Types of Steam

The following types of steam are typically used in pharmaceutical manufacturing operations:

- Plant Steam – steam produced by the feed of potable water or equivalent to an industrial-type boiler. Corrosion/degassing control additives may be used in the boiler system maintenance. Typically, this steam is used for not direct contact process heating and is supplied from a commissioned system.
- Chemical Free Steam (CFS) – steam produced from pretreated water with no volatile boiler additives. Non-volatile boiler additives should meet the FDA Generally Recognized as Safe (GRAS) listed additives or other equivalent international standard where applicable. Typically, CFS is used for humidification and not for product/process contact operations, and is supplied from a commissioned system.
- Process Steam – direct impact steam that once condensed, meets the quality characteristics of potable water. Typically, this steam is used in manufacturing areas for direct injection heating and sterilization and is supplied from a system that has been verified.
- USP Pure Steam – condensate is equivalent to WFI and adheres to TOC, Conductivity, and Endotoxin pharmacopeial mandates; Microbials are NOT sampled as microbials are not viable in pure steam. (See USP <1231> Section 3.1.4 [1].)

Where applicable, steam utilized for sterilizing should also meet the requirements of European Standard BS EN 285 Sterilization. Steam sterilizers. Large sterilizers [24] for autoclave use. These requirements may not be applicable for equipment SIP.

6.2.2 Commissioning and Verification

The commissioning and verification process for process contact steam systems generally follow the same documentation requirements and sequencing documented previously for water systems. Audits and verifications for steam systems also follow previously recommended practices for water systems. However, steam systems have minor variations or additions to the requirements.

6.2.3 Process Steam Commissioning and Verification

- The system should be documented (e.g., as-built record drawings and P&ID IV)
- Condensate should be removed from the distribution system to maintain steam quality.
- The system should be constructed of materials that are non-additive, non-adsorptive, and non-reactive so as not to impact the quality of the process steam for generation or distribution.
- Typically, process steam (condensate) is tested for compliance with drinking water at a predetermined interval. However, levels of CQAs, such as TOC and conductivity, may be used as indicators of possible contamination. These levels may be established during PQ and further assessed during ongoing monitoring. Levels of these quality indicators may be tested during the three phases of PQ.

6.2.4 Process Steam PQ

PQ for process steam typically follows the following, if PAT is not used

- Phase 1 may be considered as a demonstration phase. Typical levels of quality indicators of possible contamination are assessed in anticipation of Phase 2.
- Phase 2 PQ testing consists of not less than three samples for these indicators from each sampling location. Sampling frequency interval for each sample from each sampling location is not less than 24 h and not more than 7 days.
- For Phase 3 and ongoing monitoring, sample the system monthly from rotating sampling locations for the indicators of possible contamination, including the supply and POU furthest from the point of supply.

6.2.5 Pure Steam Commissioning and Verification

Pure steam should be generated by vaporization in a unit specifically designed for the purpose. Feed water to the generation unit should be verified to meet the requirements of the manufacturer to ensure that the steam condensate will meet the relevant specifications. Where the feedwater is not WFI, it is prudent to monitor endotoxin levels in the system feedwater.

Commissioning (IV and OV) should include:

- Confirmation of the trap manufacturer's minimum upstream piping length for condensate cooling at the design steam pressure.
- Confirmation that there is a condensate trap immediately prior to pressure reducing valves. Note that pressure reducing valves do not handle condensate well especially in the higher flow range. Condensate can cause erratic pressure operation of a pressure reducing valve.
- Verification confirmation that the specified materials and elastomers have been used.
- Confirmation that the installed piping/tubing system meets the specified slope for condensate removal.
- Confirmation that condensate traps are installed at all low points in the system.
- Component confirmation of valve configuration to ensure condensate removal from the body cavity and drainage through the valve.

Maintenance Procedures:

- Verification of procedures should include a verification of a maintenance procedure for condensate trap periodic operational check and maintenance program.
- Commissioning of SIP systems should include confirmation of temperature transmitter location. The recommended location of the SIP cold spot temperature transmitter is above the trap manufacturer's minimum upstream piping length for condensate cooling for the design steam pressure.
- Verification of the correct operation of all condensate traps through visual inspection of condensate removal.
- Pure steam generator capacity testing is recommended; this is typically conducted as part of the FAT, where all of the instrumentation required is available and it is possible to carry out an accurate test.
- Pure steam distribution system capacity testing is recommended consistent with the generator capacity. This can be accomplished by operating the design maximum loads, e.g., all autoclaves confirming that adequate pressure is available at each use point.

6.3 Pure Steam Quality [56]

The feedwater for a pure steam system must meet local potable water standards at a minimum. The steam quality, measured as condensate, must meet the relevant specifications for WFI, excluding microbiological requirements, but including endotoxins. There are national standards and guidelines defining the engineering specifications for steam generation plant and distribution systems, such as those from the American Society of Mechanical Engineers (ASME) [30], and *ISPE Baseline Guide: Water and Steam Systems (Third Edition)* [8]. These include material specifications, dimensions/tolerances, surface finish, material joining, and QA procedures.

There are also specifications established through EN 285, Sterilization. Steam Sterilizers. Large Sterilizers [24], for pure steam quality when used for sterilizers:

- Supply steam must have a dryness value of no less than 0.95.
- Obtain no more than 3.5 ml of gases per 100 ml of condensate.
- Superheat to be less than 25 K when expanded to atmospheric pressure.

The requirement for steam quality tests has long been a topic of discussion. Questions arise from a lack of understanding the origins of steam testing and how the principles apply to different sterilizer load types: non-porous and porous.

The heat penetration for a non-porous load comes from the heat transfer to the item through the outer surfaces; usually the challenge is that in the time taken to heat up the thermal mass, any steam superheat would be dissipated by the mass, and the impact of NCGs would be limited. A porous load, like a filter or a garment, presents a different challenge. In this case, NCGs could create a problem by lodging in the load and acting as an insulator, preventing heat transfer to the inner parts of a load item.

6.4 Steam Quality Testing [56]

Currently the standards for steam quality testing are defined in a number of documents, shown in Table 6.1; however, each differs slightly.

Table 6.1: Steam Quality Testing Standard Documents [56]

Steam Quality Testing Standard	Document Title	Notes
ANSI/AAMI ST79: 2017 [57]	Comprehensive guide to steam sterilization and sterility assurance in health care facilities	Higher requirement for steam dryness compared to EN 285 [24]
EN 285:2015+A1:2021 [24]	Sterilization. Steam sterilizers. Large sterilizers	Covers Europe and the UK
Parenteral Drug Association (PDA) Technical Report TR1 [58]	Validation of moist heat sterilization processes; Cycle design, development, qualification and ongoing control	TR 61- Steam-in-Place does not have any relevant content
Parenteral Drug Association (PDA) Technical Report TR48 [59]	Moist heat sterilizer systems: Design, commissioning, qualification and maintenance	
USP-NF 2024, Issue 1 ⁸ [1]	Monograph for Pure Steam	The level of steam saturation or dryness, and the amount of NCGs are to be determined by the Pure Steam application

The pharmaceutical/biotechnology industry typically references EN 285 [24], which has the requirements listed in Table 6.2.

Table 6.2: EN 285 Requirements [56]

Parameter	Steam Dryness Value	NCGs	Superheat
EN 285:2015+A1:2021	> 0.95 w/w	≤ 3.5 ml/100 ml	> 25 K
v/v = volume per unit of volume; w/w = weight per unit of weight. For laboratory autoclaves, > 0.90 w/w is considered acceptable.			

EN 285 [24] and HTM 2010 [60] refer to dryness as a unitless value. The term w/w is properly used when describing the dryness fraction. It should be noted that the NCG limit has changed in EN 285 and is no longer represented by a percentage.

Note that the EN 285 also describes the test locations, equipment required, and test methods.

Various aspects of the system design and installation impact the system's ability to meet the requirements; therefore, it is imperative to design and install systems properly. Each quality requirement is discussed in the following sections.

⁸ This is changed from the source to list the latest USP version.

6.5 Steam Dryness [56]

Steam dryness is important, because wet steam can cause wet loads with consequent risks to sterility if products are stored wet. Wet steam also has less enthalpy than dry steam, so a greater quantity is required to provide the equivalent heat energy. Moisture droplets can also damage pipework and valves.

The dryness value is primarily a function of the distribution system design/demand at the time of testing. Testing for dryness around the steam distribution system at critical points is a valuable way of confirming that the steam system has a competent design and is well maintained.

6.5.1 Dryness Fraction

The term “dryness fraction” implies an absolute and exact mathematical measurement of the mass of water contained in a given mass of steam. The term “dryness value,” however, is used to describe the amount of moisture present based on the EN 285/HTM 2010 [24/60] methodology, which is not exact.

A well-designed steam generator produces steam with a dryness fraction of 1. Many generators include or recommend that a separator is fitted immediately downstream of the generator outlet to ensure such a dryness fraction is achieved.

By definition, 100% dry saturated steam with a dryness fraction of 1 is steam at its condensing point and, as such, in a transient condition. Any heat loss generates condensate and the steam becomes wetter. Maintaining steam as dry as possible requires good engineering design and practice throughout the steam distribution system.

6.5.2 Condensate

As soon as the steam leaves the generator, it is distributed through a metallic pipework system kept hot by the heat gain from the steam condensing on the distribution system walls. The pipework is kept at a temperature close to the steam temperature, although there is a small temperature loss due to the boundary layer. Some of the condensate from the pipe walls is picked up by the steam being transported through the system. There will be more condensation if the insulation is poor or damaged, and more condensate will be carried by the steam, if the system has a poor fall, slope, or if it is inadequately drained. If the steam consumption increases, there will be less condensate proportionally. Typically, steam systems have a small rivulet of condensate that is removed by steam traps placed throughout the piping system.

A well-designed distribution system has a slope of not less than 100 mm per 10 m of pipe (1:100) in the direction of steam flow, with steam traps installed in pockets at 30 to 50 m (90 to 150 ft) intervals to remove any condensate that has formed.

It is good engineering practice to fit a separator just before the connection to the sterilizer to ensure as much condensate is removed as possible. Condensate in any pools can be picked up by higher steam velocities as they occur through the sterilizer cycle. Additionally, an in-line separator can remove any entrained water/condensate. Suspended water droplets are impinged on a series of baffles before flowing via gravity into a drainable outlet trap. Separators installed immediately before or after the valve are a simple but effective solution and typically remove more than 99% of condensate.

Installation verification should confirm that the installation meets the relevant specifications to ensure reliable operation.

6.5.3 Test Sample

Per EN 285 [24], the quality test sample should be taken from the center of the pipe and be representative of the steam quality used in the system. However, it is not representative of the overall steam quality at that point, and as such it is known as the dryness value. Overall steam quality can be measured using a static mixer to mix all the condensate together with the steam and measuring close to the mixing point.

In this case, the sample is representative of the overall system steam condition at that sampling point and is known as the dryness fraction. With well-functioning steam traps and separators, condensate at the bottom of the pipe will be removed, so a measurement from the center of the pipe may be considered representative.

6.5.4 *Dryness as a Diagnostic*

Accurately measuring the steam dryness fraction can be a very useful diagnostic of the overall health of a steam system. A low fraction value can be caused by:

- Damaged/degraded insulation on the steam distribution system
- Inadequate drainage of the distribution system due to insufficient fall/slope
- Pipe sagging
- Dead legs (sections of piping that do not allow steam to flow)
- Steam velocity (demand)
- Insufficient drainage slope on steam headers
- Malfunctioning steam traps/separators, preventing effective condensate removal
- Inadequate/poorly located steam traps, preventing effective condensate removal
- Clogged steam filters (if they are used), causing an unusually high pressure drop that reduces steam flow
- Poor steam generator maintenance or operation

6.5.5 *Impact of a Low Dryness Fraction*

Wet steam can cause wet loads with consequent risks to sterility if products are stored wet. Wet steam has less enthalpy than dry, so a greater quantity is required to provide the equivalent heat energy. Moisture droplets can also damage pipework and valves.

The dryness value is primarily a function of the distribution system design/demand at the time of testing. Testing for dryness around the steam distribution system at critical points is a valuable way of confirming that a steam system has a competent design and is being well maintained.

6.5.6 *Commissioning, Testing, and Monitoring*

Testing is required for steam sterilizers. If the pipework design is similar in terms of the design (length of pipe run/fall, steam trap types and location) per sterilizer connection, then only a reading on the index run is necessary. But as the pipework typically varies for each connection, each autoclave connection is typically tested for dryness value during commissioning/qualification.

6.6 Non-Condensable Gases [56]

NCGs are gases that are entrained in the steam during generation. Air and other NCGs act as insulators and should be minimized in pharmaceutical steam systems. Such impurities offer a highly effective barrier to steam penetration and heat transfer, resulting in a reduced load temperature or absence of moisture at the interface with the load. With porous load (such as gowns), the gas may prevent penetration of the load, which could mean lower temperatures for system components or process equipment, potentially leading to incomplete sterilization.

Thermostatic steam traps are placed within the distribution system in positions where air is prone to collect, such as the terminal points of the main and large branches of the steam header. Working on the basis that air is heavier than steam in the distribution system, the traps separate and remove the NCGs to improve the quality of the steam. Although this is true under static conditions, when steam is flowing, NCGs travel in the direction of flow. Good practice is to also fit air vents at the end of branches and at system high points, because excessive levels of air may slow down the discharge of condensate. Excess water from subcooled condensate can cause insufficient sterilization temperatures.

NCGs are a function of the feedwater quality and the effectiveness of any degassing system. Whether through preheating feedwater and allowing it to vent, or through the use of a separator on the generator, the levels will vary depending on feedwater quality and on the system state and flow rate. It should also be noted that levels may be higher after a period of nonuse (e.g., overnight). Even small amounts of NCGs can accumulate in the steam distribution system and subsequently can be pushed out as a large volume on start-up. This is mitigated in a well-designed system by ensuring the steam distribution system has proper air venting arrangements.

For SIP system (including a tank or lyophilizer), the efficacy of the design for the vents and system drainage are confirmed during commissioning. The location of the cold spot is typically identified and used for ongoing monitoring or periodic performance assessment. This is considered adequate control and it is not necessary to confirm NCG levels for a steam system feeding an SIP system.

6.6.1 High NCG Value Causes

High NCG values can be caused by a number of conditions:

- Air ingress into the distribution system (e.g., if the system is shutdown overnight, a vacuum forms as it cools, potentially pulling in gases). A preoperational cycle on the autoclave can help manage this by flushing the system with steam
- Inadequate venting of the steam distribution (or, in the case of SIP for a pipework system, inadequate venting of the pipework system)
- Inadequate deaeration of the steam generator feedwater/degassing of the steam
- Leaking glands on steam valves that allow the compressed air used for the valve actuators to enter into the steam system

6.6.2 High NCG Impact

NCGs are released when the steam condenses. For an autoclave, this is at the interface with the load. Therefore, as more steam comes in to fill the void created by the change in volume created by steam condensing, more gas is released. The gas creates insulating pockets, preventing the surface of the load from reaching the desired temperature. Degasification either at or downstream of the steam generator can easily rid the system of NCGs. There has never been a product quality issue cited due to NCGs; however, NCGs will not appear in a root cause analysis if there is no knowledge or understanding of them. A typical scenario is where there is a failure(s) of a Biological Indicator (BI) and the response is to increase sterilization time way in excess of what should be necessary to inactivate the BI.

6.6.3 Commissioning, Testing, and Monitoring

NCGs are a function of the feedwater quality and particularly of temperature, system state and flow rate, the distribution system, venting design, and demand at the time of testing. Testing should be completed for each sterilizer prior to qualification according to the procedure described in EN 285 [24] to confirm compliance and identify the worst-case location.

If a particular location (typically the index run) on a system consistently gives higher results than other points, that single point should be verified annually, with all points verified if the result at that point is a failure or marginal. (This strategy assumes that the original testing was carried out under normal operating conditions; therefore, the readings are typical.)

For an SIP system (including a tank or lyophilizer) commissioning typically maps the system to ensure that there is adequate venting and drainage to obtain uniform temperature distribution. The location of the cold spot is typically identified and used for ongoing monitoring or periodic performance assessment.

6.7 Superheated Steam [56]

Superheated steam is steam at a temperature higher than its vaporization point at the absolute pressure where the temperature is measured. It can therefore cool by a certain amount without changing state from a gas to a mixture of saturated vapor and liquid [61]. Superheated steam heats or cools convectively, whereas condensing steam heats directly by giving up its latent heat of vaporization. The heat transfer coefficient of the two mechanisms is very different.

For example, humans breathe superheated air. At atmospheric pressure, the vaporization point for air is -194.35°C (-317.83°F). If the ambient air temperature is 20°C , then the air is superheated by 214.35°C (385.83°F) [61].

The heating convective coefficient at atmospheric pressure is likely to be in the range of $5\text{--}100\text{ W/m}^2\text{C}$ depending on steam velocity [62]. For steam condensing on a flat vertical surface, the value is $4,000^{\circ}\text{C}$ to $11,300\text{ W/m}^2\text{C}$. Superheated steam is therefore up to 11 times less effective than saturated steam as a heating agent. Although energy transfer plays no part in the F0 (minimum time-temperature see USP <1229.1> [1]) calculation, the rate of the transfer affects the time required to reach the desired temperature. Fortunately, the amount of energy in superheat in a typical pharmaceutical plant steam system is small and easily dissipated by any wetness or heat transfer to the load requiring sterilization, resulting in saturated steam.

This can be illustrated with the following example: 100% dry (dryness fraction of 1) steam at 7.0 barg (101.5 psig) is passed through a pressure reducing valve to reduce the pressure to 1.037 barg (15.05 psig). Such a pressure reduction is isenthalpic and an adiabatic calculation shows that the steam temperature after pressure reduction would be 149.76°C compared to a saturated temperature at 1.037 barg of 120.8°C – in other words nearly 29°C of superheat. See Table 6.3 for other examples.

Although this might seem a large number, it represents less than 3% of the energy available from condensing the low-pressure steam in the sterilizer. In most installations, there is enough residual “wetness” to absorb this energy or sufficient “heat leakage” in the piping before the sterilizer to dissipate any superheat.

Sterilization is achieved from the transfer of heat energy contained in saturated steam through condensation when the load temperature is raised sufficiently to inactivate bioburden loads, proteins, and other potential pathogens; sterilization occurs because of the presence of temperature and moisture. The energy level and rate of its transfer does not play a part in the sterilizing effect. Also, sterilization occurs in fluid loads in the absence of latent heat. Where the steam is superheated, the heat transfer during the initial cooling phase from the superheated steam temperature to the saturation temperature is not as efficient as it is after the steam cools to the saturation temperature and where condensation occurs—the condensate improves the heat transfer raising the temperature, with the time at temperature sterilizing the material.

6.7.1 Potential Issues

Certain conditions may present a problem and must be watched for:

- Steam that is dry saturated (or close to it) can be subject to a significant pressure drop.

Note: The risk increases with drier steam and larger pressure drops.

- A jacket temperature or pressure that is too high in an autoclave can effectively superheat the steam as it enters the autoclave.
- Steam flowing through a small orifice or a tight-radiused direction change between its source and the chamber or equipment can cause a large pressure reduction or steam velocity increase with no pipework to allow superheat to dissipate after the fitting/orifice (which could be a valve).
- The system cannot maintain adequate pressure. For a simple system, steam is generated at a pressure slightly higher than that required by the user to allow for the pressure losses in the distribution system. For a large installation, the distribution pressure is typically significantly higher to ensure that there is adequate pressure to supply demand when multiple users call for steam at the same time. In this type of design, a pressure reducing valve is used on the supply to the use points.
- A poorly designed system reduces the pressure above a ratio (n) of 2:1 and not provide an adequate length of pipework for the steam to equilibrate before going to the control valve. Excessive pressure reduction can result in the steam generating significant superheat. With the more usual ratios, the quality of the steam changes (the process is isenthalpic (i.e., the enthalpy value remains constant), hence the changes in the steam quality).

Table 6.3: Examples of Superheat Values [56]

Pressure/Dryness Fraction	
Before Pressure Drop	After Pressure Drop
5 BarA/0.95	3.2 BarA/0.96
5 BarA/0.98	3.2 BarA/0.99
5 BarA/0.95	2.1 BarA/0.97
5 BarA/0.98	2.1 BarA/1.0
5 BarA/0.98	1.0 BarA/at 10°C of superheat

Good design limits the stage pressure reduction per stage (not more than 2:1 per HTM 2010 Part 2 paragraph 7.20 [60]) and allows an adequate length of pipe for the steam to reach equilibrium before it is added to the autoclave chamber. Per HTM 2010 Part 2 [60], “where the supply pressure at the inlet to the sterilizer would exceed the maximum value specified by the manufacturer, a pressure reducing system and separator should be fitted to the supply pipe at least 3 meters (approximately 10 ft) from the sterilizer. Heat loss from the section between the pressure reducing system and the sterilizer will help prevent superheating.”

Note that pure steam generators typically do not have the capability to produce superheated steam. The water is heated and evaporates, passing through a separator to prevent moisture droplets from being carried over with the steam. The steam from the generator is ideally dry saturated steam (with a dryness fraction of 1). The system is not designed to add heat to the steam or to create superheat.

6.7.2 Excess Superheat Impact

For a pharmaceutical system, the quality of the steam from the generator is generally consistent: superheat is a function of the distribution system design/flow rate.

For an SIP system (including a tank or lyophilizer), there is adequate pipework or metalwork for any superheat to be reduced. Due to the significant ratio of surface area to volume of the system, commissioning typically also temperature-maps the system to ensure there is adequate venting and drainage to obtain uniform temperature distribution; because of this, it is not considered necessary to measure superheat.

6.7.2.1 Typical PQ Considerations for Pure Steam Systems

The quality testing for USP Pure Steam systems generally follow the same testing requirements and sequencing documented previously for water systems though where the system is fed by a qualified WFI or PW system the duration may be reduced as there are less variables [61]. Typically, sampling strategies and programs for new systems or major revisions and sampling strategies/programs for minor revisions to existing steam systems follow the previous recommendations for direct impact water systems. However, due to the sterilizing properties of steam systems, the quality testing for minor revisions may be reduced based on an assessment of any risks introduced by implementing the modification.

An understanding of the steam quality attributes and process requirements allows a better definition of the testing required; see Table 6.4.

Table 6.4: Proposed Test Requirements for Commissioning/Qualification and Ongoing Monitoring [56]

System Type	Commissioning/Qualification				Monitoring			
	Chem/endotoxin	NCG	Dryness	SHT	Chem/endotoxin*	NCG	Dryness	SHT
SIP	X	N/A	N/A	N/A	Q	N/A	N/A	N/A
LYO	X	N/A	N/A	N/A	Q	N/A	N/A	N/A
Autoclave	X	X	X	X	Q	Q*	A	N/A
Stopper Processor	X	X	X	X	Q	Q*	A	N/A
A = annually; LYO = lyophilizer; N/A = not applicable; Q = quarterly; SHT = Superheat. Q* = quarterly initially until data is available to support reduced testing; this assumes the steam generator feedwater is routinely monitored.								

- Testing the pure steam quality is recommended as described in EN 285/ISO17665 [24, 55] where the steam is used for an autoclave, with dryness and NCGs for SIP applications. However, POU of lower criticality may not be tested (with appropriate justification) depending on the use of the steam.
- Pure steam generators typically include a condensate cooler for obtaining samples and continuously monitoring the TOC and conductivity of the condensate. It is recommended to obtain the quality sample of the condensate from the pure steam generator from the same condensate cooler utilized for continuous monitoring.
- Where testing of steam to meet EN 285 [24]/ISO 17665 [55] (i.e., dryness, superheat, and NCGs) is required, piping should have the appropriate configuration and fittings – details are provided in the ISO standard. Testing should be conducted at the locations per the standard.
- PQ testing is typically performed in three phases. Condensate chemical quality is usually tested for the same chemical attributes as that of WFI. Endotoxin content also is limited to that of WFI microbial testing is not required – note this does not exclude endotoxin testing.
- The frequency of tests for physical attributes should be based upon potential risk. An example testing program is:
 - Phase 1 – (start-up) approximately three days – sample each POU at least once. Sample each pure steam generator outlet at least once.
 - Phase 2 – (system consistency/stability) approximately one week – sample each POU at least once. Sample each generator more than once.
 - Phase 3 – (deviations) approximately four weeks – sample the system weekly rotating POUs to cover all POUs in a four week period. Sample the generator outlet weekly.

6.7.2.2 Ongoing Monitoring

The sampling strategy should be established, typically monthly sample from generator outlet, outlet furthest from the generator, and another outlet, changing this sample point each month.

Maintenance activities should include the requirement for an additional test if the routine breaks the system integrity.

The use of the steam impacts the sampling strategy, for an autoclave, superheat levels can be tested during commissioning. Because the pipework to each system varies slightly, each autoclave should be tested during commissioning. For an autoclave or SIP system, superheat is usually effectively monitored through the comparison of the temperature and pressure function in the automation. Alarm systems are in place to highlight a significant mismatch. A sample test regime proposed is shown in Table 6.4 supported by a risk assessment; see Table 6.5 and supporting Tables 6.6, 6.7, and 6.8.

Table 6.5: Sample Risk Assessment Used to support Proposed Sample Strategy [56]

Potential Failure Mode	Potential Effect of Failure	Severity	Potential Cause	Likelihood	Design Controls	Operational Controls	Detection	Detection	Recommendation	Risk Level
Clean Steam does not meet specification at point of use. (Chemistry, Endotoxin and Physical Properties)	Ineffective sterilization in steam sterilizers (autoclaves, stopper processers)	5	High non-condensable gases	3	Clean steam generator supplied with pre-heated water/fitted with degasser function	Monitoring of degassing critical parameter (temperature of feedwater or evaporator temperature in case of integral deaerator)	Commissioning/Qualification testing of capability, and routine monitoring of evaporator or feedwater water temperature (depending on degasser type)	3	Initial performance testing at generation and supply to each sterilizer. Routine monitoring at worst case point through annual testing at furthest autoclave or end of header.	Low
			Excess Superheat	1	Piping design – limited pressure drop per stage with adequate pipework for fluid equilibration	Company SOPs require the steam supply to be qualified as part of the sterilizer commissioning/qualification	Confirmation of superheat levels during commissioning/qualification		Initial performance testing at steam generator and supply to each sterilizer. Re verification if changes are made to local piping.	
		5	Low Dryness (Excess water droplets)	3	Generation system incorporates droplet separator. Distribution system incorporating drainage (falls), trapping, and specified levels of insulation, with a pocket and steam trap on the connection to the sterilizer.	Company SOPs require the steam supply to be qualified as part of the sterilizer commissioning/qualification. Annual system survey to detect changes or deterioration in distribution piping insulation, and confirm tagging is in place.	Confirmation of superheat levels during commissioning/qualification. Periodic testing at representative sample point	3	Testing at generation and supply to each sterilizer. Routine monitoring at worst case point through annual testing at furthest autoclave or end of header.	Low
	Product contamination due to high bacterial endotoxin in steam.	5	High endotoxin feedwater, carried over into steam	5	Generation system supplied with endotoxin controlled feedwater.	Company SOPs require the steam supply to be qualified as part of the sterilizer commissioning/qualification.	Quarterly sampling of feedwater for endotoxin.	1	Quarterly sampling of feedwater for endotoxin.	Low

Table 6.5: Sample Risk Assessment Used to support Proposed Sample Strategy [56] (continued)

Potential Failure Mode	Potential Effect of Failure	Severity	Potential Cause	Likelihood	Design Controls	Operational Controls	Detection	Detection	Recommendation	Risk Level
Clean Steam does not meet specification at point of use. (Chemistry, Endotoxin and Physical Properties)	Product contamination due to high bacterial endotoxin in steam (continued)	5	Ineffective separation and removal of endotoxins at generation	3	Multi stage impurity separation in clean steam generator, and blowdown to provide consistent feed quality	Routine verification (annual) of blowdown volume	Routine monitoring of pure steam condensate	3	Routine monitoring of pure steam condensate	Low
(continued)	Product contamination due presence of organic carbon	5	High TOC in feedwater, carried over into steam	3	Generation unit supplied with TOC controlled feedwater	On-line TOC monitoring of feedwater	Alarm from on line monitoring instrument	1	Continuous monitoring of the system feedwater.	Low
		5	Ineffective separation and removal at generation	3	Multi stage impurity separation in clean steam generator, and blowdown to remove impurities	Routine verification (annual) of blowdown volume	Alarm from on line monitoring instrument	1	Continuous monitoring of the generation system condensate, routine testing at representative sample point(pipework index(longest) run)	Low
	Product contamination due to high conductivity	5	High conductivity in feedwater, carried over into steam	5	Generation unit supplied with conductivity controlled feedwater	On-line Conductivity monitoring of feedwater	Alarm from on line monitoring instrument	1	Continuous monitoring of the system feedwater.	Low
		5	Ineffective separation and removal at generation	3	Multi stage impurity separation in clean steam generator, and blowdown to remove impurities	On-line Conductivity monitoring of feedwater	Alarm from on line monitoring instrument	3	Continuous monitoring of the generation system condensate, routine testing at representative sample point(pipework index(longest) run)	Low

Table 6.6: Risk Assessment Scoring [56]

Rating	SEVERITY of the effect of failure (System/Equipment)	Likelihood of OCCURRENCE	Ability to DETECT the failure
9	Severe – Serious impact to QA of the output of the system/ equipment and impact to final product quality attribute	Frequent – Failure is almost inevitable Consistent failures observed	Absolutely uncertain – Existing controls cannot detect the failure No controls are in place
7	Major – Significant impact to QA of the output of the system/ equipment and possible impact to final product quality attribute	Likely – Failure is likely and will occur in most circumstances. Repeated failures observed	Remote – Remote chance that controls will detect the failure. A control may be in place but is untested or unreliable
5	Moderate – Possible impact to QA of the output of the system/ equipment and no impact to final product quality attribute	Occasional – Failure is probable at some time and has been observed	Moderate – A moderate chance that the control will detect the failure
3	Minor – Minor impact to QA of the output of the system/ equipment and no impact to final product quality attribute	Unlikely – Failure could occur at some time. Only isolated incidents observed	High – Very likely that the control will detect the failure
1	Insignificant – No Impact to QA of the output of the system/ equipment and no impact to final product quality attribute	Remote – Failure is extremely unlikely. No history of failure	Almost certain – The control will detect the failure in almost every instance

Table 6.7: Severity Ratings [56]

		Severity Rating				
		1 Insignificant	3 Minor	5 Moderate	7 Major	9 Severe
Likelihood Rating	9 – Frequent	Medium	Medium	High	High	High
	7 – Likely	Low	Medium	High	High	High
	5 – Occasional	Low	Medium	Medium	High	High
	3 – Unlikely	Low	Low	Medium	Medium	High
	1 – Remote	Low	Low	Low	Low	Medium

Table 6.8: Detection Ratings [56]

		Detection Rating				
		1 Almost Certain	3 High	5 Moderate	7 Remote	B Nil
*from Table 6.7						
Rating*	High	Low	Medium	High	High	High
	Medium	Low	Low	Medium	High	High
	Low	Low	Low	Low	Medium	Medium

7 Appendix 3 – System Classification

This appendix provides examples of system classifications for a typical high purity water system. Figures 7.1 – 7.4 are included to clarify terminologies used within this Guide related to loop outlet valve, POU valve, POU, and sample point. See Appendix 9 (Glossary) for further definitions.

Figure 7.1: Example of Compendial Water Loop Connected to Process Equipment with Additional Piping [26]

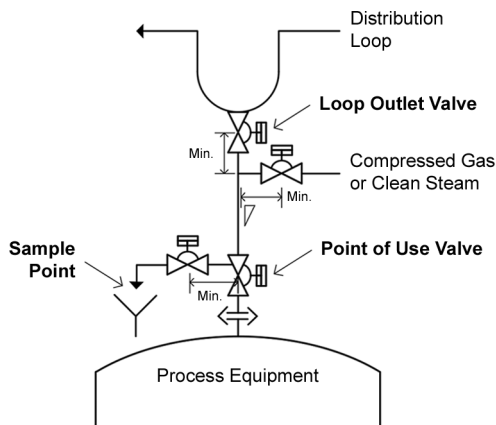


Figure 7.2: Example of Compendial Water Loop Connected to Process Equipment Directly [26]

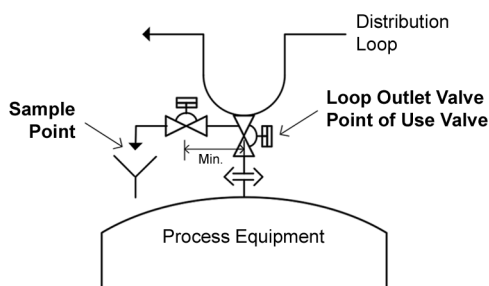


Figure 7.3: Example of Compendial Water Loop without Direct Process Equipment Connection [26]

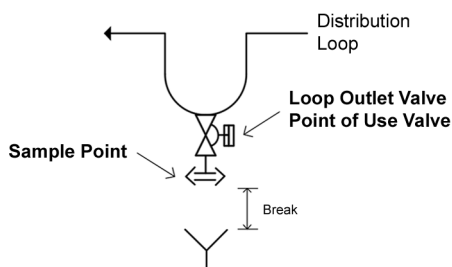
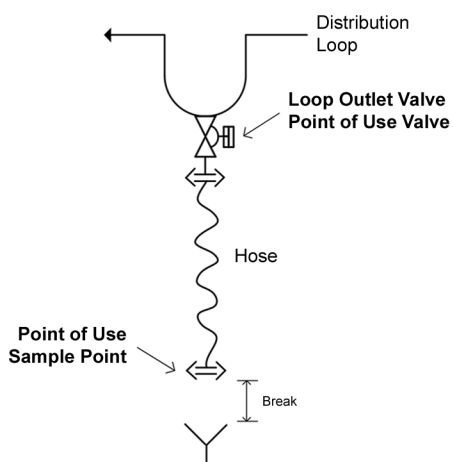


Figure 7.4: Example of Compensial Water Loop with Hose without Direct Process Equipment Connection [26]



8 Appendix 4 – Typical Specifications for Water and Steam Systems

8.1 Introduction

The requirements for a new system should be defined in a URS; therefore, if no URS is available, it should be created as a first step. If a URS is available, it should be assessed to determine if changes are needed. If company procedures allow the use of addenda, any changes or updates to the URS may be included in an addendum. (This may help to keep the main document easy to understand.

The first draft of the URS should be developed at the beginning of a project. After this, the URS document should be maintained throughout its lifecycle.

The designer (usually an architect and engineering company or specialist supplier) uses the URS and supporting company specifications and standards to develop the specifications for the system. Commissioning provides documented evidence that the installation and operation meet these specifications.

8.2 User Requirements

The URS gives the critical user requirements by describing what the water or steam system must do, but does not describe how the requirement is to be achieved, allowing the designer to propose the most effective solution. The URS should not duplicate information in the company specifications or standards, or the commercial documentation, thus ensuring that the overall process is as efficient as possible. Note that the specifications may be specific to markets or the facility they serve, but they can also be defined on a global basis. Examples of requirements are shown in Sections 8.2.1 – 8.2.8.

8.2.1 Purified Water/WFI – Quality Requirements

- WFI water or distillate from the WFI generation train should meet the most onerous requirements of the USP [1], Ph. Eur. [2], JP [3], China P [6], India P [63], Br. P [7], MP [64], and/or WHO [4] specification for WFI. Product from the PW generation train should meet the most onerous requirements from USP, Ph. Eur., JP, China P, Br. P, MP, and/or WHO specifications for PW.
- The treatment train should be capable of removing ammonia to the detection limit of the measurement instrument.
- TOC and conductivity instruments used for the system should meet the specifications given in the USP, Ph. Eur., JP, China P, or WHO.
- General methods, defined in USP, Ph. Eur., JP, China P, or WHO should be used for establishing trends in the number of CFUs observed in the routine microbiological monitoring of WFI and PW.
- The distribution system should provide WFI/PW quality water meeting the specifications at all system POUs.
- In-line TOC and conductivity measurements should be the primary methods to approve WFI quality compliance.
- Distribution piping systems should be designed without dead legs (areas of piping where the flow stops and cannot be drained). Reynolds number in the recirculating piping section of the distribution loop should be sufficient for fully turbulent conditions at all times of system operation.
- Distribution system, if deemed hot, should be able to maintain water $\geq 70^{\circ}\text{C}$ ($\geq 158^{\circ}\text{F}$) at all times of operation.

8.2.2 Process Water – Quality Requirements

- “Process water,” delivered to POU, should meet the quality specification for US EPA NPDWR drinking water [54] or appropriate national drinking water specifications. Additionally, no coliforms should be present. Finally, microbial colony forming level shall be equal to or less than 500 CFU/ml.
- The appropriate national drinking water quality testing methods should be used to determine compliance with the drinking water specification.

8.2.3 USP Pure Steam – Quality Requirements

- Pure steam from the generator, upon condensation, should meet the same chemical and endotoxin quality specifications as WFI.

Note: microbials are not measured in Pure Steam systems.

- Pure steam from the generator should have a dryness value of not less than 0.9 for porous loads for autoclaving, a dryness value of not less than 0.95 for metal loads for autoclaving, superheat content, which does not exceed 25°C (77°F) and non-condensable gas content not exceeding 3.5% by volume. Some steam applications do not require non-condensable gas limits.

8.2.4 Process Steam – Quality Requirements

- “Process steam,” delivered to points of use and once condensed, should meet the quality specification for US EPA NPDWR drinking water, or appropriate national drinking water specifications.
- Any additives for boiler operation should be FDA “food” grade additives meeting 21 CFR 173.310 [68] or equivalent.
- “Process steam,” delivered to POU, should contain superheated content, which does not exceed 20°C (68°F).

8.2.5 PW/WFI – Operational Requirements

- For the PW distribution loop, the system should be able to maintain ozone levels at 0.04 ppm or higher in water in the storage vessel.

8.2.6 USP Pure Steam – Operational Requirements

- Example: The pure steam generator should provide 454 kg/h (1000 pounds/h) of pure steam at 4.2 kg/sq cm (60 psi) at discharge of the generator.
- Generator blowdown should be between 5% and 15% of generator capacity during normal operation.

8.2.7 Process Steam – Operational Requirements

- Example: “Process steam” distribution system should simultaneously provide 45 kg/h (100 pounds/h) of process steam at 2.8 kg/sq cm (40 psi) to six use points.
- An on-line conductivity/temperature meter should be used to measure the conductivity of the “process steam” condensate.

8.3 Detailed Design Specification/Functional Design Specification

The DDS provides the information necessary to install and set up the system, specifying how to build the direct impact water or steam system. The designer/specialist supplier should also develop a FDS describing how the components of the system will work, and how they will operate as a system. Typically, compliance with the DDS and FDS is verified during commissioning. Sections 8.3.1 – 8.3.4 are typical items included in the DDS for water or steam systems.

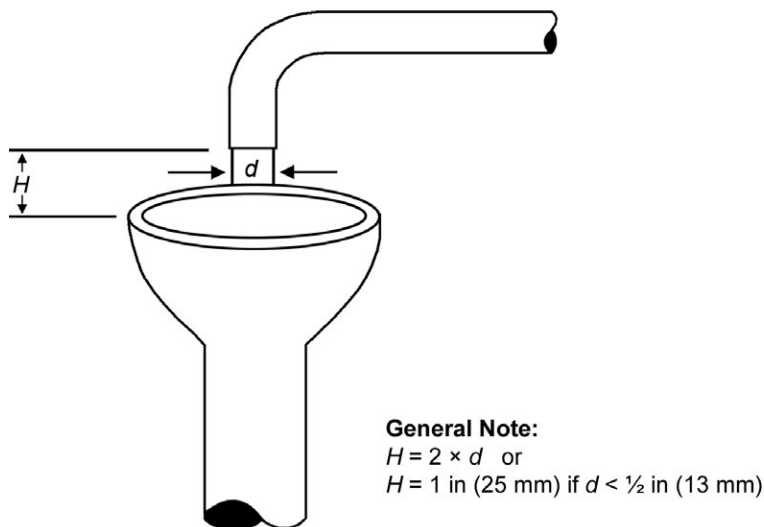
8.3.1 Purified Water/WFI

- Process/product-contact surfaces for tanks, piping/tubing, valves, pumps, and other components that contact the WFI should be manufactured from 316L SS and have Ra finishes per Section SF of the ASME BPE [36].
- All branch valves, which separate branch piping from the main loop, should be installed per the ASME BPE [36] requirements for “dead legs.”

8.3.2 Process Water

- The drawings for the process water distribution system should be modified to ensure that they represent as-built status during construction.
- Air gaps should be used to separate (provide back flow protection) process water POU and drains, as shown in Figure 8.1.

Figure 8.1: Drain with Air Gap (Break)



8.3.3 USP Pure Steam

- Provide an isometric drawing for the systems.
- Pure steam piping distribution system should be constructed of 316L SS.
- All pipework should drain to low points fitted with steam traps to remove condensate.
- Provide welding logs and borescoping

8.3.4 Process Steam

- The drawings for the process water distribution system should be modified to ensure that they represent as-built status during construction.
- Provide a certified drawing for the portable steam condensate cooler.
- All pipework should drain to low points fitted with steam traps to remove condensate.

9 Appendix 5 – Sampling and Testing Plan

In the C&Q of pharmaceutical water and steam systems, the precise sampling and rigorous testing of both microbial and chemical parameters are critical. Commissioning and pre-validation water testing of the pretreatment, generation, and distribution systems is an exercise in GEP. This activity contributes to the development of operational and maintenance practices that are confirmed during qualification and provides baseline information for future operations.

The testing activities encompass chemical testing for specific unit operations and microbial testing to identify the microbial bioburden. The duration of commissioning activities varies with system complexity and size. Once the entire system is operational, the effluent of each unit operation should be monitored according to the qualification plan to establish baseline values. Individual sampling sites should be chosen based on a risk-based assessment of the POU per the actual production or process.

These procedures are essential for mitigating risks associated with biofilm formation and chemical contamination, ensuring compliance with stringent pharmaceutical standards. As water is introduced into the system, microbial control measures such as sanitization need to be established. Microbial sampling should be performed between qualification phases to assess the efficiency of the microbial control measures employed. This ensures that the process remains under stringent control, across all purification steps, and within the distribution system where biofilm formation can pose a risk.

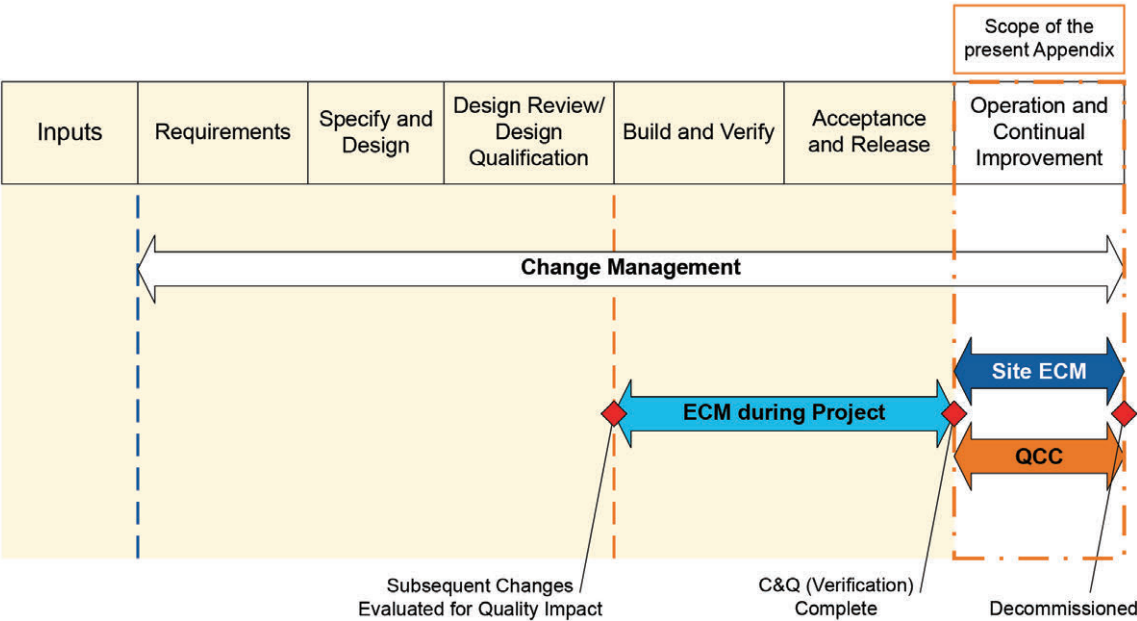
Sampling techniques, sampling points of use, and maintaining sampling regimens are described in the *ISPE Good Practice Guide: Sampling for Pharmaceutical Water, Steam, and Process Gases* [47].

10 Appendix 6 – Change Management During the Operation of Pharmaceutical Water and Steam Systems

Pharmaceutical water and steam systems require change management programs in place during the C&Q project execution lifecycle and during the operation of the system until decommission. Change management programs are a regulatory requirement for direct impact systems, which are crucial to ensure the potential risks to patient safety and product quality are properly addressed. These programs may include other benefits such as facilitating cross-departmental collaboration, and provide a clear accountability and summary of the change to build stakeholders confidence, inter alia.

The purpose of this appendix is not to navigate through the change management processes during the C&Q initial project execution lifecycle, since this part is covered in Chapter 2, but to give relevant information about specific steps to consider for the C&Q for changes in an operational pharmaceutical water and/or steam system.

Figure 10.1: Change Management Application during C&Q Project Execution Lifecycle



Specific information regarding change management processes in pharmaceutical manufacturing premises, including the purpose and benefits, the principles and concepts, and the two different modes of change management proposed; ECM and Quality Change Control (QCC), can be found in the *ISPE Baseline Guide: C&Q (Second Edition)* [5].

10.1 Considerations

The design, configuration and operation of pharmaceutical water and steam systems is particular and specific for every site, as is the range of changes that can be made to the systems. Consequently, it is not possible to define a straightforward strategy for commissioning and validating the systems after modifications that covers all options. Nevertheless, it is possible to provide some common or general points to consider. The scope of the C&Q strategy for the different changes should be agreed upon between the different involved roles, such as SMEs, process owners, and the QU (if direct impact systems are affected).

Identical changes (typically called “like-for-like”) may have a special consideration and may not require a formal change management process. To classify a change as like-for-like, a robust assessment should be in place. Given the nature of some like-for-like changes, there may be a set of predefined actions, such as sampling, to be completed before the system is returned to service. It is essential to evaluate the absence of impact on direct impact systems or the product, while also considering the predefined actions applicable to specific changes when determining whether a change qualifies as like-for-like.

Changes affecting not direct impact systems may not require a change management program in place per a regulatory perspective, but it is still highly beneficial to have an established one. QU involvement is not necessary in these changes.

Flexibility is a core principle of change management systems, enabling them to adapt to the wide range of system changes possibilities.

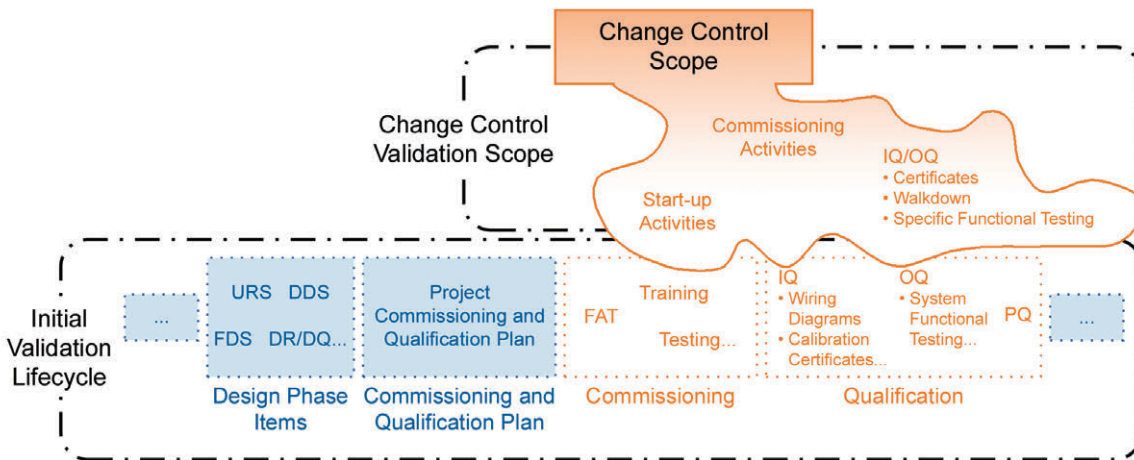
10.2 System Validation

10.1.1 Defining the Validation Steps

As part of the change management program it should be identified, among other attributes, the specific commissioning and validation phases and activities affected and required to be performed. Determining in advance which system CQAs/CPPs/CAs/CDEs may be impacted is key to determining these activities. Additionally, assessing the scope and magnitude of the change plays a crucial role in determining the extent of every required activity.

The different validation life cycle tasks (see Chapter 2) are subject to be repeated or partially repeated. Not necessarily all the initial C&Q stages are to be repeated due to a change, and as with an initial validation, it is the responsibility of every organization to determine the different testing structure along with the titles of the record documents. For example, is the system URS affected due to the change? Does it make sense to execute a DQ for a minor modification of a system or not? Does it make sense to conduct all the testing in a single record or is it better in multiple ones? These are examples of questions that, considering the regulations and internal organizational policies, the change control team should assess.

Figure 10.2: Qualitative Representation of Validation Tasks to Be Performed Due to Change Control



Note: Figure 10.2 conceptually represents the validation scope for a change control in contrast to the system initial validation lifecycle. It is the responsibility of the organization to assess the change control activities for the different changes.

10.2.2 Installation Testing

The installation testing phase needs to demonstrate that the system modifications are correctly installed and meet the defined design elements, industry standards and regulations. As part of the C&Q strategy it is essential to determine not only which modifications or elements are newly installed in the system, but also to assess which already existent and validated ones are affected for the new change installation. For instance, consider the situation where it is necessary to install a new element in existing pipework. Validated elements (e.g., valves, sensors) may need to be dismantled and reinstalled. These elements might be affected during this process, thus a correct Installation testing verification of them could be necessary.

It is essential to understand which are the Installation testing requirements for the system and for the performed modification (general requirements for these systems in Chapter 2 of this guide), and to understand which ones are affected.

Examples:

Installing a new POU for the system might require an update of drawings and electrical documentation, a walk down execution, multiple certificates verification (surface, materials, welding, passivation, etc.), an update of the maintenance plan, a hydrostatic pressure test, among others.

In contrast to the first example, a modification of the operation of the system (for example a modification in alarms level) might only have an affectation to the system documentation during the installation testing phase.

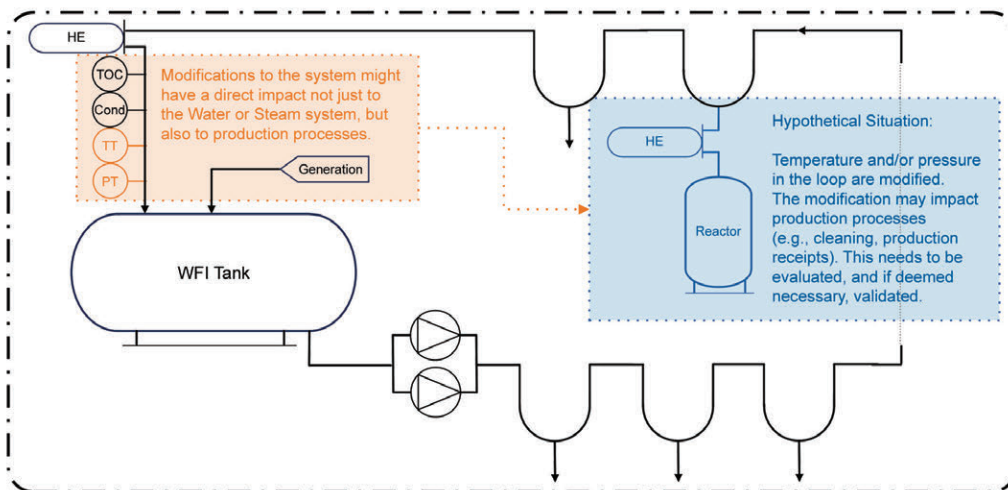
10.2.3 Operational/Functional Testing

Some changes to the system affect its functionality. Hence, it is required to verify that these functionalities are working as expected and meet the system CAs/CDEs requirements, when applicable.

Changes affecting the operation of the system can increase the complexity of the implementation of the change once it is approved, if they are directly related to the installation. In consequence, the commissioning and validation efforts might also be increased. It is of special consideration to understand how familiar people are applying the modification to the system. Is it the same supplier who did the initial design of the system? Is it another one? How old is the system? How standard or personalized is the equipment? Depending on the outcome, and the degree of uncertainty, the validation efforts might require adjustment.

Pharmaceutical water and steam systems are commonly used in conjunction with different process systems. When applying modifications to these systems, it is not only important to consider their effects, but also their effect on the systems serving as utilities. An example is the modification of the temperature or the pressure of a WFI loop, which might affect production processes. These are points need to be identified, and if determined necessary, to be verified as well.

Figure 10.3: Water and Steam Systems Modifications and Their Possible Effect on Manufacturing Processes



Example:

Consider a scenario in which the organization plans to modify the thermal sanitization parameters of a PW loop by reducing both the time and temperature. The goal is to enhance energy efficiency and minimize system downtime. It is fundamental to test that the new sanitization parameters are achieved as expected, since this has a direct impact to the microbiological parameters of the system (CQA). Likewise, at some point during the commissioning and validation of the change it is necessary to directly verify that these microbiological parameters are correct over time.

In defining the Operational/Functional testing, it is key to evaluate the effects and functionality to CAs and/or CDEs, and therefore to CPPs and/or CQAs.

10.2.4 Performance Qualification Testing

Once the Installation verification and Operational/functional phases are completed, depending on the change, it may be required to perform PQ Testing. To determine this, it is essential to assess whether the modification could impact the system's CPPs or CQAs, and to ensure the system can reliably meet the established requirements. Not only physical modifications to the system or modifications to the functionalities of the system can have an effect on the CPPs or CQAs, but also modifications in how the system is operated.

As stated in Chapter 2, a traditional PQ sampling program for pharmaceutical water systems is developed in three phases, reducing the sampling frequency between phases as confidence in the system stability and reliability increases. In an operational system where it is proven that the system is reliable and continuously provides water with the expected quality, modifications on the system affecting CQAs or CPPs should not directly involve repeating the three PQ phases, and it might be enough to conduct a few days of sampling verification. Depending on the organizational preferences, the sampling can be done as part of a specific PQ or directly as a monitoring task as part of the change control system. Situations where major changes are applied, where the reliability of the system after the change and over time is not well known and/or the seasonality is required to be evaluated, may involve two or three phases of PQ evaluation. If PAT is used instead of PQ, continual monitoring with PAT application obviates the need for PQ. Pure Steam systems do not have microbial risks due to high temperatures; therefore, the sampling phase is typically significantly shorter than in water systems.

As important as determining the PQ phases to perform is determining the sampling frequency and locations. As changes can be very location specific, some sampling locations may represent better evidence of the system performance after the change. It is necessary to consider the different natures of distribution in water systems for chemical and biological CQAs. PAT techniques can have an important role in this strategy (for more information about PAT techniques, refer to Chapter 3). This sampling evaluation should be part of the specific strategy for a change.

11 Appendix 7 – Sample Turnover Package Requirements

Turnover packages provide documented evidence that the system has been installed in accordance with the specifications (installation verification) as well as detailed information to help set up the system's operation.

Some companies create a separate installation verification file with the original record documents, keeping a copy in the turnover package, so that the turnover package is maintained as a complete set of system records.

Typical contents for a turnover package include:

- URS
- Risk assessments/list of CAs
- Spares list – recommended spares
- Spares list – consumables
- Material certification/certificates of conformity
- Construction records
- Quality inspection records
- DDS
- FDS
- As-built record drawings and specifications
- Parts list
- Calibration certificates

This information may include supplier recommended instrument calibration and maintenance schedules, pump operation instructions, safety requirements, etc.

Typically, construction records or TOP include:

- Copy of the detailed design specifications
- As-built drawings (may be markup copies initially)
- Engineering change records (where applicable)

- Material certification:
 - Heat certificates for metals
 - Certificates of conformity for nonmetallic materials and instruments
 - Certificate of Conformity (CE) for European adherence
 - Record of surface finishes
- Weld purge gas certification
- Welder's certification
- Weld procedures
- Weld coupons
- Weld inspection reports/videos
- Weld log
- Pressure test
- Cleaning and passivation report/test results
- Calibration certificates for system instruments, the welding equipment, and construction QC test instruments

Turnover Package

- General information
 - Purchase order plus change orders
 - Specifications and datasheets
 - Drawings (P&IDs, supplier-provided, shop, record)
 - Punchlist(s)
 - Start-up and commissioning (proposed and executed plans with results and reports)
 - Warranties
 - System release memo before start-up, commissioning, or validation; this should include the pre-start-up safety report

- Equipment
 - System documentation (surface finish reports, welding procedures and qualification, welder certification, weld map). Consult ASME BPE [36] for more detailed information.
 - System documentation (specifications, datasheets, bill of materials, drawings, performance charts, certifications, inspection and pressure test reports, recommended spare parts list, installation guides and OEM manuals, ASME certificates, test records and calculations, USP <87> and <88> documentation [1], 21 CFR Part 177 [28] documentation, and ASME BPE [36].)
 - SAT (proposed and executed plans with results and reports)
- Piping
 - Piping system documentation (isometric drawings, stamped seismic details, submittals and material test reports and certifications, surface finish tests and certifications)
 - Piping/tubing weld documentation (weld procedures, isometric drawings, weld logs, welder certification, weld inspection, gas certifications)
 - Pipe/tube brazing/soldering documentation (procedures and inert gas certificates)
 - Installation documentation/field verification (piping slope, pressure test, flushing/cleaning, and passivation reports, isometric drawings with slope verification)
 - Installation documentation/field verification (piping/tube slope and hydrostatic pressure test reports and as-built or record drawings)
- Electrical
 - Single line drawings
 - Electrical submittals
 - Electrical equipment checkout electronic equipment installation
 - Declaration of Electrical Safety/Electrical Safety Certification
 - Motor checklist
 - Pump operation instructions
- Instruments
 - Instrument index (instrument list, I/O list)
 - Instrument submittals (specifications, datasheets, loop diagrams, O&M manuals, calibration, control valve field verification reports)
 - Instrument punch list
 - Instrument field verification
 - Supplier recommended instrument calibration and maintenance schedules

- Controls
 - Control specifications
 - Controls submittals
 - Operator interface
 - Control panel
 - Software submittals (software development standards, design description, operator interface panels, screen prints, executed test plans, electronics, and hard program copy)
 - Sequence of operation (executed testing procedure)
 - Controls checkout reports (loop check, field power source, input/output, alarms)

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13 Appendix 9 – Glossary

13.1 Acronyms and Abbreviations

ATMP	Advanced Therapy Medicinal Products
BET	Bacterial Endotoxin Test
BI	Biological Indicator
BOD	Basis of Design
C&Q	Commissioning and Qualification
CA	Critical Aspects
CDE	Critical Design Element
CE	Certificate of Conformity (EU)
CEDI	Continuous Electro De-ionization
CFU	Colony Forming Units
CIP	Clean in Place
CP	China Pharmacopoeia
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
DDS	Detailed Design Specification
DI	Deionized Water
DQ	Design Qualification
DR	Design Review
ECM	Engineering Change Management
EDI	Electrodeionization
EMA	European Medicine Agency (EU)
EPA	Environmental Protection Agency (US)
EQP	Engineering Quality Process
FAT	Factory Acceptance Test
FB	Brazilian Pharmacopoeia
FDA	Food and Drug Administration (US)
FDS	Functional Design Specification
FS	Functional Specification
GEP	Good Engineering Practice
GMP	Good Manufacturing Practice
GRAS	Generally Recognized as Safe (US FDA)
GxP	Good “x” Practice
HMI	Human-Machine Interface

HSE	Health, Safety, Environmental
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IFC	Issued for Construction
IOV	Installation/Operational Verification
IQ	Installation Qualification
IT	Information Technology
IV	Installation Verification
JP	Japan Pharmacopoeia
NCG	Non-Condensable Gases
NIST	National Institute of Standards and Technology
NPDWR	National Primary Drinking Water Regulations (US)
OEM	Original Equipment Manufacturer
OQ	Operational Qualification
ORP	Oxidation Reduction Potential
OSHA	Occupational Safety and Health Administration (US)
OV	Operational Verification
OWBA	Online Water Burden Analyzer
P&ID	Piping and Instrumentation Diagram
PAT	Process Analytical Technology
PC	Process Control
PDA	Parenteral Drug Association
PFD	Process Flow Diagram
PLC	Programmable Logic Controller
PM	Preventive Maintenance
POU	Point of Use
PP	Polypropylene
PPE	Personal Protective Equipment
PQ	Performance Qualification
PV	Process Variable
PVDF	Polyvinylidene Fluoride
PW	Purified Water
QA	Quality Assurance
QbD	Quality by Design
QC	Quality Control
QCC	Quality Change Control
QMS	Quality Management System

QRM	Quality Risk Management
QS	Quality Systems
QU	Quality Unit
RACI	Responsible, Accountable, Consulted, Informed
RMM	Rapid Microbial Monitoring
RO	Reverse Osmosis
RSD	Relative Standard Deviation
RTM	Requirements Traceability Matrix
SAT	Site Acceptance Test
SIP	Steam-In-Place
SME	Subject Matter Experts
SOP	Standard Operating Procedure
SP	Set Point
SPP	System Process Parameter
SQA	System Quality Attribute
SRA	System Risk Assessment
TOC	Total Organic Carbon
TOP	Turnover Package
URS	User Requirements Specification
US	United States
USP	United States Pharmacopeia
UV	Ultraviolet
VC	Vapor compressor
WFI	Water for Injection
WHO	World Health Organization

13.2 Definitions

Action Level

Levels or ranges (actual attribute values) that, when exceeded, indicate that a process has drifted from its normal operating range. Exceeding an action level indicates that corrective action should be taken to bring the process back into its normal operating range.

Alert Level

Levels or ranges (actual attribute values) that, when exceeded, indicate that a process may have drifted from its normal operating condition. Alert levels constitute a warning and do not necessarily require corrective action.

Active Pharmaceutical Ingredient (API)

Any substance or mixture of substances intended to be used in the manufacture of a drug (medicinal) product and that, when used in the production of a drug, becomes an active ingredient of the drug product. Such substances are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease or to affect the structure and function of the body.

As-Built Drawings (Record Drawings)

Construction drawings that represent the physical condition of the plant or system at turnover from the designer or installer at satisfactory operation. These documents supplement and complement the system manuals and protocols.

Calibration

A comparison of an instrument measurement standard or instrument of known accuracy to detect, correlate, report, or eliminate by adjustment any variation in the accuracy of the unknown standard or instrument.

Change Management System

A systematic approach to proposing, evaluating, approving, implementing, and reviewing changes.

Commissioning

A well planned, documented, and managed engineering approach to the start-up and turnover of facilities, systems, and equipment to the end user that results in a safe and functional environment that meets established design requirements and stakeholder expectations.

In this Guide, the term commissioning describes the GEP activity used to provide documented evidence that a system is installed and operates to meet the specifications.

Commissioning Plan

A plan that defines the facilities, systems, and equipment that will be commissioned based on agreed system boundaries. It also defines how they will be commissioned and what the responsibilities for the associated staff personnel are.

Compendial

Official; purported to comply with USP, EP, or JP.

Control Loop

A group of devices that compare the process variable to a required process condition defined as a set point. Should the process variable deviate from the desired Set Point (SP), the controller will perform a control algorithm based on the error between PV and SP that will then adjust a final control device (valve, actuator, heater, motor, damper, etc.) to bring the PV back to the desired SP.

Critical Aspects (CA) [5]

Are typical functions, features, abilities, and performance or characteristics necessary for the manufacturing process and systems to ensure that CPPs are consistently controlled, and other requirements are met (e.g., regulatory requirements). They should be identified and documented based on scientific product and process understanding.

Critical Component

A component within a system where the operation, contact, data, control, alarm, or failure may have a direct impact on the quality of the product.

Critical Design Element [5]

A design feature or function of an engineered system that is necessary to consistently produce products with the desired quality attributes.

Critical Process Operating Parameter

A processing variable (temperature, pressure, level, pH, etc.) that may influence the final water or steam quality characterization or impurity profile.

Critical Process Parameter (CPP)

A process parameter whose variability impacts a critical quality attribute and therefore needs to be controlled to ensure the process produces the desired quality. (ICH Q8(R2) [14])

A critical process parameter remains critical even if it is controlled.

Critical Quality Attribute (CQA)

A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. (ICH Q8(R2) [14])

Attributes of the water or steam product usually relate to the identity, purity, or quality of the water or steam product. Some attributes for water and steam are contained levels of TOC, conductivity, microbial activity, etc.

A CQA is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality. CQAs are generally associated with the drug substance, excipients, intermediates (in-process materials) and drug product.

Critical Operating Data (COD)

Subset of process critical operational data (typically, measured values of critical quality attributes) that exists (measured or system generated) in electronic measurement and control systems. This data may depict the identity, purity, or quality of the water or steam product.

Design Qualification (DQ) (also called Enhanced Design Review)

Documented verification that the proposed design of the facilities, equipment, or systems is suitable for the intended purpose.

Detailed Design Specification (DDS)

The Detailed Design Specification (DDS) gives the detailed design requirements. The DDS specifies **how to build** the direct impact water or steam system. Typically, the DDS is tested or verified during commissioning or qualification.

Direct Impact System

A system, which is expected to have a direct impact on final water or steam quality. These systems are designed and commissioned in line with good engineering practice, and in addition, are subject to qualification practices that incorporate the enhanced review, control, and testing against specifications or other requirements necessary for GMP compliance.

Drinking Water (USP [1])

Drinking water is not covered by a compendial monograph but must comply with the critical quality attributes of the US EPA NPDWR or comparable regulations of the European Union, World Health Organization (WHO), or Japan. It may be derived from a variety of sources including a public water utility, a private water supply (e.g., a well), or a combination of more than one of these sources. Drinking water may be used in the early stages of chemical synthesis and in the early stages of the cleaning of pharmaceutical equipment. It is the prescribed source feed water for the production of pharmaceutical waters. As seasonal variations in the critical quality attributes of the drinking water supply can occur, processing steps in the production of pharmaceutical waters must be designed for this characteristic.

Enhanced Design Review (EDR)

A documented review of the design, at an appropriate stage in a project, for conformance to operational and regulatory expectations.

Factory Acceptance Test (FAT)

The partial commissioning and qualification of equipment and/or systems prior to their shipment from the fabricator's site.

Final Device Loop

A group of devices that manipulate the process in a manner that affects the process variable.

Functional Design Specification (FDS)

The Functional Design Specification (FDS) gives the functional design requirements. The FDS says how the direct impact water or steam system will perform its function. Typically, the FDS is tested or verified during commissioning or qualification.

Good Engineering Practice (GEP)

Established engineering methods and standards that are applied throughout a project's "lifecycle" to deliver appropriate, cost-effective solutions.

High Purity Water

Water conforming to USP Monographs or equivalent.

Impact Assessment

The process of evaluating the impact of the operating, controlling, alarming, and failure conditions of a system on the quality of a product.

Inspection

The process by which the construction and installation is verified as in accordance with the detailed design, specified construction standards and materials, and any relevant legal or regulatory requirements relating to these areas.

Installation Qualification (IQ)

For direct impact systems, the documented verification shows that all aspects of a facility, utility or equipment that can affect product quality adhere to approved specifications (e.g., construction, materials) and are correctly installed.

Installation Verification (IV) [5]

The *activity* of reviewing, inspecting, testing, checking, auditing, or otherwise establishing and documenting whether items, processes, services, or documents conform to specified requirements.

Loop Check

Different than a loop test, loop check is a process to verify that the components of a measurement loop, control loop, or final device loop, not only are wired correctly, but to confirm that they are “operating as designed.” The loop check can include defining the loop performance benchmark and providing a method for monitoring the performance over time.

Loop Test

Procedure to verify that a group of instruments (transmitters, controllers, recorders, alarms, marshaling panels, input/output modules and any other devices) perform together correctly to achieve a process control strategy.

Measurement Loop

A group of devices that are connected together to infer a process variable (e.g., flow, temperature, pressure, level, etc.), and convert this process variable into a process measurement signal.

Measurement Uncertainty (MU)

Data analyses which provide information to establish the extent of confidence limits bounding a measurable quantity.

No Impact System

A system that will not have any impact, either directly or indirectly, on product quality. These systems are designed and commissioned following good engineering practice only.

Non-Critical Component

A component within a system where the operation, contact, data control, alarm, or failure will have an indirect impact or no impact on the quality of the product.

Online Water Burden Analyzer (OWBA)

An online method and instrument to determine microbial presence, enumeration, or viability. Different methodologies may be employed. Acceptance of the OWBA is referenced in USP <1223> [1] and Ph. Eur. 5.1.6 [2].

Operational Qualification (OQ)

For direct impact systems, the documented verification shows that all aspects of a facility, utility, or equipment that can affect product quality operate as intended throughout all anticipated operating ranges.

Operational Verification (OV)

The activity of reviewing, testing, checking, or otherwise establishing and documenting that a system operates according to written and pre-approved specifications throughout all specified operating ranges.

Ozone

Ozone is a very strong gaseous oxidizing agent. It is used in deionized water systems to kill bacteria and to reduce, by oxidation, the amount of TOC in the water. Ozone is O₃ and due to reaction with other things rapidly becomes oxygen (O₂). Therefore, it has a short but effective oxidizing potential. It can be destructive to ion exchange using membrane filters and other plastic materials in the system. Ozone can be toxic to human beings. Regulatory standards for Ozone exposure can be found in OSHA standards: 29 CFR 1910.1000 Table Z-1 [65].

Process Analytical Technology (PAT) (ICH Q8(R2) [14])

A system for designing, analyzing, and controlling manufacturing through timely measurements (i.e., during processing) of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality.

Performance Qualification (PQ)

For direct impact systems, the documented verification that all aspects of a facility, utility, or equipment that can affect product quality, perform as intended meeting predetermined acceptance criteria.

Periodic Review (PR)

A documented assessment of documentation, procedures, records, and performance to ensure that facilities, equipment, and systems continue to be fit for purpose and safety regulatory compliance requirements. The frequency of periodic review is dependent upon the complexity, criticality, and rate of change.

Potable Water

Water that is suitable for drinking.

Pure Steam (formerly known as Clean Steam)

Pure steam is water that has been heated above 100°C and vaporized in a manner that prevents source water entrainment. It is prepared from water complying with the US Environmental Protection Agency National Primary Drinking Water Regulations, or with drinking water regulations of the European Union or Japan, or with WHO drinking water guidelines. It contains no added substance. The level of steam saturation or dryness, and the amount of NCGs are to be determined by the Pure Steam application. When condensed, the pure steam condensate meets the quality specification for WFI.

Purified Water

USP Purified Water: Prepared from water complying with the critical quality attributes of drinking water with conductivity in accordance with Stage 1, 2, and 3 tests and conductivity tables. Total organic carbon limit is at 0.5 mg/l. typically less than 100 CFU/ml (10,000 CFU/100 ml) for microbiological acceptability.

Process Contact Surfaces

Surfaces of process components that, under normal design operating conditions are in direct contact with, or have the potential to come in contact with raw materials, in-process materials, Active Pharmaceutical Ingredients (APIs), clean utilities (e.g., WFI, pure steam, CIP solutions, process gases), and where there is a potential for a component (e.g., stoppers) surface to impact drug product(s) strength, identity, safety, purity, and quality.

Product-Contact Surfaces

Surfaces of process components that are in direct contact with/or have the potential to come in contact with drug product(s) where drug product(s) are defined by owner/user. Examples of product-contact surfaces may include the interior surfaces of bioreactors, process tanks, storage vessels, transfer piping/tubing, pumps, chromatography columns, and recirculating segments of CIP systems.

Qualification

The process to demonstrate the ability to fulfill specified requirements.

Qualification Summary Report

Report summarizing the results of protocol execution.

Quality (ICH Q6A [12])

The suitability of either a drug substance or drug product for its intended use. This term includes such attributes as the identity, strength, and purity.

Quality by Design (QbD) (ICH Q8(R2) [14])

A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.

Quality Control

Process or group responsible for coordinating the activities associated with analytical test planning and execution

Quality Unit (QU) (also known as Quality Assurance)

The activity or group responsible for ensuring that the facility and systems meet GMP requirements.

Quality System (ICH Q9 [15])

The sum of all aspects of a system that implements quality policy and ensures that quality objectives are met.

Reverse Osmosis (RO)

A process that reverses (by the application of pressure) the flow of water in the natural process of osmosis so that it passes from the more concentrated to the more dilute solution. This is one of the processes used to reduce the ionic TDS, TOC, and suspended materials of feed water through a semipermeable membrane leaving dissolved and suspended materials behind. These are swept away in a waste stream to drain.

Risk (ISO/IEC Guide 51 [66])

The combination of the probability of occurrence of harm and the severity of that harm.

Risk Control (ISO Guide 73 [67])

Actions implementing risk management decisions.

Rapid Microbial Monitoring (RMM)

The use of real or near real-time methods to determine microbial presence, enumeration, or viability. Determination of the acceptance of this technology is referenced in USP <1223> [1] and Ph. Eur. 5.1.6. [2]

Sequence of Operation

A detailed description of the system start-up, normal operation, and cleaning (as applicable), process monitoring, data acquisition and archive, alarm conditions and response, and shutdown.

Site Acceptance Test (SAT)

Inspection and/or dynamic testing of the systems or major system components to support the qualification of an equipment system conducted and documented at the manufacturing site.

Specifications

Documentation which clearly and explicitly defines the system requirements, codes, and standards to be followed during fabrication and construction, test requirements, acceptance criteria, and the associated deliverables

Standard Operating Procedure (SOP)

Written and approved procedures to ensure that activities are performed the same way each time. A comprehensive SOP program must be in place in any regulated organization.

System

An organization of engineering components, which have a defined operational function, (e.g., piping, instrumentation, equipment, facilities, computer hardware, computer software, etc.).

System Boundary

A limit drawn around a system to logically define what is and is not included in the system.

System Process Parameter (SPP)

A term used for process variables that have an impact on a system Quality Attribute (SQA) and, therefore, should be monitored or controlled to ensure the system product obtains the desired quality.

This is also known as an Equipment Process Parameter.

System Quality Attribute (SQA)

This is a physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired system product quality.

System Quality Attribute and System Process Parameter

The terms used in the *ISPE Baseline Guide: C&Q (Second Edition)* [5] are CQA and CPP, as defined in ICH Q8(R2) [14]. Some companies prefer to use these terms for product but not for qualification.

Where equipment is used for a number of products, or, as in this case where the system is a utility system, the CQAs/ CPPs may not be the same for all products (as defined in the regulatory filings). In this case, a platform approach to the system risk assessment may be used that address all the relevant CQAs and CPPs and the ICH definition refers to product.

For this reason, during C&Q, a company may choose to use different terminology, modifying the definitions from ICH Q8(R2) to encompass process and utility systems; typical terms seen include:

This is also known as an Equipment Quality Attribute.

Turnover Package (TOP)

A collection of pertinent design, construction, supplier, and operational documentation. This collection of documentation is used for the qualification and process validation activity, as well as reference and single source information for the life of any particular system, process, or piece of equipment.

User Requirement Specification (URS)

A description of the requirements of the facility in terms of product to be manufactured required throughput and conditions in which the product should be made.

Validation

Establishing documented evidence which provides a high degree of assurance that a specific process will consistently produce a product meeting its predetermined specifications and critical quality attributes.

Validation Lifecycle

Listing of a series of events within the life of a direct impact manufacturing system which are identified as important to create or maintain product quality over the life of the direct impact system.

Validation Master Plan

A high-level document, which establishes an umbrella validation plan for the entire project and is used as guidance to the project team for resource and technical planning.

Verification (ISO [69]) (1) (ASTM [70]) (2)

(1) Confirmation, through the provision of objective evidence that specified requirements have been fulfilled. (2) A systematic approach to verify that manufacturing systems, acting singly or in combination, are fit for intended use, have been properly installed, and are operating correctly. This is an umbrella term that encompasses all types of approaches to assuring systems are fit for use such as qualification, commissioning and qualification, verification, system validation, or other.

Water for Injection (WFI)

USP water for injection. Prepared from water complying with the critical quality attributes of drinking water. Prepared produced by distillation or a process that is equivalent or superior to distillation in the removal of chemical and microorganisms. Conductivity in accordance with Stage 1, 2, and 3 tests and conductivity tables. Total organic carbon limit is at 0.5 mg/l. Typically, less than 10 CFU/100 ml for microbiological acceptability. Less than 0.25 EU/ml.



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